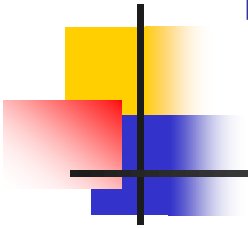


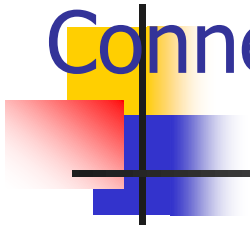
# 2. Network Models

# Design issues for the Layers



- Mechanism for identifying senders and receivers
- Rules for data transfer (Simplex, Half Duplex, Full Duplex)
- Error control mechanism
- Flow control Mechanism
- Data sequencing.
- Multiplexing and Demultiplexing Mechanism
- Routing

# Connection oriented and connectionless services



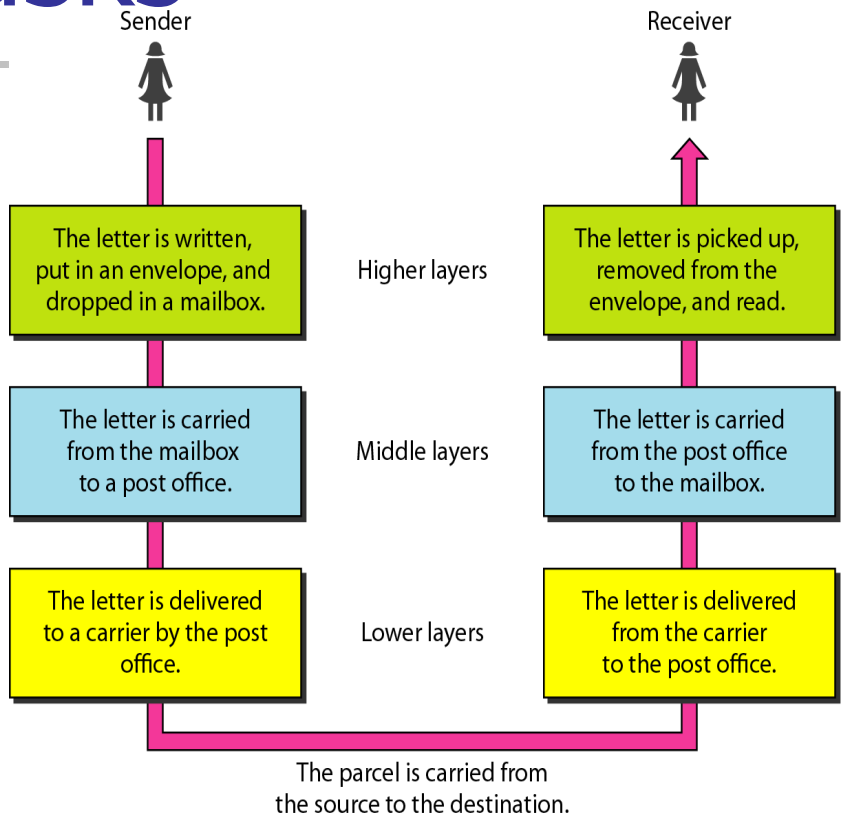
- Connection oriented services
  - Establish a connection
  - Use the connection
  - Release the connection
- Connection-less services
  - Each Message carries a full destination address
  - Routed message INDEPENDENTLY

# Service Primitives

| Primitive | Meaning |
| --- | --- |
| Request | An entity wants the service to do some work |
| Indication | An entity is to be informed about an event |
| Response | An entity wants to respond to an event |
| Confirm | The response to an earlier request has come back |

# 2.1 Layered Tasks

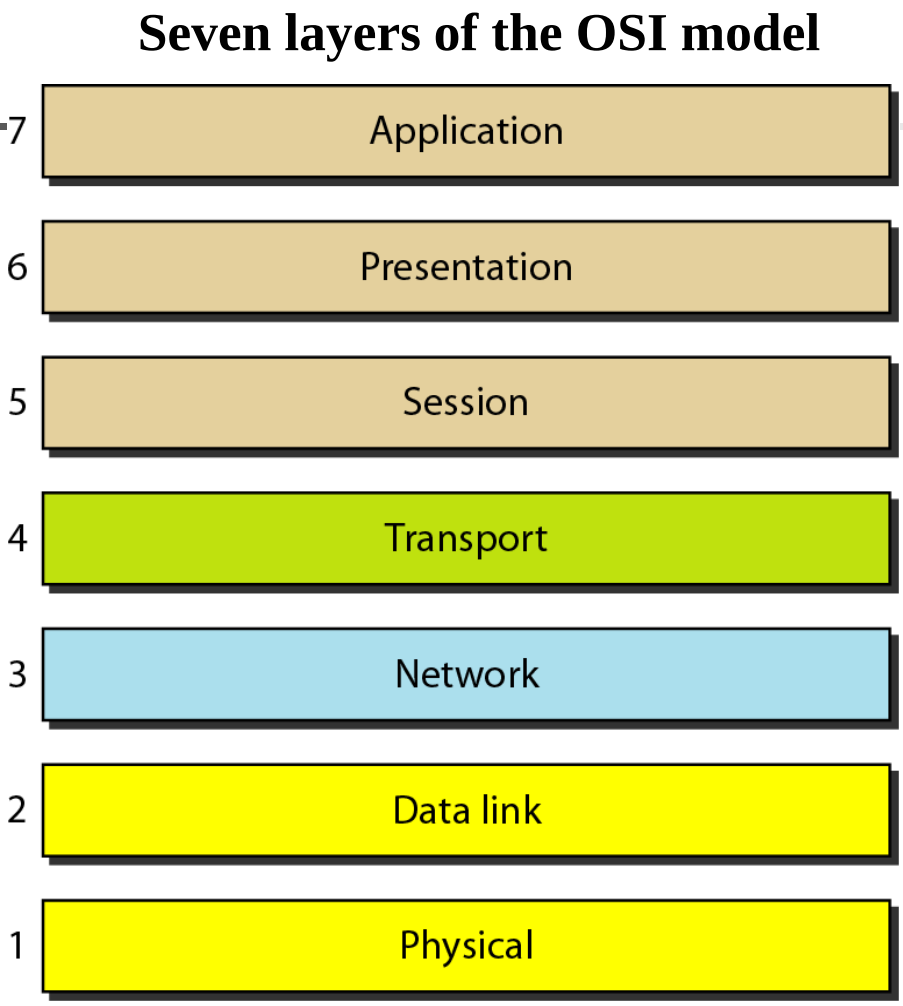
- **Sender, Receiver, and Carrier**
- **Hierarchy – tasks done in an order**
  - **Preparation**
  - **Sending**
  - **Delivering**
- **Services**
  - **Higher layer uses lower layer service**
  - **Lower layer uses the service of the carrier**



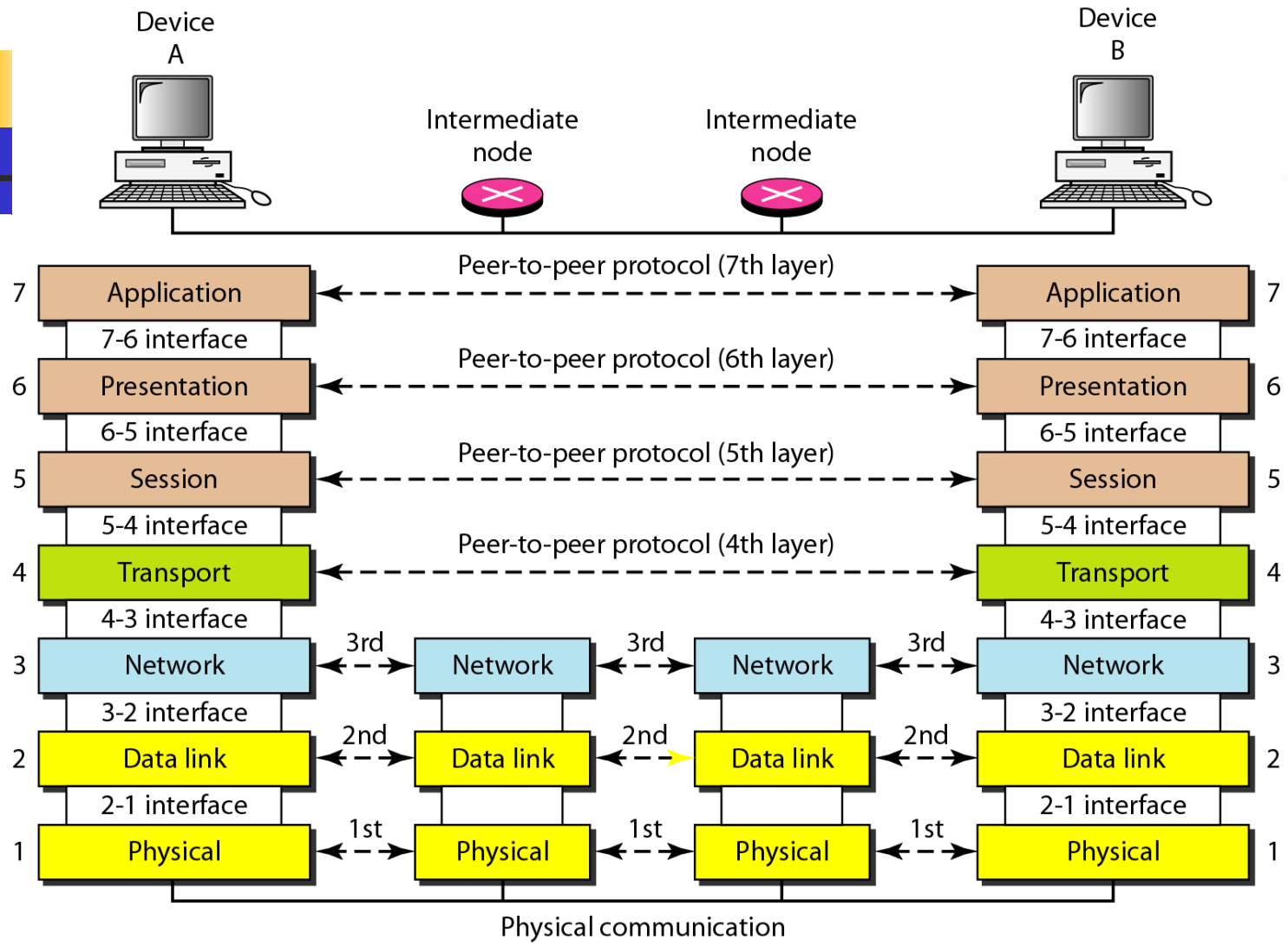
# Principles applied to arrive at the seven layers are

- A layer should be created where a different level of abstraction is needed
- Each layer should perform a well defined function
- The function of layer should be chosen with an eye toward defining internationally standardized protocol
- The layer boundaries should be chosen to minimize the information flow across the interface
- The number of layer should be large enough that distinct function need not be thrown together in the same layer out of necessity and small enough that the architecture does not become unwieldy

# 2.2 OSI Model


Seven layers of the OSI model
7
Application
6
Presentation
5
Session
4
Transport
3
Network
2
Data link
1
Physical

# The interaction between layers in the OSI model

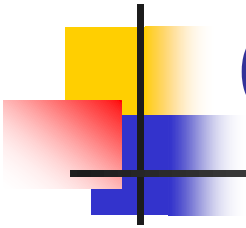


# Organization of the Layers

**Network Support Layers**

- **Including layers**
  - **Layer 1 - Physical Layer**
  - **Layer 2 - Data Link Layer**
  - **Layer 3 - Network layer**
- **physical aspects of transmitting data**
  - **Electrical specifications**
  - **Physical connections**
  - **Physical addressing**
  - **Transport timing**
  - **Reliability**

# Organization of the Layers (cont.)



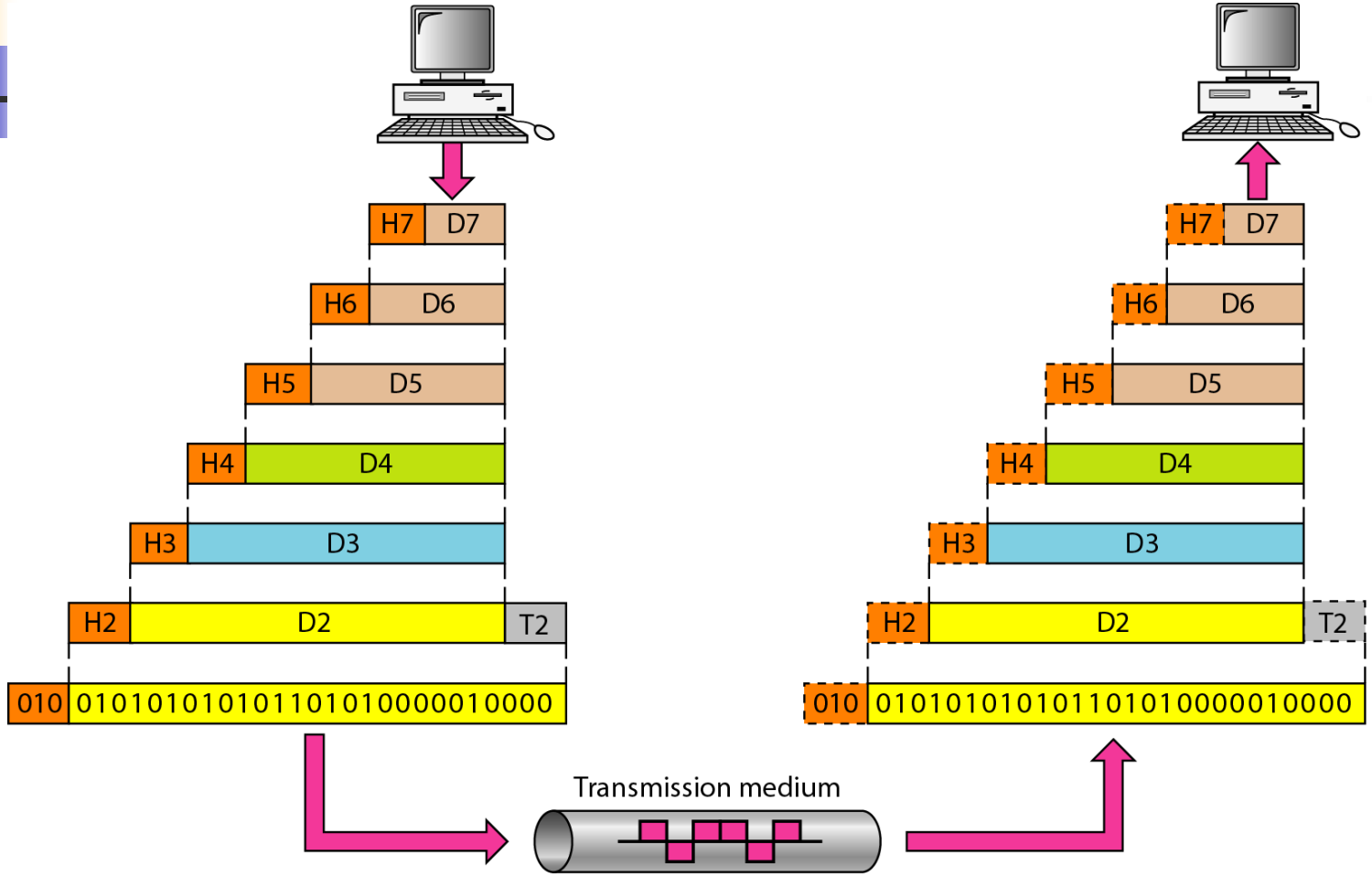
**Transport Layer**

- Including Layer 4 – transport layer

**User Support Layer**

- Allows interoperability among unrelated software systems
- Including layers:
  - Layer 5 – session layer
  - Layer 6 – presentation layer
  - Layer 7 – application layer

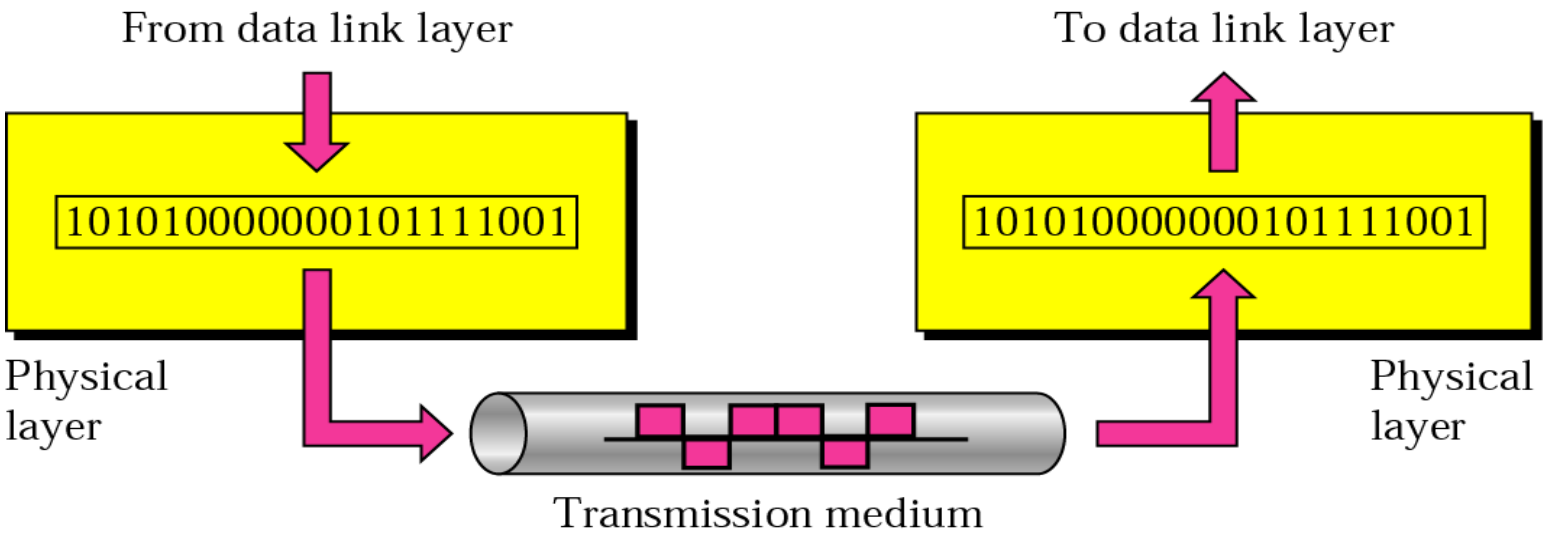
# An Exchange Using Internet Model



# Physical Layer

- **Interface & Transmission Media**
  - **Electrical Specification**
  - **Mechanical Specification**
- **Responsible for movement of individual bits from one hop (node) to the next.**
- **Define functions and procedures for data transmission**
  - **Physical characteristics of interfaces and medium**
  - **Representation of bits**
  - **Data Rate:** No.of bits/sec
  - **Synchronization of Bits**
  - **Line configuration:** POINT TO POINT OR MULTIPOINT
  - **Physical topology**
  - **Transmission mode**

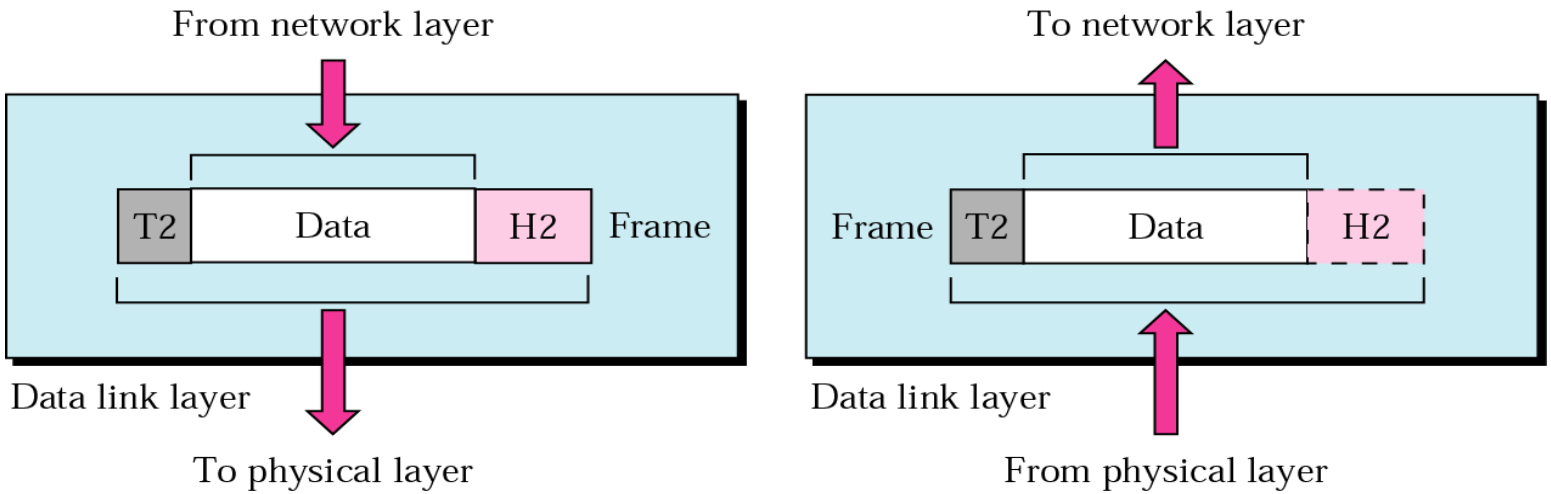
# Physical Layer



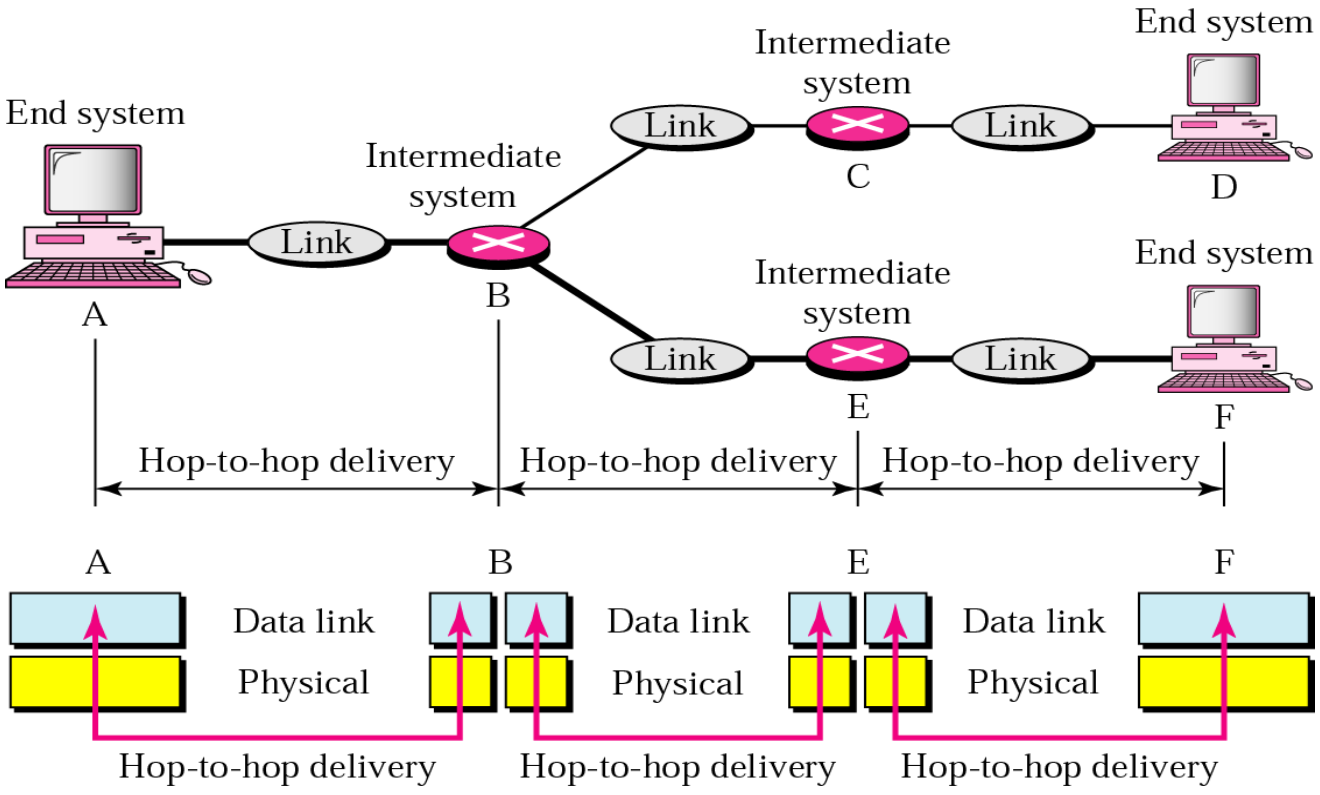
# Data Link Layer

- **Provide a reliable Link**
- **Error free to the upper layer (network layer)**
- **Responsible for moving frames from one hop (node) to the next.**
- **Major responsibilities**
  - **Framing**
  - **Physical addressing**
  - **Flow control**
  - **Error control**
  - **Access control**

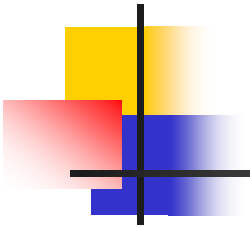
# Data Link Layer (cont.)



# Data Link Layer – Hop-to-hop (Node-to-Node) Delivery

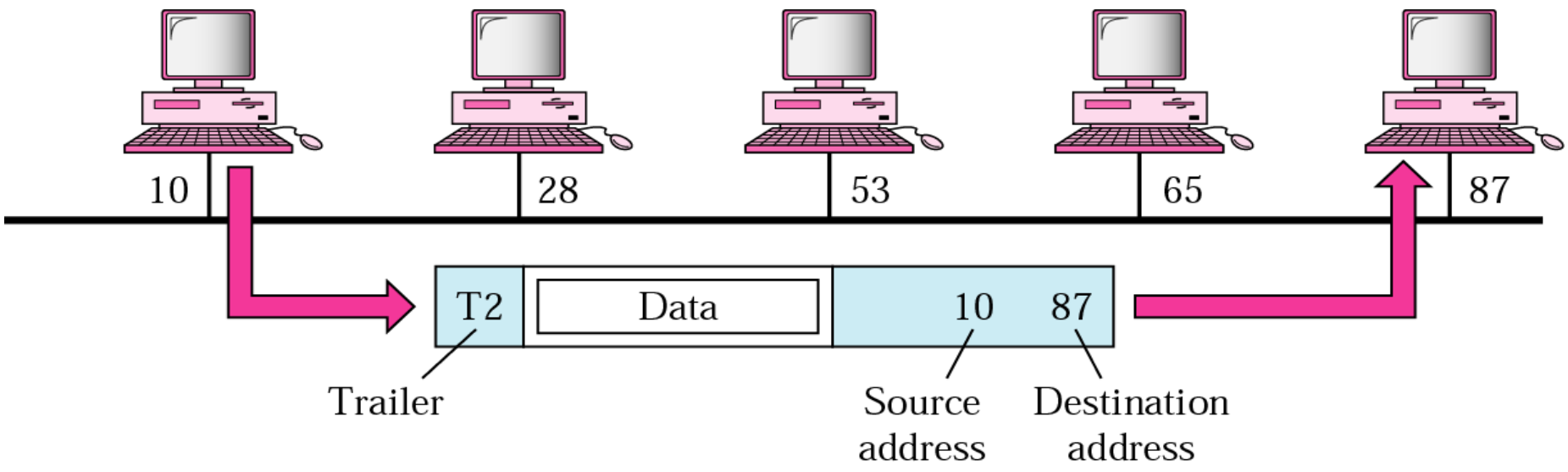


# *Example 1*

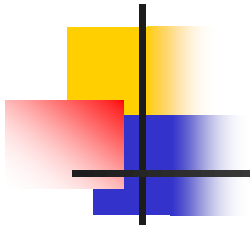


In a node with physical address 10 sends a frame to a node with physical address 87. The two nodes are connected by a link. At the data link level this frame contains physical addresses in the header. These are the only addresses needed. The rest of the header contains other information needed at this level. The trailer usually contains extra bits needed for error detection

*Example 1*

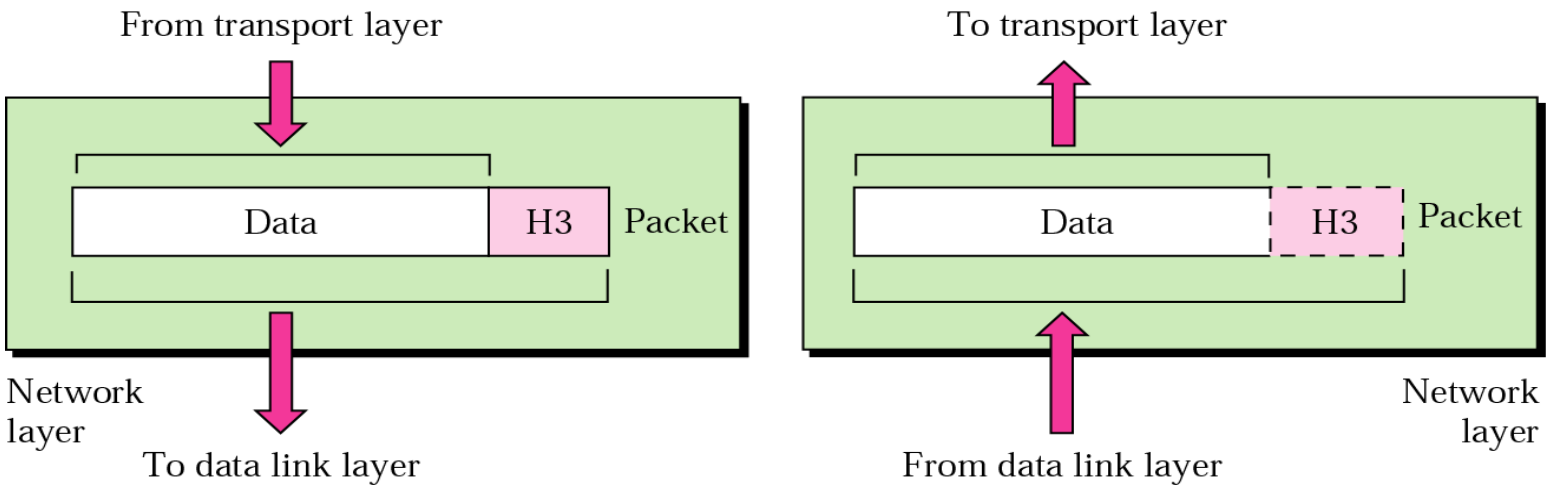

10
28
53
65
87
T2
Data
10
87
Trailer
Source address
Destination address

# Network Layer

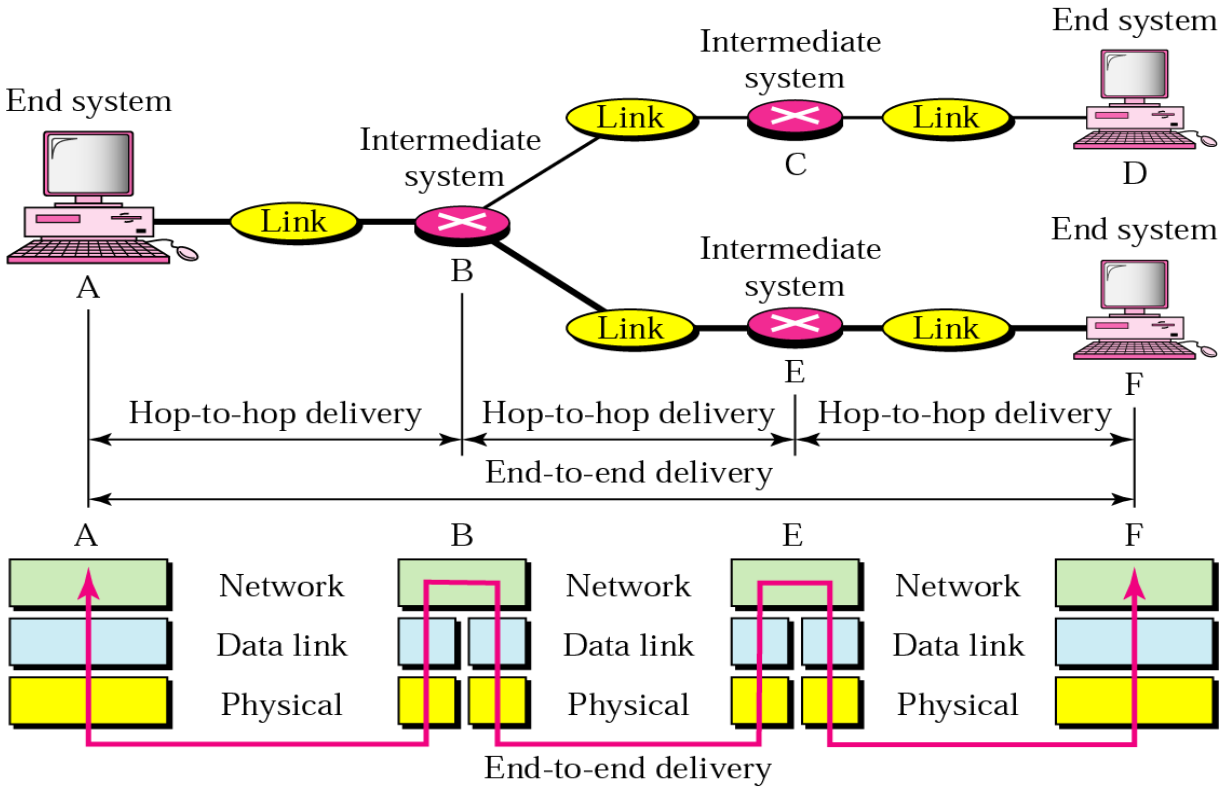


- **Source to destination delivery of a data packet across multiple networks**
- **Responsible for the delivery of individual packets from the source host to the destination host.**
- **Major responsibilities**
  - **Logical addressing (IP)**
  - **Routing**
    - **When independent network or links are connected**

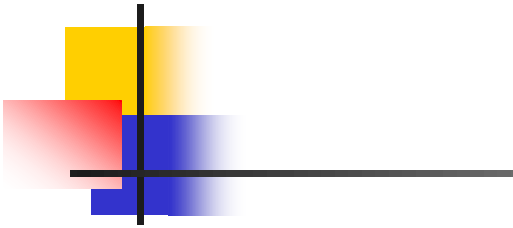
# Network Layer



# Network Layer – Source-to-Destination Delivery

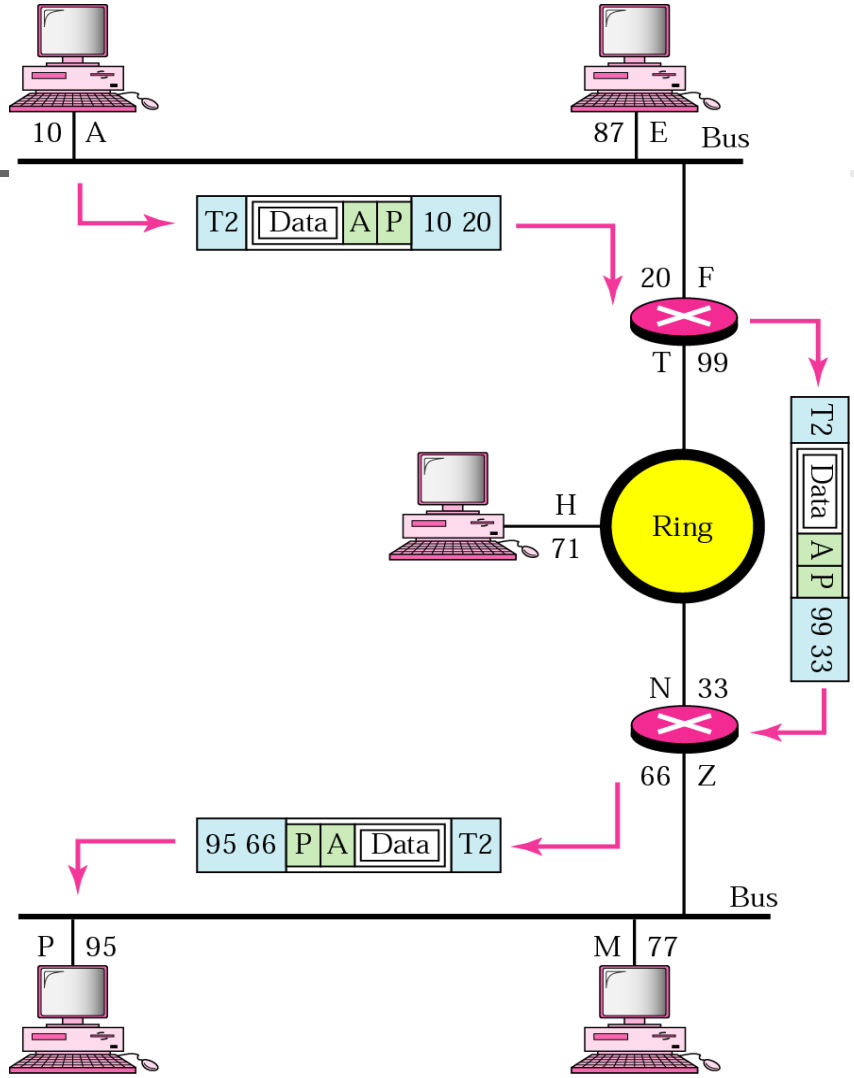


## *Example 2*

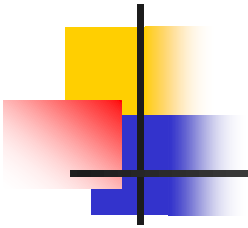


In we want to send data from a node with network address A and physical address 10, located on one LAN, to a node with a network address P and physical address 95, located on another LAN. Because the two devices are located on different networks, we cannot use physical addresses only; the physical addresses only have local jurisdiction. What we need here are universal addresses that can pass through the LAN boundaries. The network (logical) addresses have this characteristic.

## *Example 2*

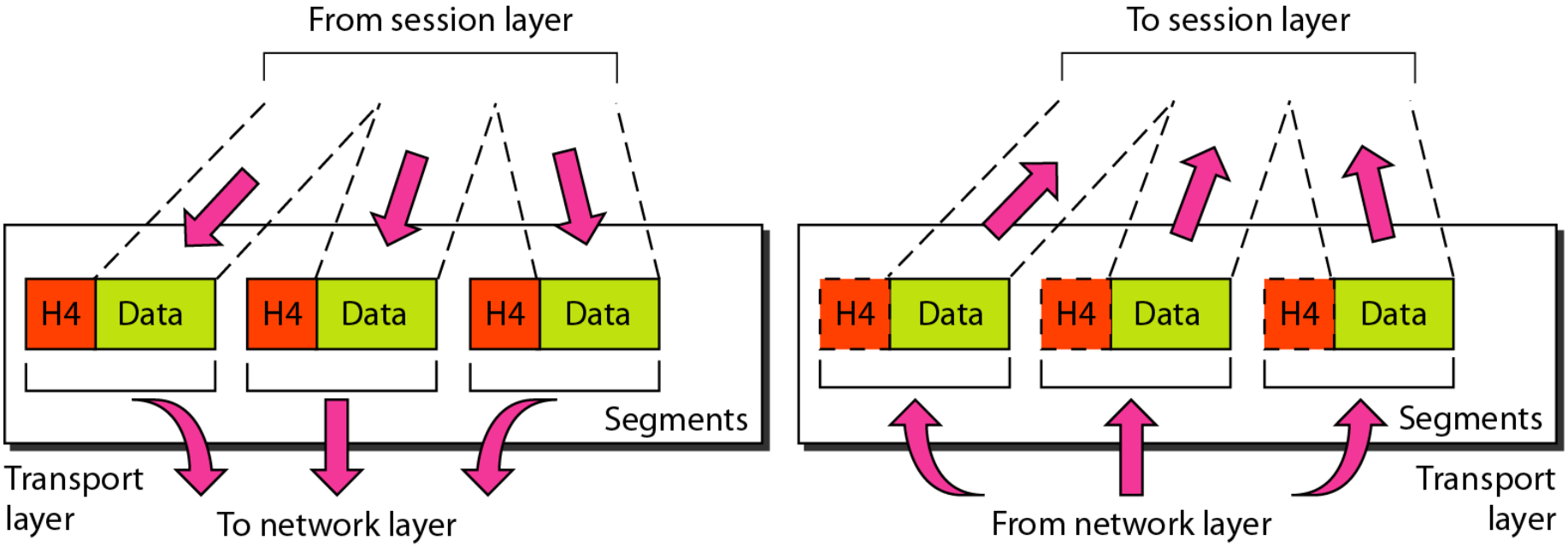

10 A
87 E
Bus
T2 Data A P 10 20
20 F
T 99
T2 Data A P 99 33
H
71
Ring
N 33
66 Z
95 66 P A Data T2
Bus
P 95
M 77

# Transport Layer

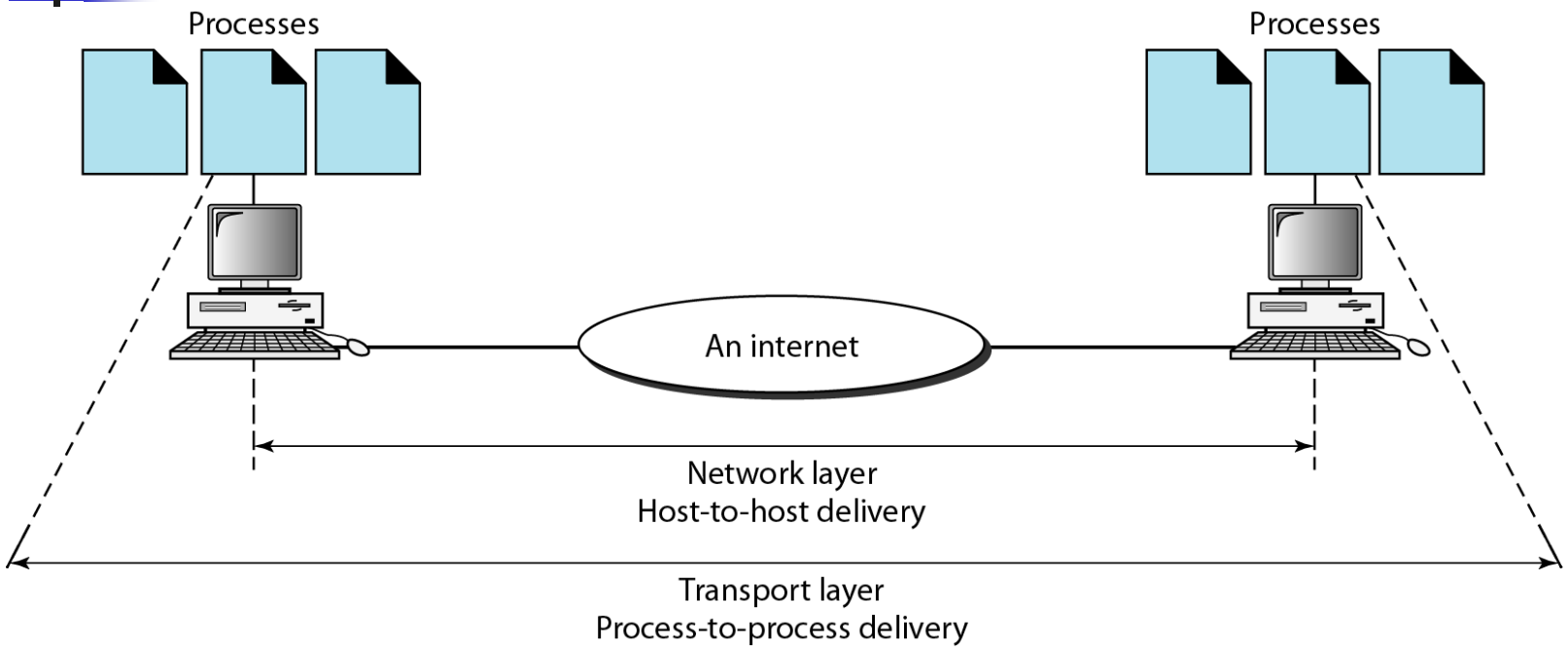


- **Responsible for process-to-process delivery of the entire message**
- **Responsible for the delivery of a message from one process to another**
- **Ensure reliable and orderly transmission of the whole message**
- **Oversee error control and flow control at the source-to-destination level.**
- **Major responsibilities**
  - **Service-point addressing**
  - **Segmentation and reassembly**
  - **Connection control (Connectionless and connection oriented services)**
  - **Flow control and error control**

# Transport Layer



# Reliable process-to-process delivery of a message



# *Example 3*

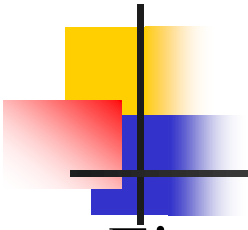
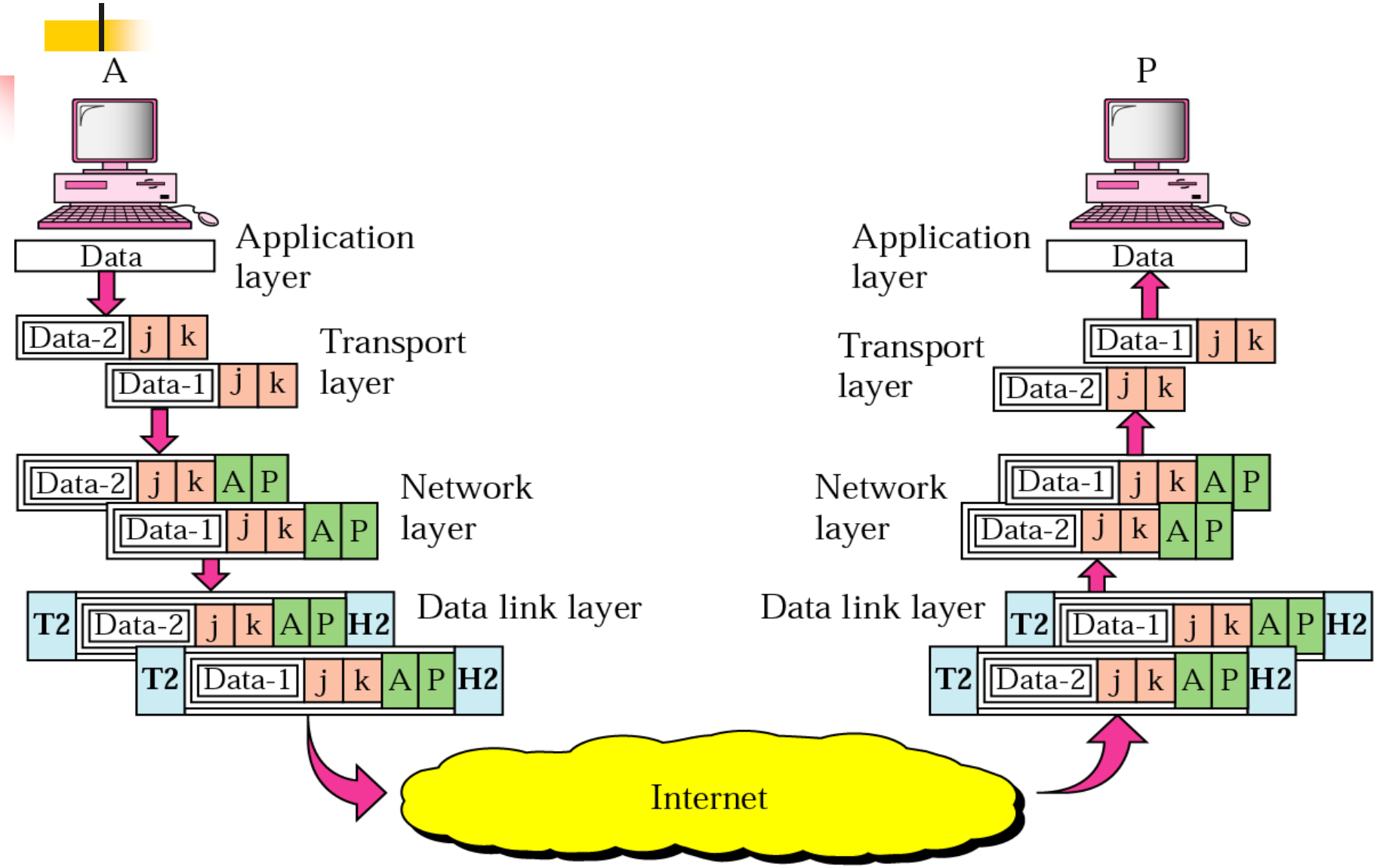
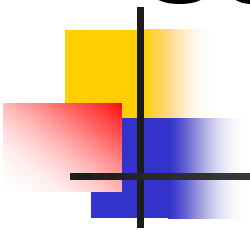


Figure shows an example of transport layer communication. Data coming from the upper layers have port addresses j and k (j is the address of the sending process, and k is the address of the receiving process). Since the data size is larger than the network layer can handle, the data are split into two packets, each packet retaining the port addresses (j and k). Then in the network layer, network addresses (A and P) are added to each packet.

**Figure 2.14** *Example 3*

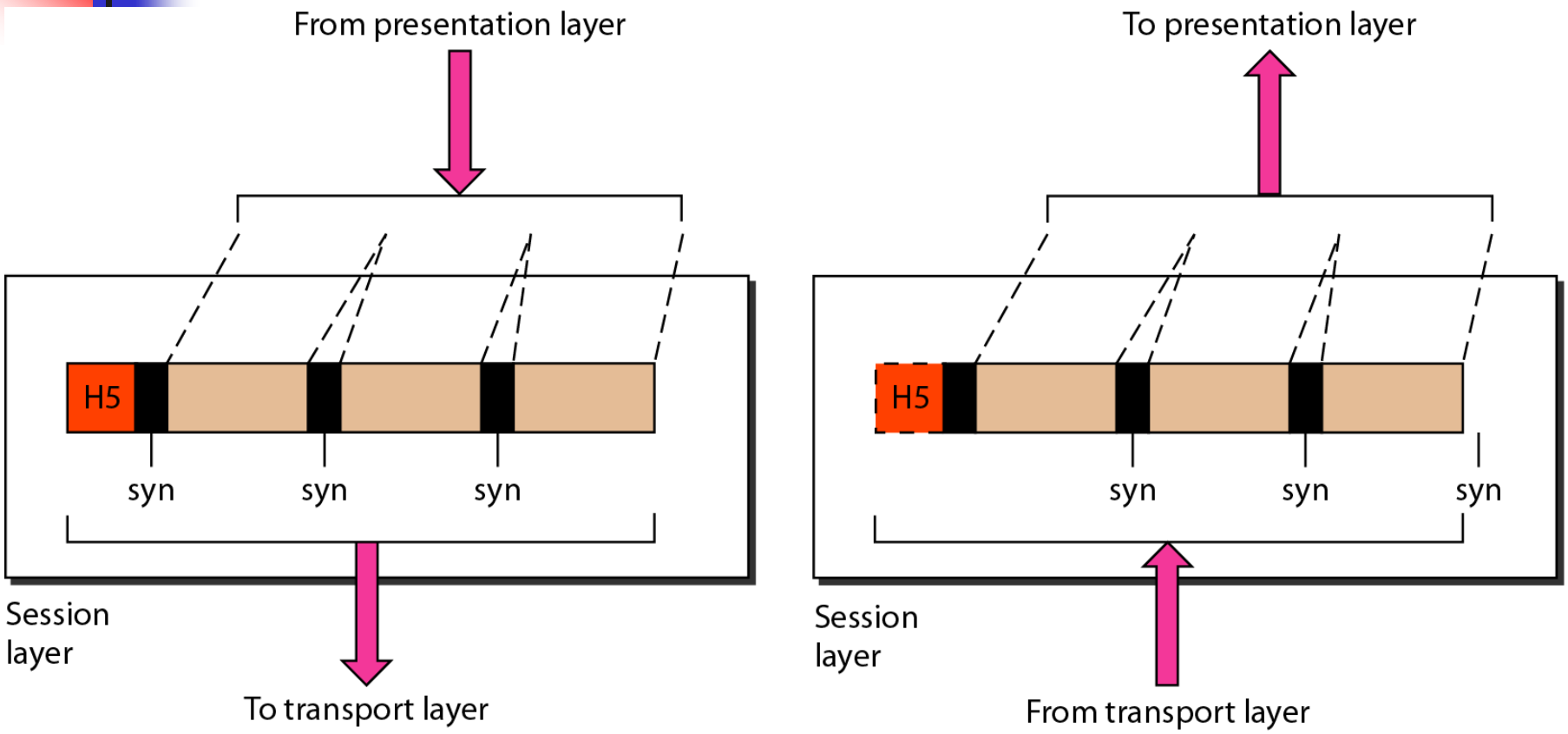


# Session Layer

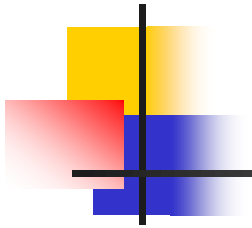


- **Establish, maintain and synchronize the interaction among communication system.**
- **Responsible for dialog control and synchronization.**
- **Major responsibilities**
  - **Dialog control**
  - **Synchronization**
    - **Allows a process to add checkpoint (synchronization) points to a stream of data**

# Session Layer

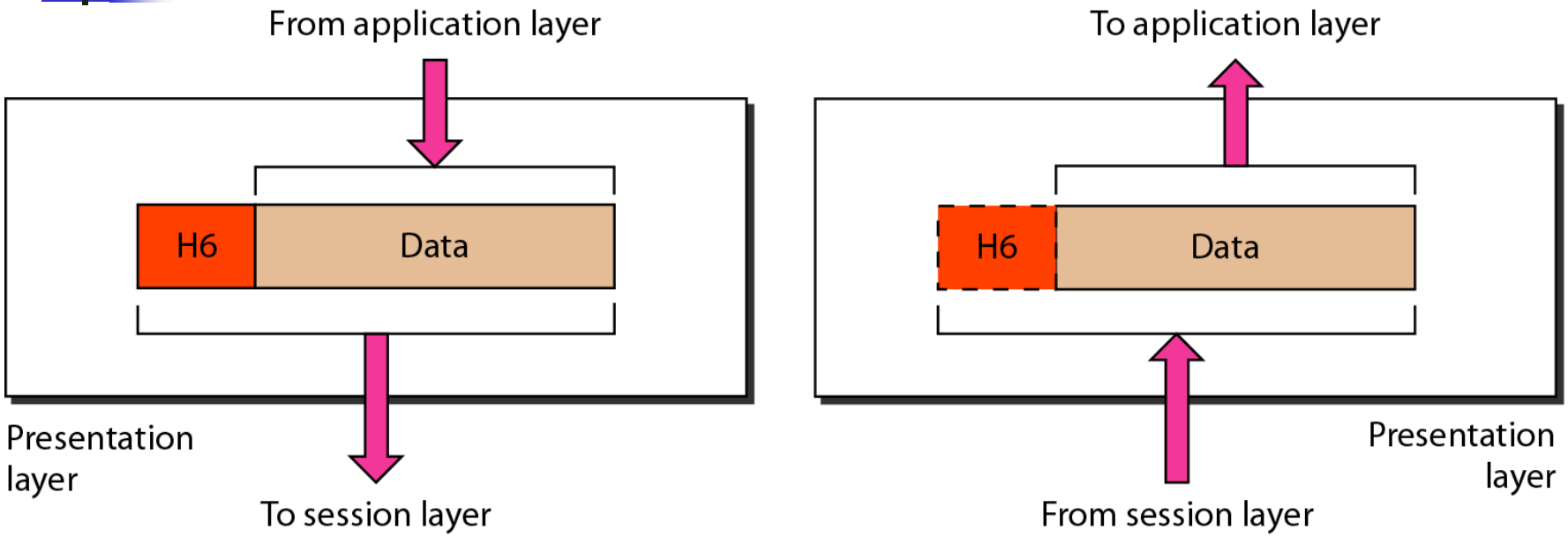


# Presentation Layer



- Syntax and semantics of the information exchanged between two systems.
- Major responsibilities
  - Translation
  - Encryption
  - Compression

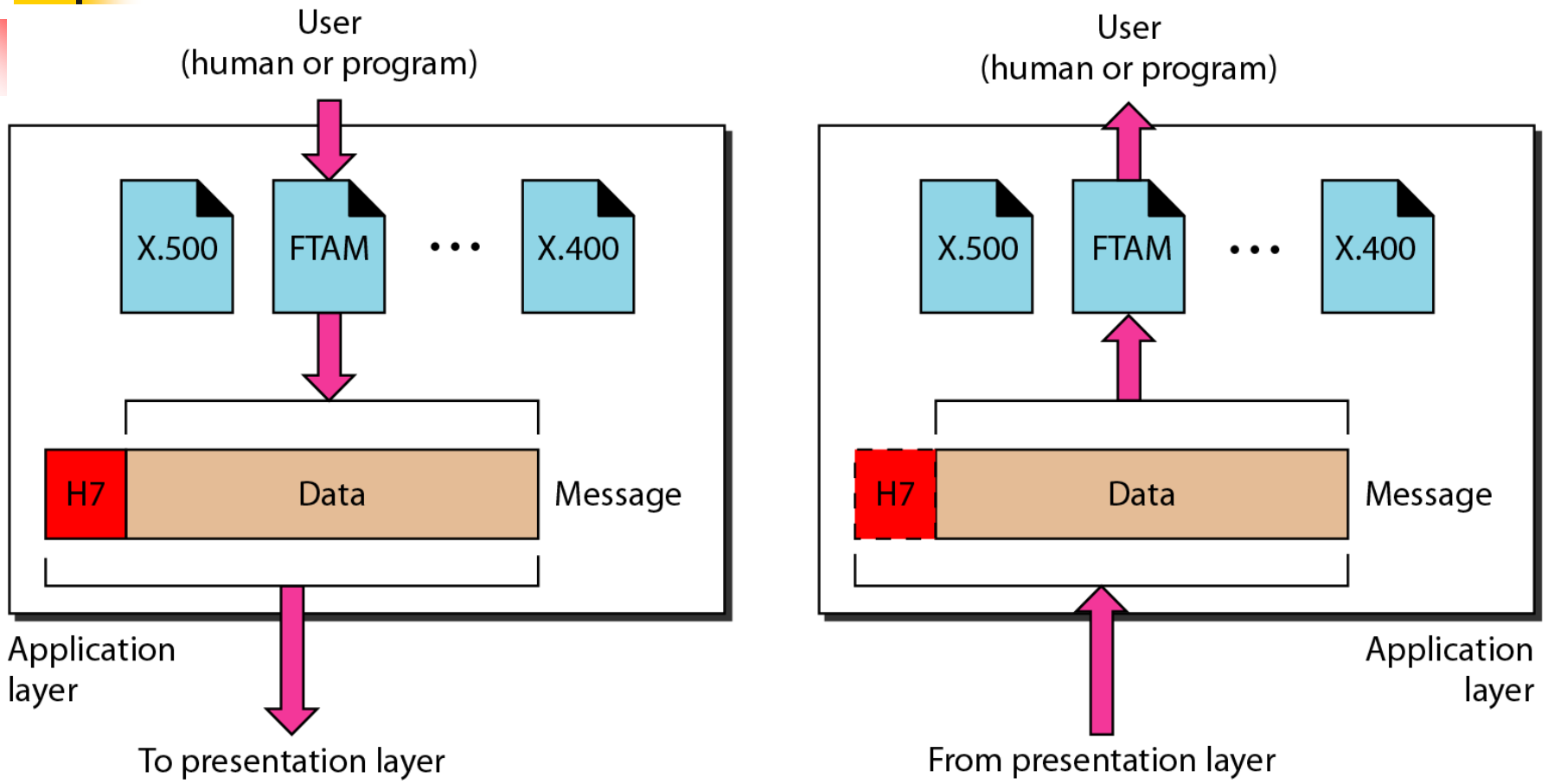
# Presentation Layer



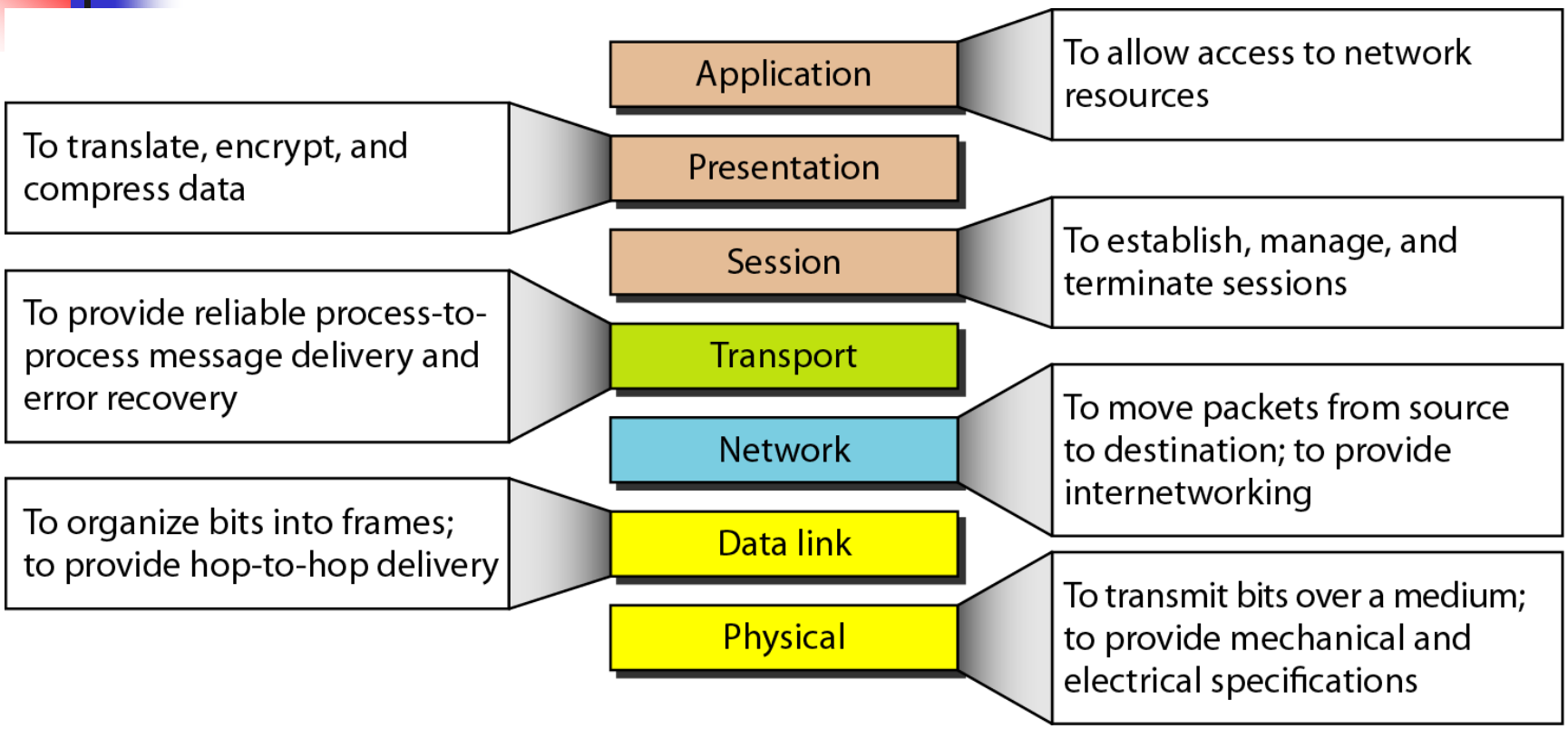
# Application Layer

- Provide user interface and support for services.
- Services provided
  - Network virtual terminal (It is a software version of a physical terminal and it allows a user to log on to a remote host)
  - File transfer, access, and management
  - Mail services
  - Directory services (it provides distributed database sources and access for global information about various objects and services)

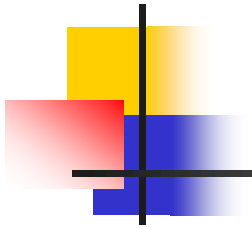
# Application Layer



# Summary of Layers

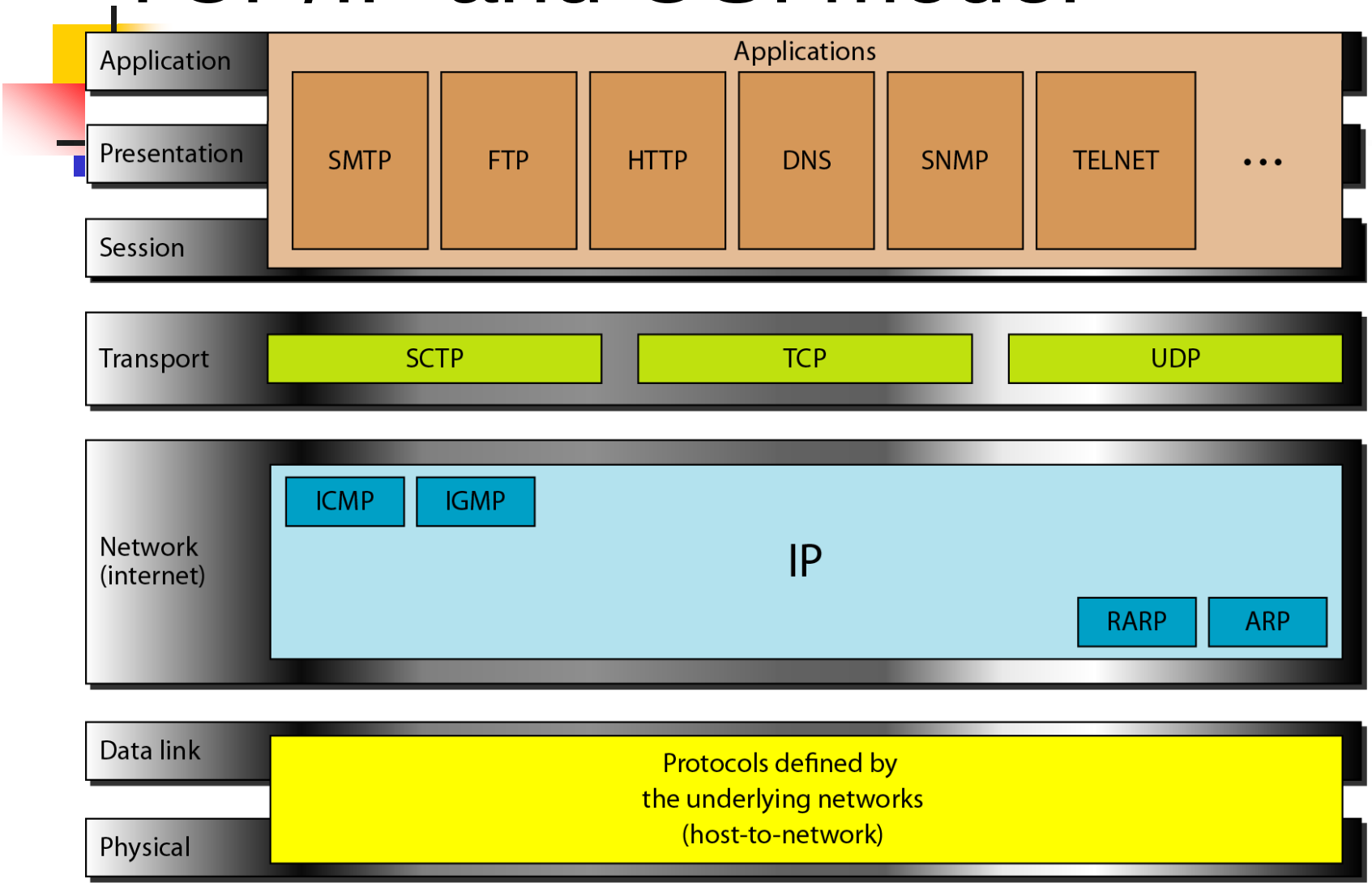


# 2.4 TCP/IP Protocol Suites

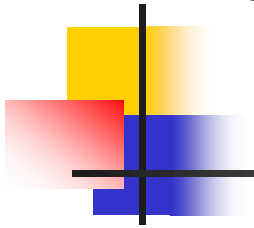


- 5 Layer Model
  - Layer 5 Application layer
  - Layer 4 Transport layer
  - Layer 3 Network layer
  - Layer 2 Data Link layer
  - Layer 1 Physical layer

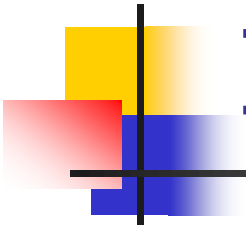
# TCP/IP and OSI model



# TCP/IP Protocol Suites (cont.)



- Physical and Data Link Layers
  - Does not define any specific protocol.
  - Support all the standard and proprietary protocols.
- Network Layer
  - IP (Internetworking Protocol)
  - ARP (Address Resolution Protocol)
  - RARP (Reverse Address Resolution Protocol)
  - ICMP (Internet Control Message Protocol)
  - IGMP (Internet Group Message Protocol)

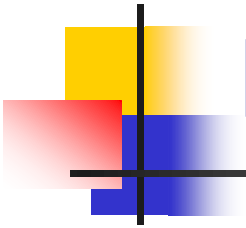


# IP (Internetworking Protocol)

- It is the transmission mechanism used by the TCP/IP protocol
- It is an unreliable connectionless protocol (best effort delivery)

  best effort means no error checking or tracking
- It supports datagram
- No facility for reordering datagram at destination

# ARP (Address Resolution Protocol)

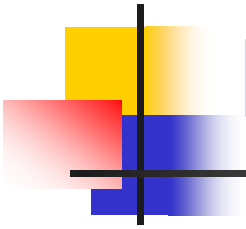


- It is used to associate a logical address with physical address
- On a typical N/W such as LAN each device on a link is identified by physical address or station address imprinted on NIC (network interface card)
- It is used to find the physical address of the node when its Internet address is known

# RARP ( Reverse Address Resolution Protocol)

- It is used to discover Internet address when it knows physical address.It is used when a computer is connected to a network for the first time

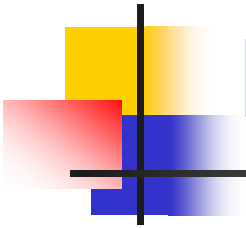
# (ICMP) Internet Control Message Protocol



- It is a mechanism used by hosts & gateways to send notification of datagram problems back to the sender

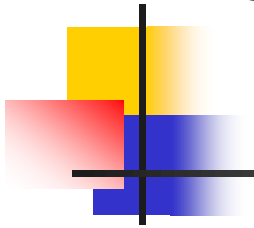
- It sends query & error reporting message

# (IGMP) Internet Group Message Protocol



- It is used to facilitate the simultaneous transmission of message to a group of reception

# TCP/IP Protocol Suites (cont.)

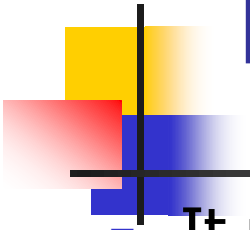


- Transport Layer
  - UDP (User Datagram Protocol)
  - TCP (Transmission Control Protocol)
  - SCTP (Stream Control Transmission Protocol)

# UDP (User Datagram Protocol)

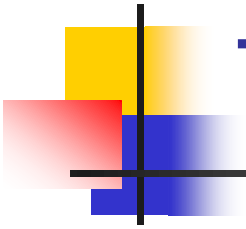
- It is a process to process protocol that adds only port addresses, checksum error control and information to the data from the upper layer.

# TCP (Transmission Control Protocol)



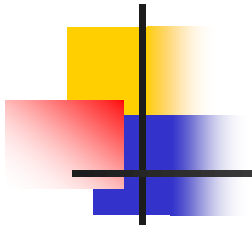
- It provides full transport layer services to the applications
- It is a reliable stream (connection oriented) transport protocol
- At Transmitter end TCP divides a stream of data into smaller units called segment.
- Add sequence number
- At the receiving end collects each datagram
- Reordering the datagram

# SCTP (Stream Control Transmission Protocol)



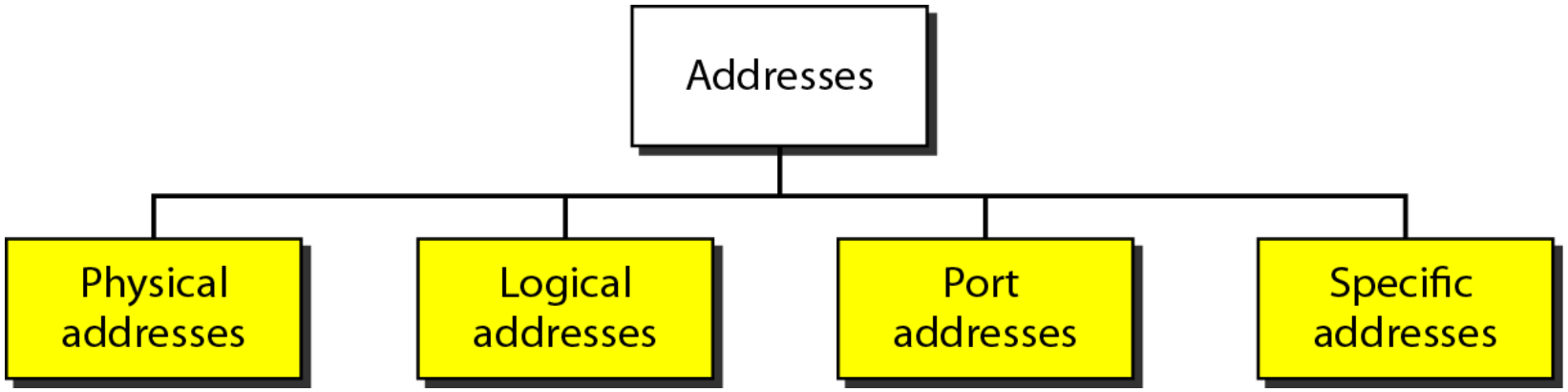
- It support for newer  applications such as voice over the Internet
- It combines the best features of UDP and TCP

# Application Layer

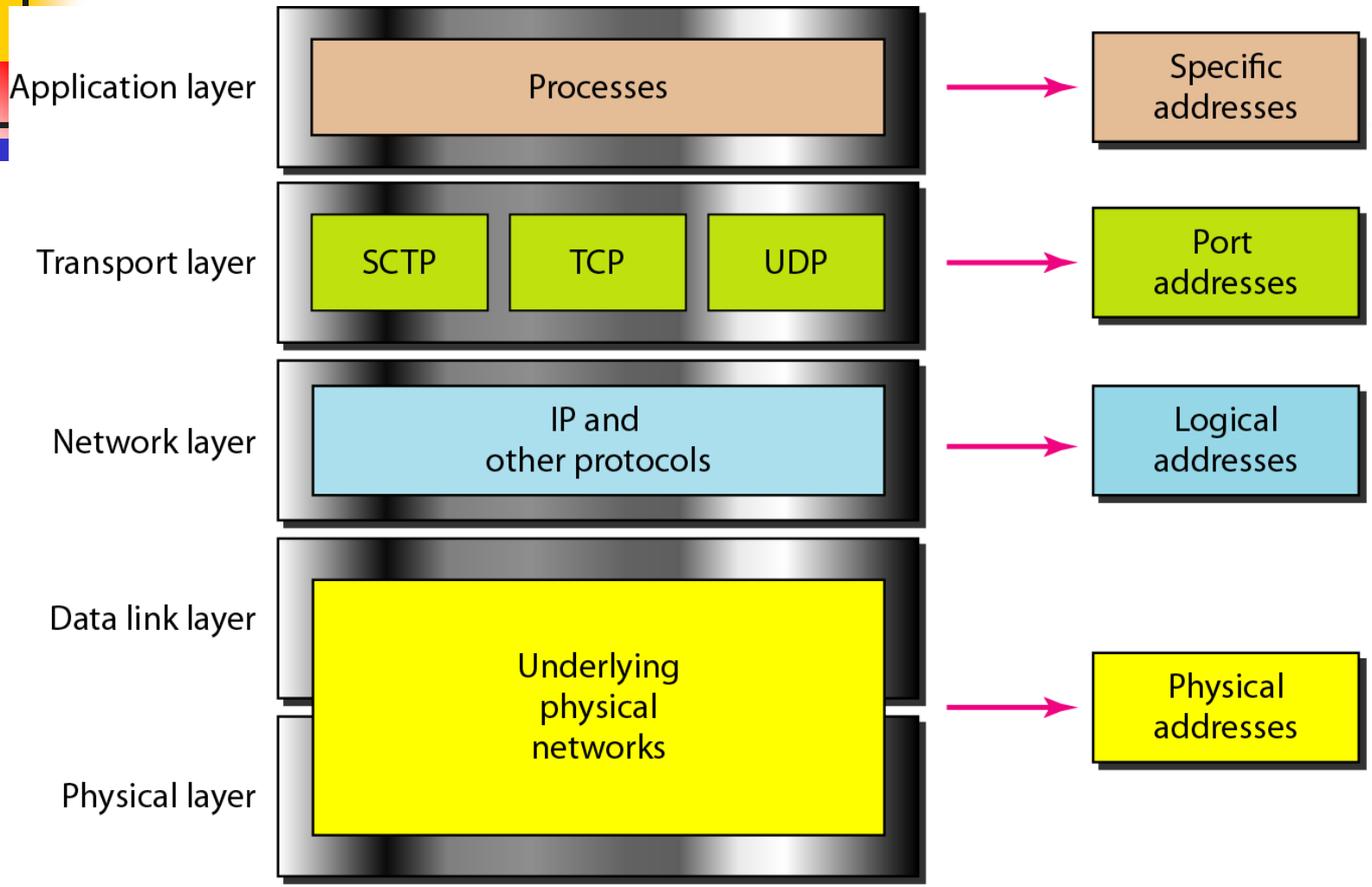


- Equivalent to the combined session, presentation, application layers in OSI

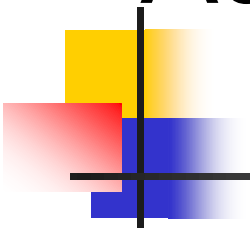
# 2.5 Addressing



# Relations of layers and addresses in TCP/IP

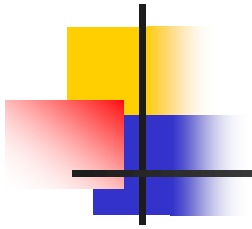


# Addressing



- Physical Addresses
  - Link address
  - Defined by its LAN or WAN
  - Included in the frame used by data link layer
  - The lowest level addresses
  - Size and format of the address depend on the network
    - Ethernet uses a 6-byte address

# *Example 2.2*

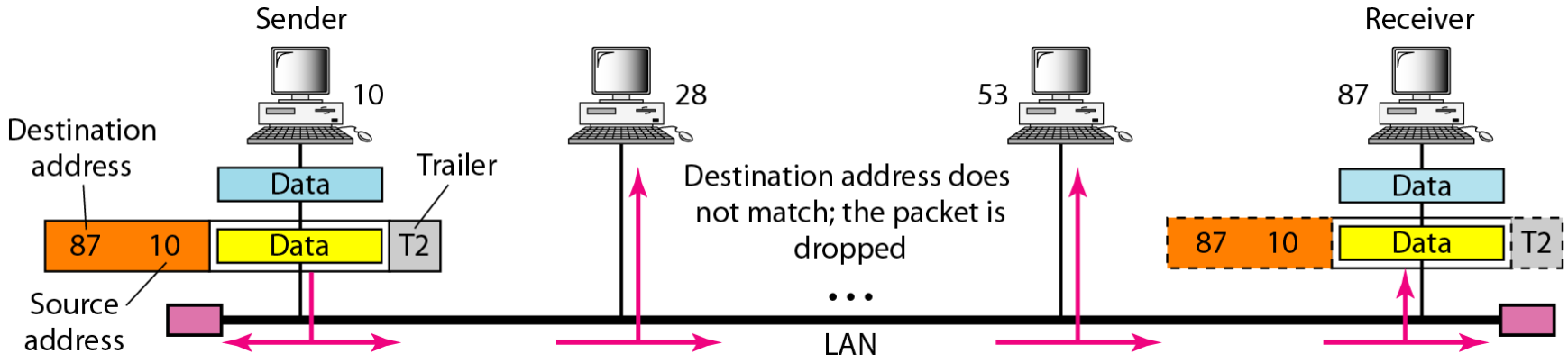


Most local-area networks use a 48-bit (6-byte) physical address written as 12 hexadecimal digits; every byte (2 hexadecimal digits) is separated by a colon.

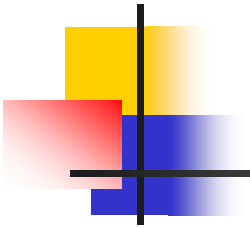
**07:01:02:01:2C:4B**

**A 6-byte (12 hexadecimal digits) physical address.**

# Physical Address

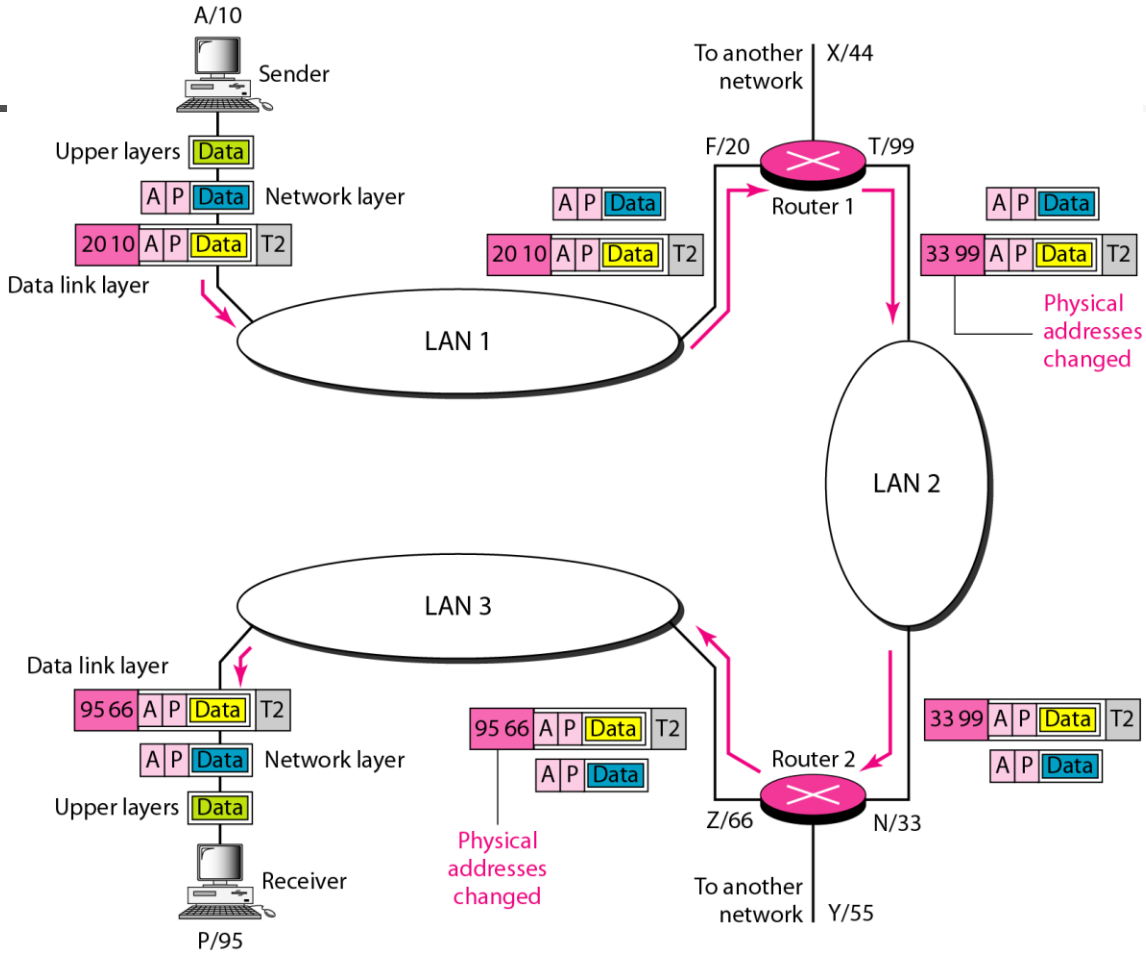


# Addressing (cont.)



- Logical Addresses
  - Necessary for communications that are independent of underlying physical networks.
  - Each host can be identified uniquely
  - Currently a 32-bit address in the Internet
  - The physical addresses will change from hop to hop, but the logical addresses usually remain the same

# IP Addresses



# Addressing

- Port Addresses
  - Internet communication
    - Process communicates with another process
    - Processes – TELNET, FTP, HTTP etc.

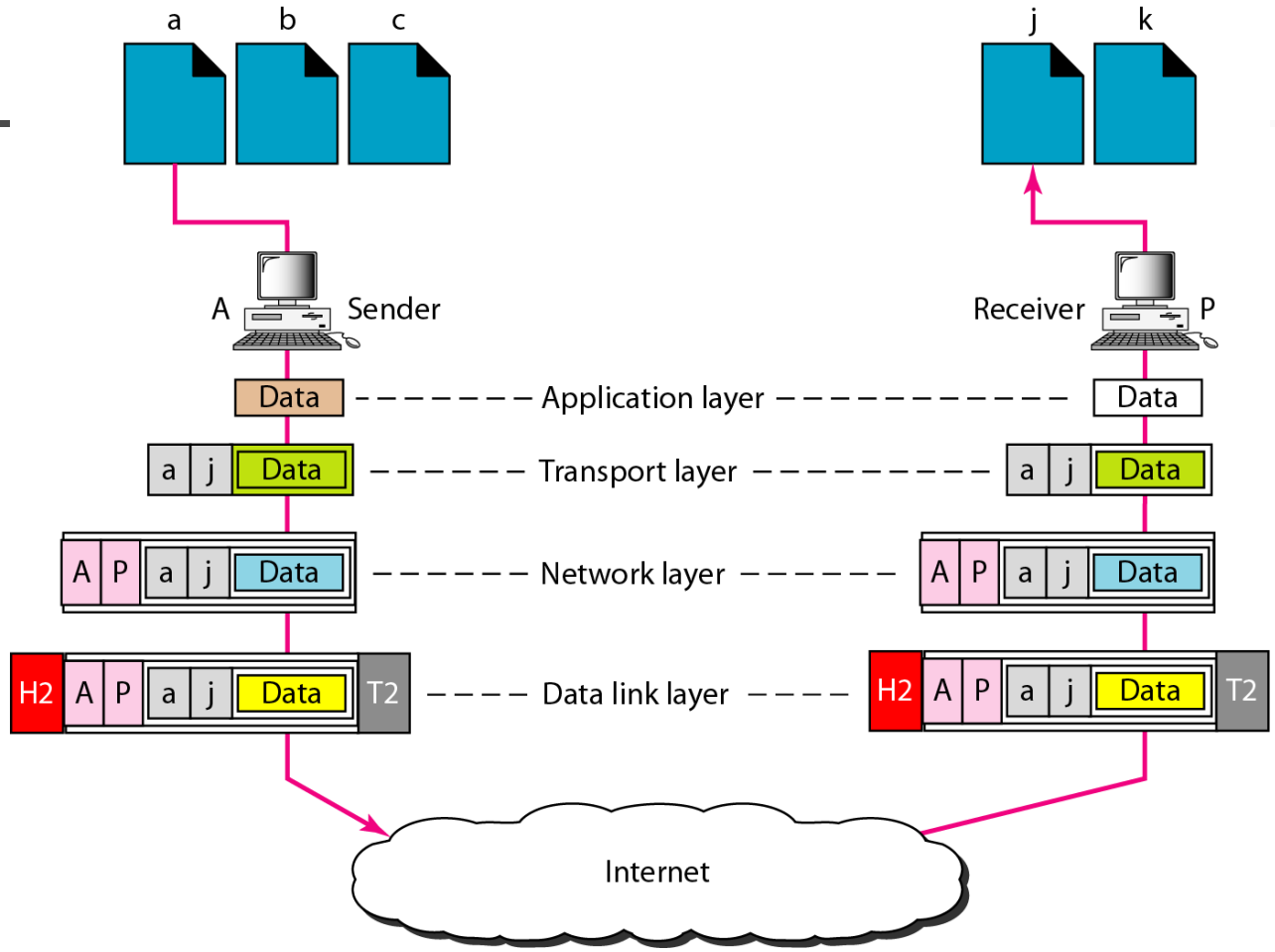
    TELNET (Terminal N/W)-A general purpose client server program that allows remote login.

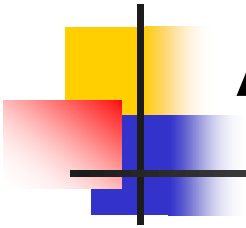
    FTP (File Transfer Protocol)-Transfer files between to sites.

    HTTP (Hyper text transfer protocol)-An application service for retrieving a web document.

  - A method to label different processes.
  - 16 bits in TCP/IP
  - Represented by one decimal number, e.g. 23, 80, 753

# Port Addresses





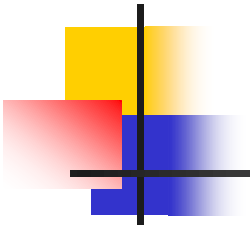
# Addressing

- Specific Addresses
  - Email address – luoh@ipfw.edu
  - Web url – http://www.ects.ipfw.edu/~luoh

# Summary

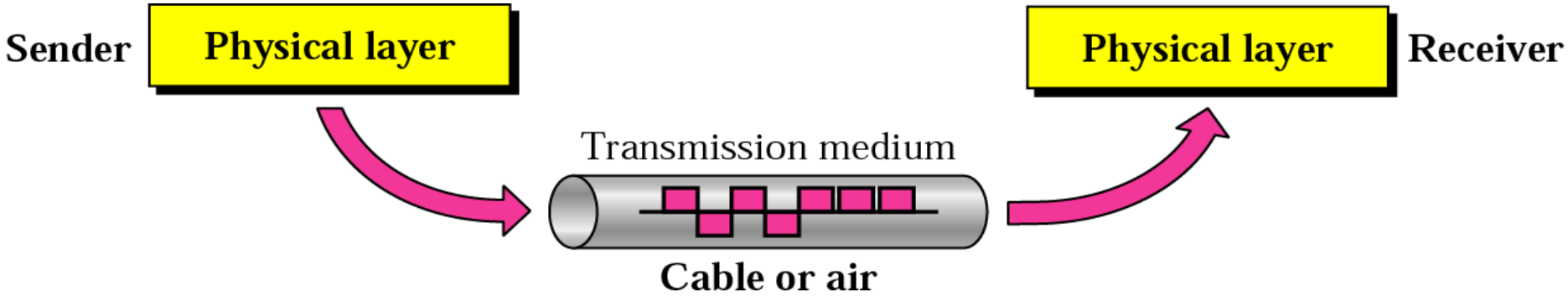


- Layered tasks
- The OSI model
- TCP/IP protocol suites
- Addressing

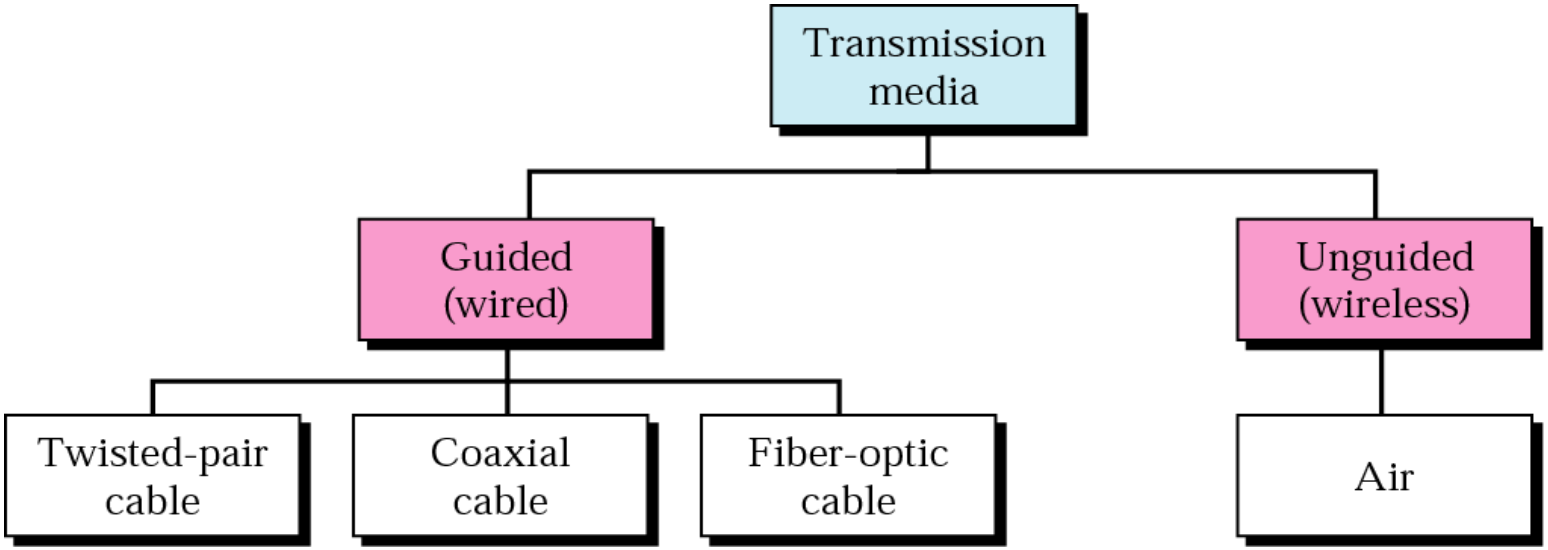


# *Transmission Media*

## *Transmission medium and physical layer*



# Classes of transmission media



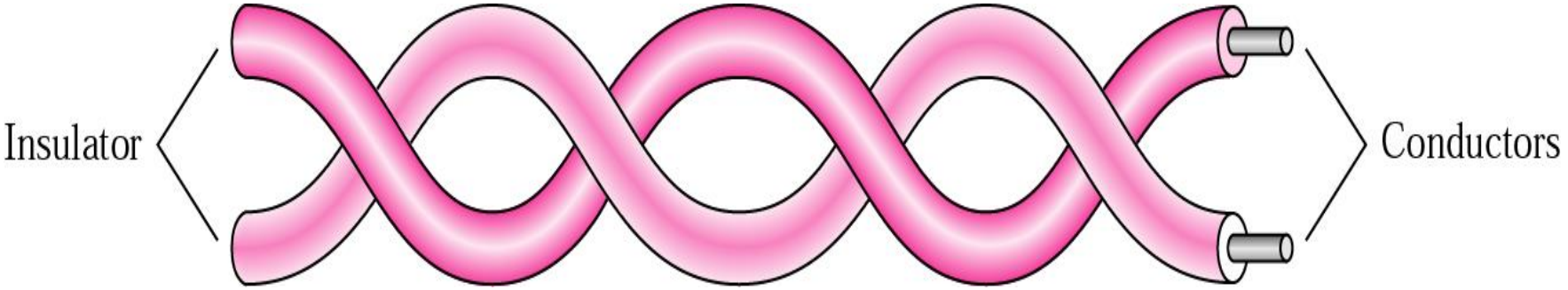
# 7.1 Guided Media

Twisted-Pair Cable

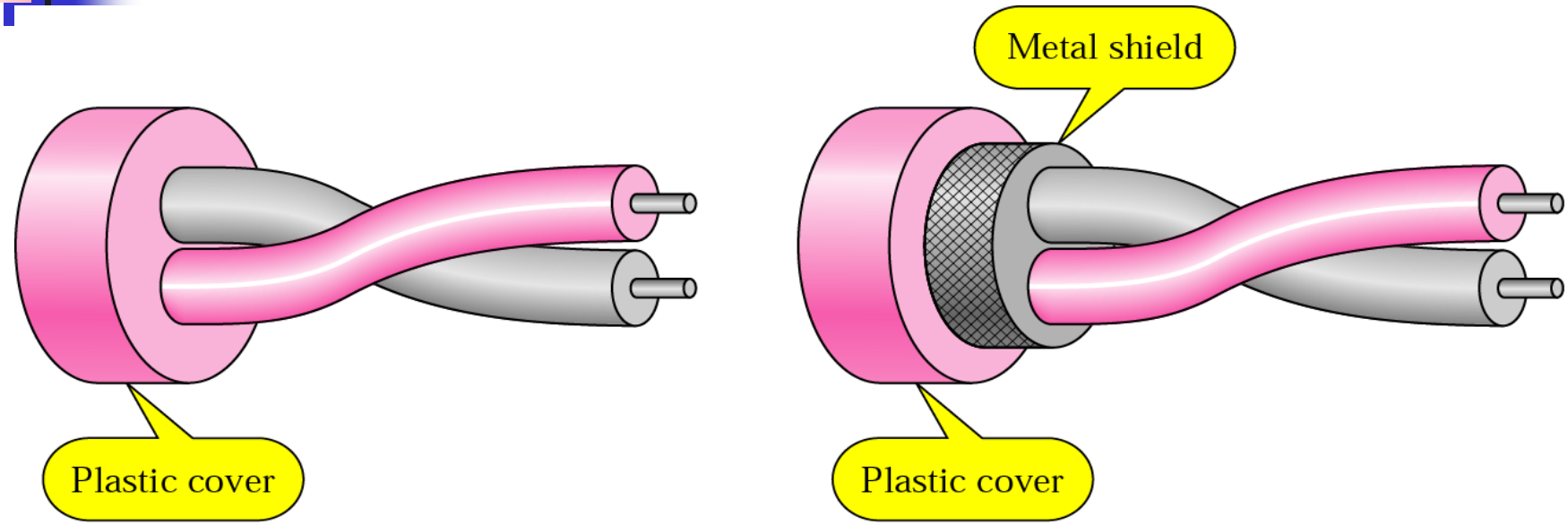
Coaxial Cable

Fiber-Optic Cable

***Twisted-pair cable***



## *UTP and STP*



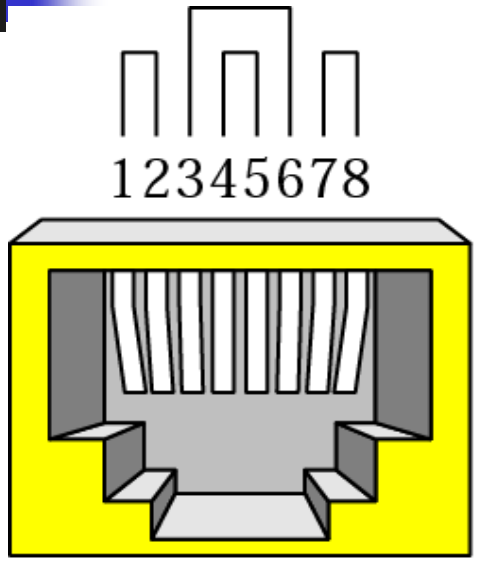

a. UTP

b. STP

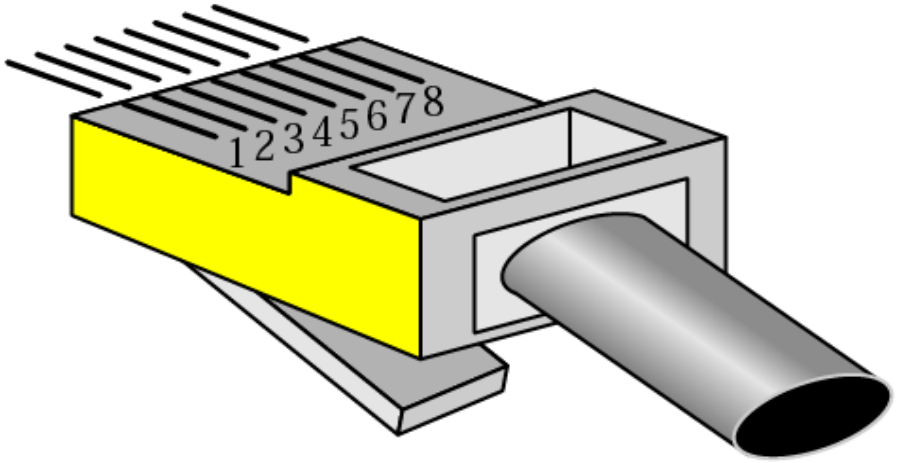
# *Categories of unshielded twisted-pair cables*

| Category | Bandwidth | Data Rate | Digital/Analog | Use |
|---|---|---|---|---|
| 1 | very low | < 100 kbps | Analog | Telephone |
| 2 | < 2 MHz | 2 Mbps | Analog/digital | T-1 lines |
| 3 | 16 MHz | 10 Mbps | Digital | LANs |
| 4 | 20 MHz | 20 Mbps | Digital | LANs |
| 5 | 100 MHz | 100 Mbps | Digital | LANs |
| 6 (draft) | 200 MHz | 200 Mbps | Digital | LANs |
| 7 (draft) | 600 MHz | 600 Mbps | Digital | LANs |

## *UTP connector*

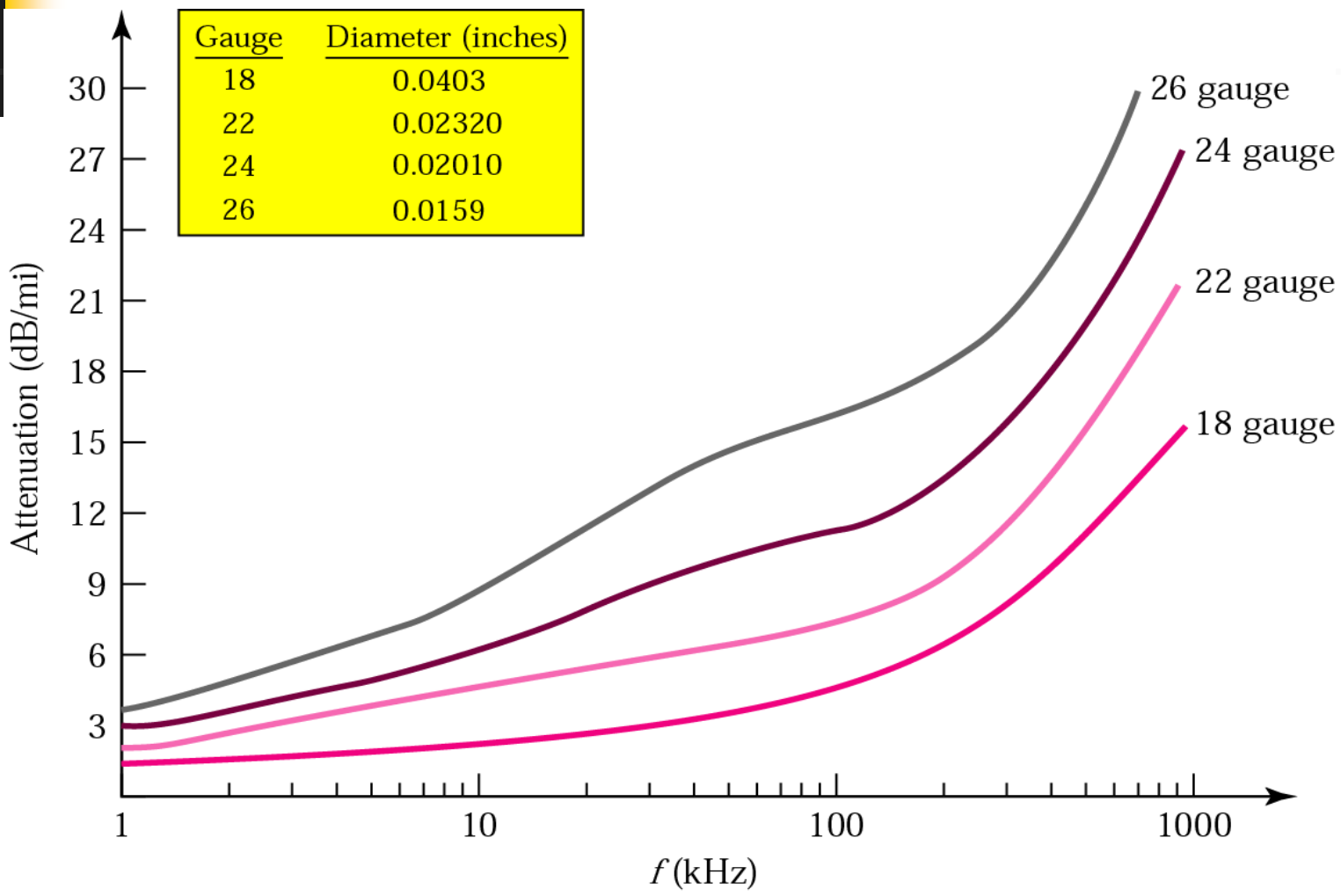



RJ-45 Female

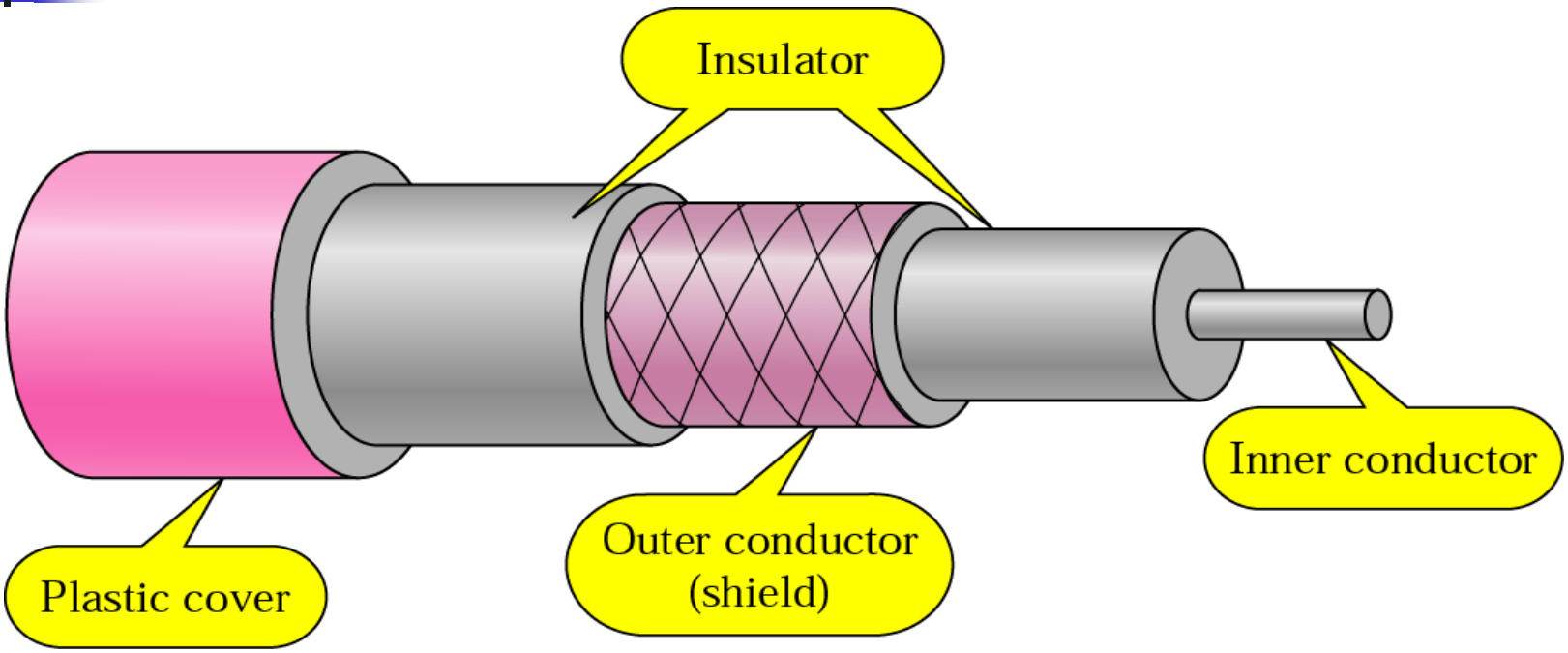



RJ-45 Male

## *UTP performance*



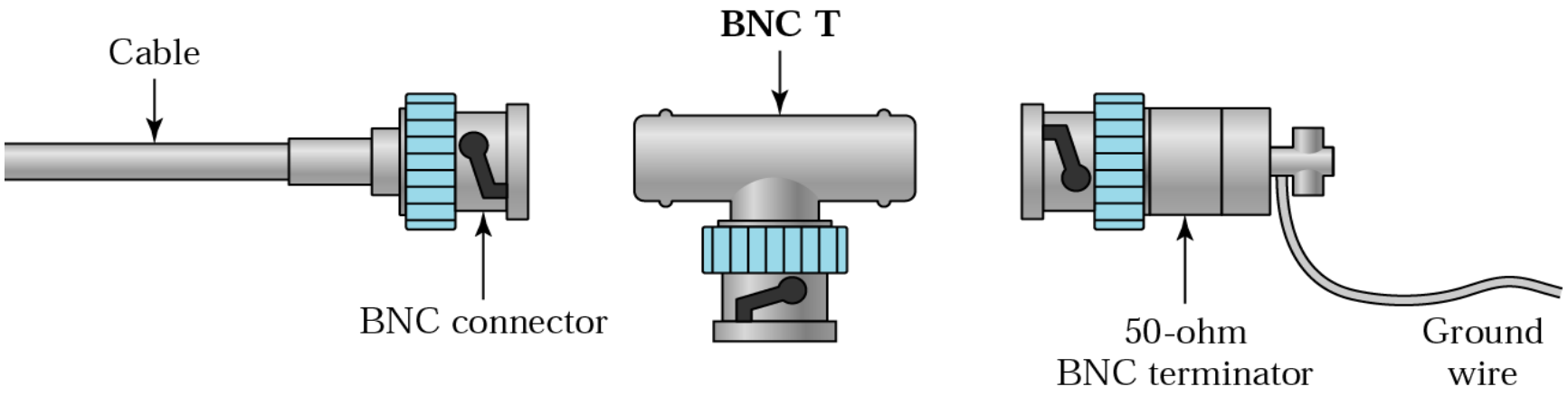
## *Coaxial cable*



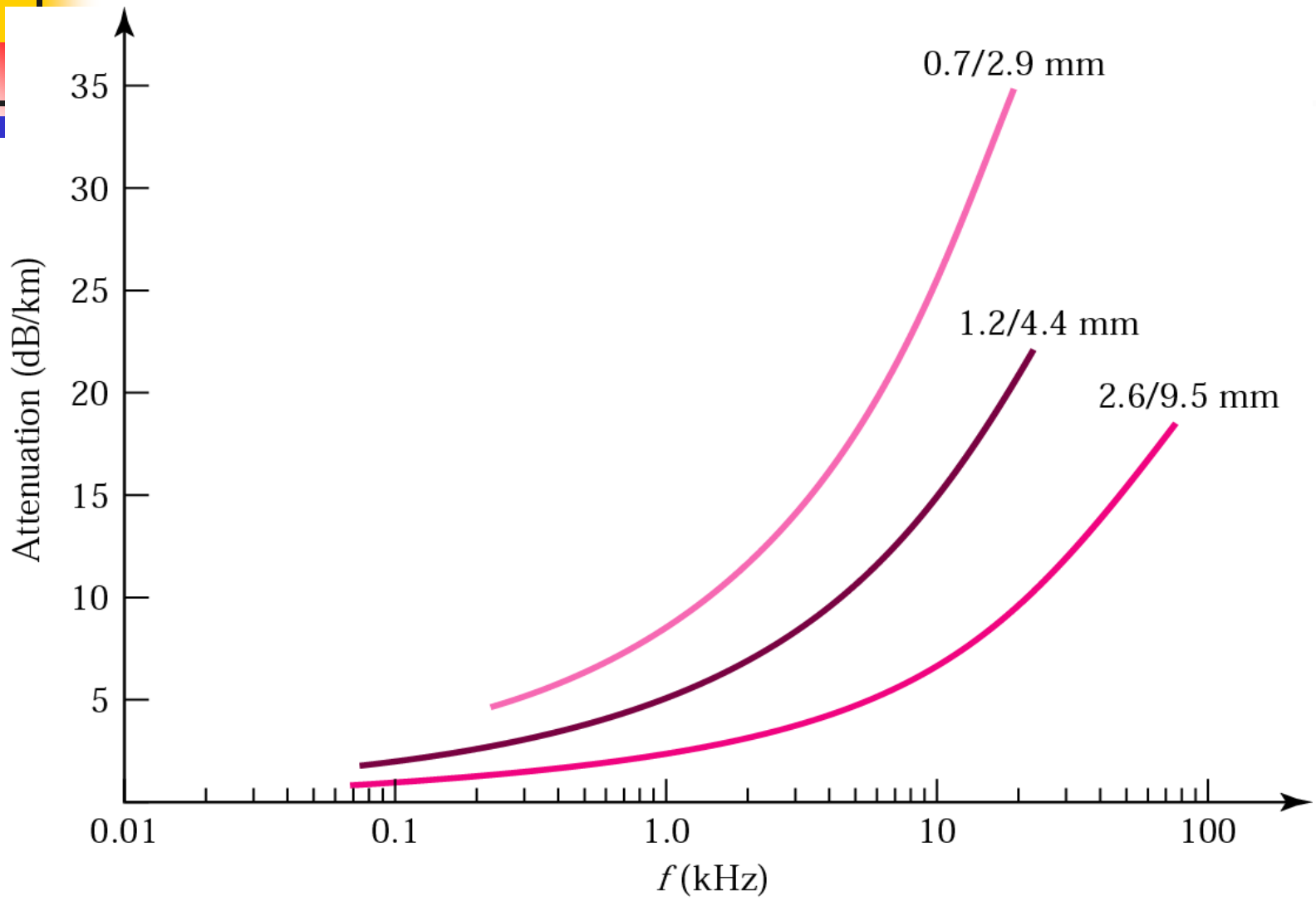
# *Categories of coaxial cables*

| Category | Impedance | Use |
|---|---|---|
| RG-59 | 75 Ω | Cable TV |
| RG-58 | 50 Ω | Thin Ethernet |
| RG-11 | 50 Ω | Thick Ethernet |

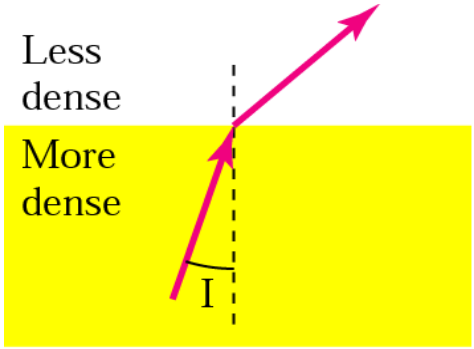
# *BNC connectors*



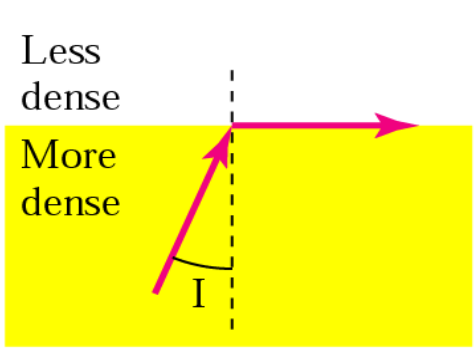
## *Coaxial cable performance*



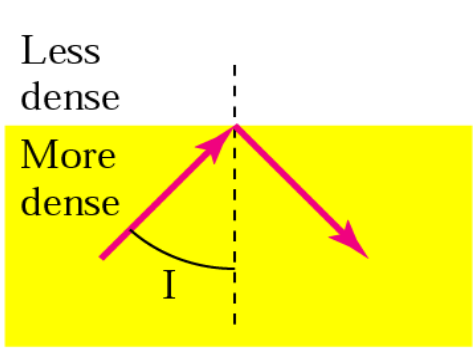
# *Bending of light ray*




I < critical angle,
refraction

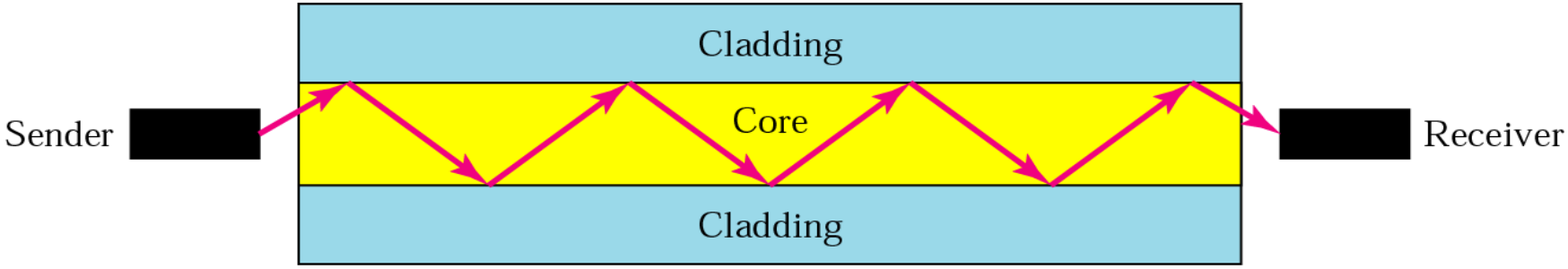



I = critical angle,
refraction

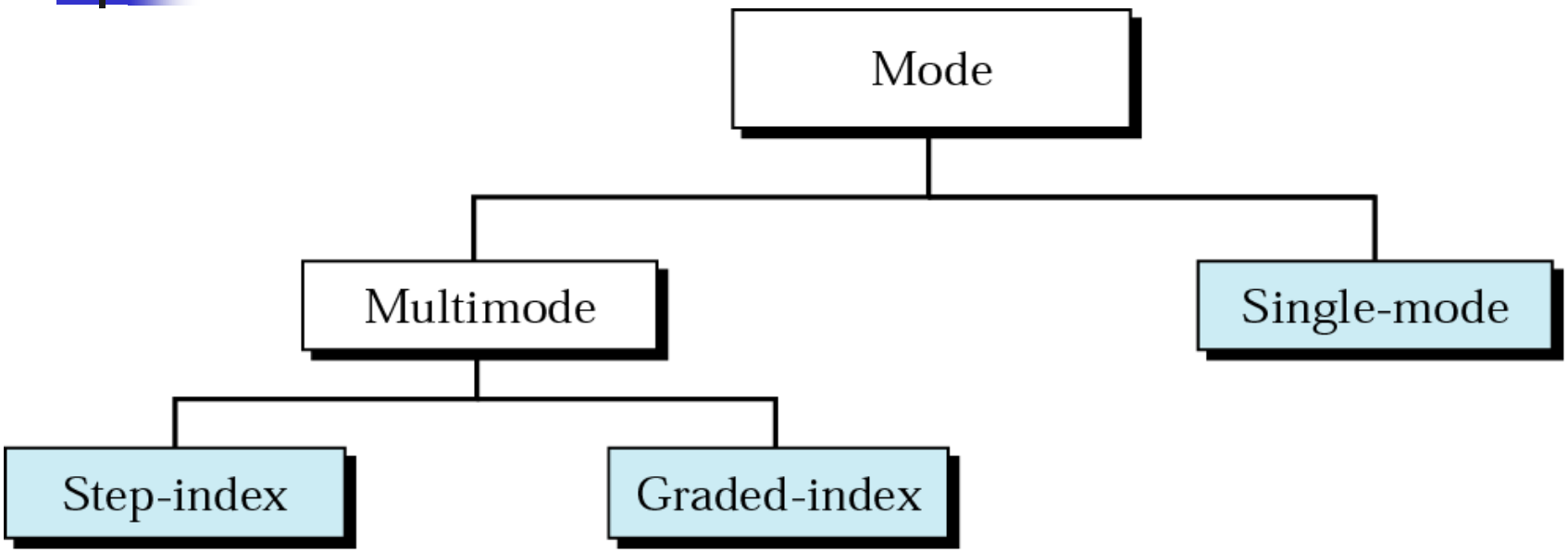



I > critical angle,
reflection

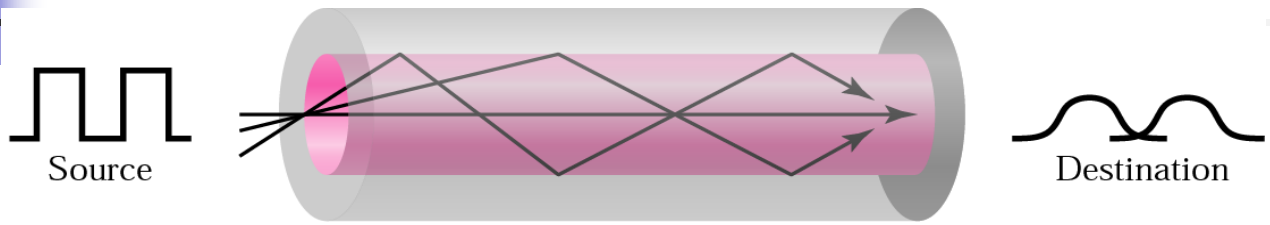
# *Optical fiber*



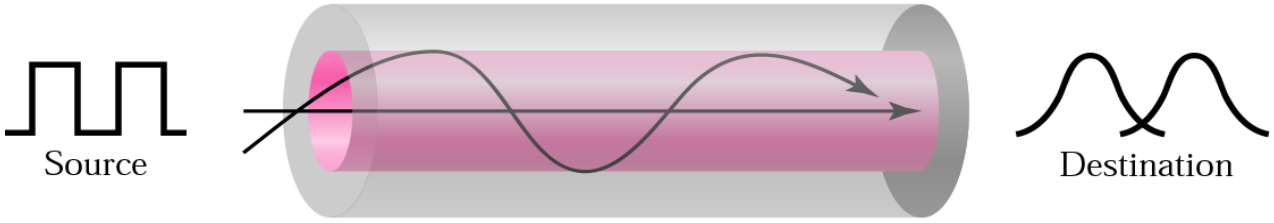
## *Propagation modes*



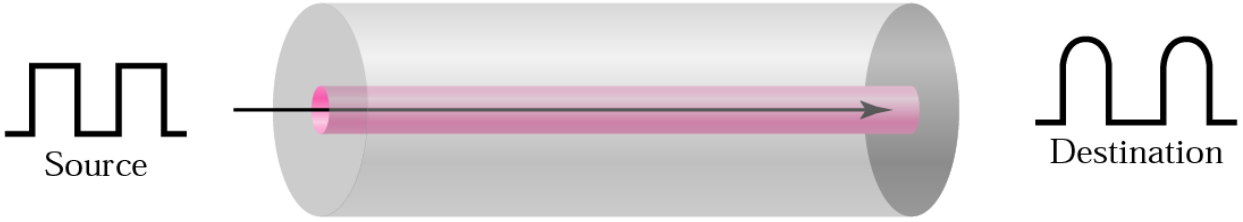
# *Modes*




a. Multimode, step-index




b. Multimode, graded-index

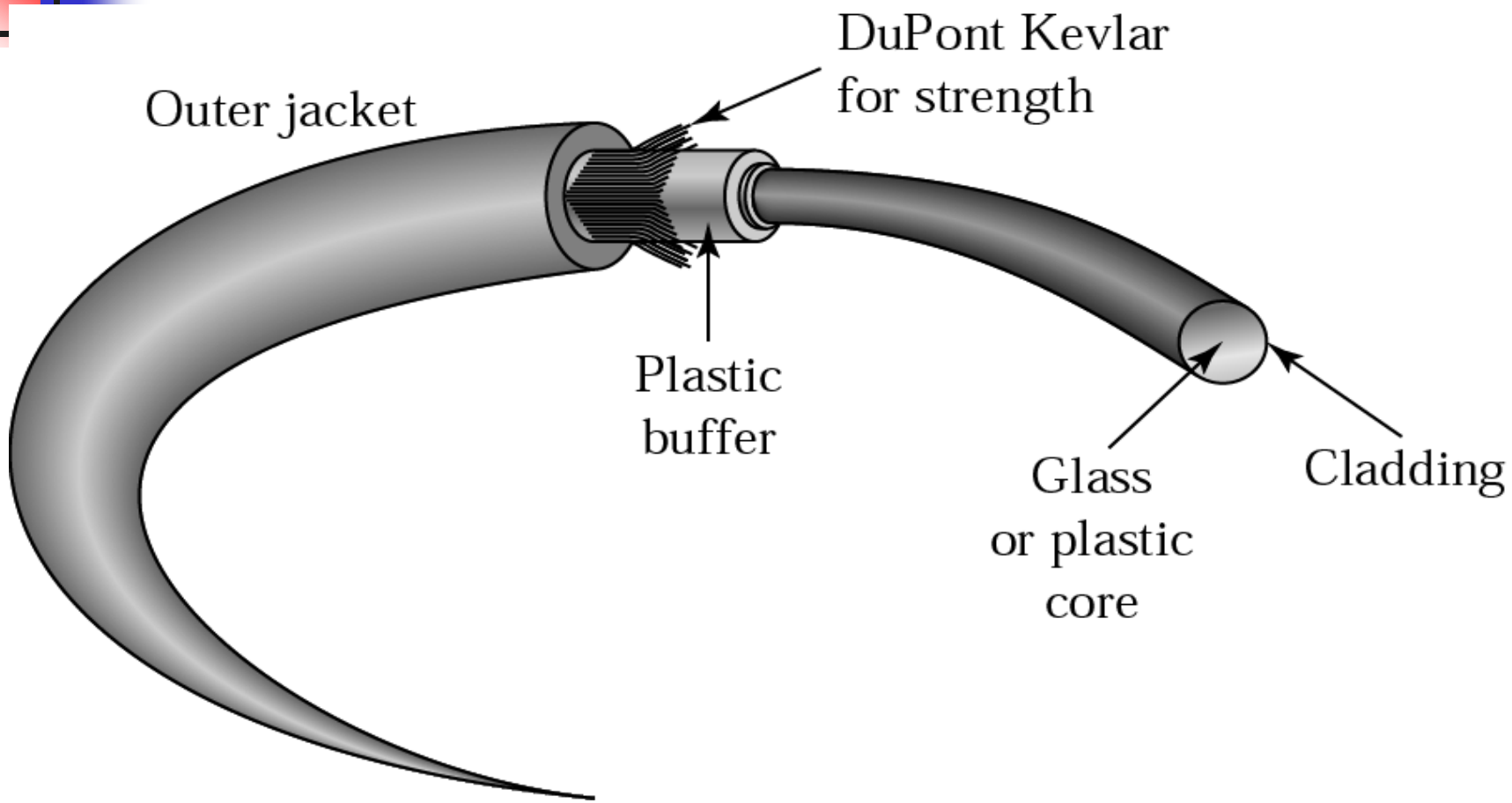



c. Single-mode

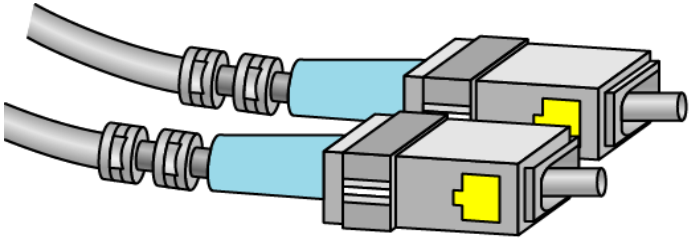
## *Fiber types*

| Type | Core | Cladding | Mode |
|---|---|---|---|
| 50/125 | 50 | 125 | Multimode, graded-index |
| 62.5/125 | 62.5 | 125 | Multimode, graded-index |
| 100/125 | 100 | 125 | Multimode, graded-index |
| 7/125 | 7 | 125 | Single-mode |

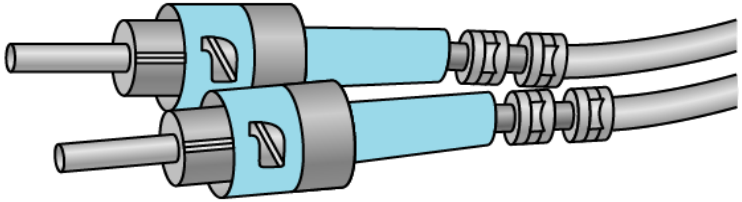
***Fiber construction***



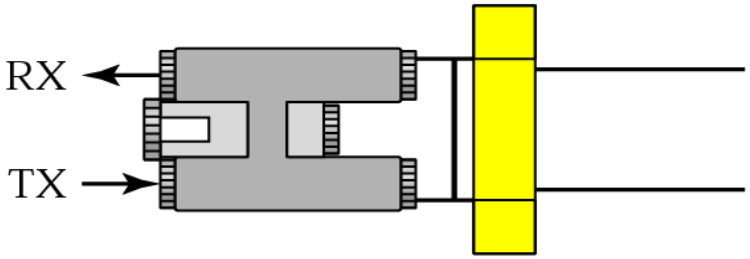
## *Fiber-optic cable connectors*



SC connector

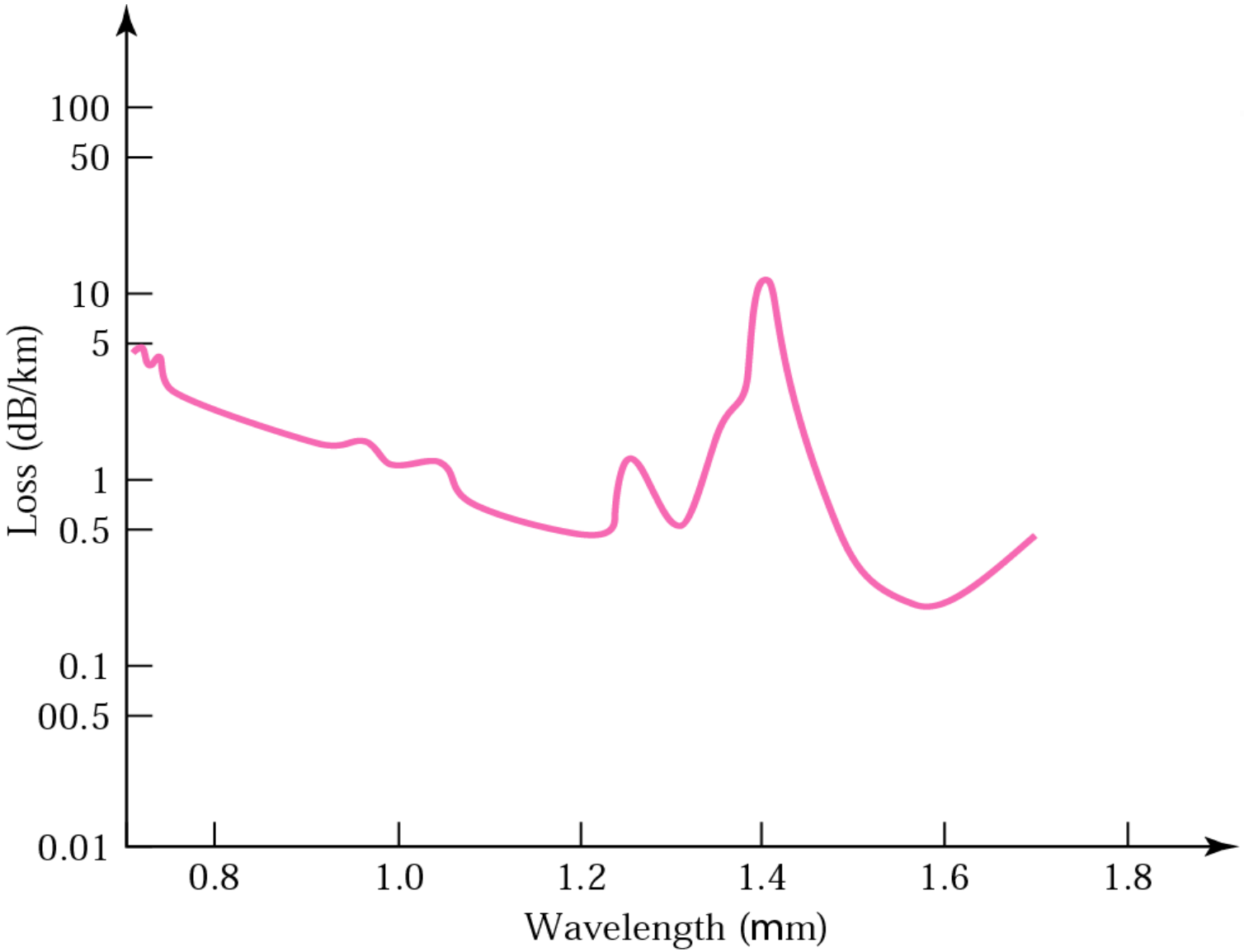


ST connector




MT-RJ connector

## *Optical fiber performance*



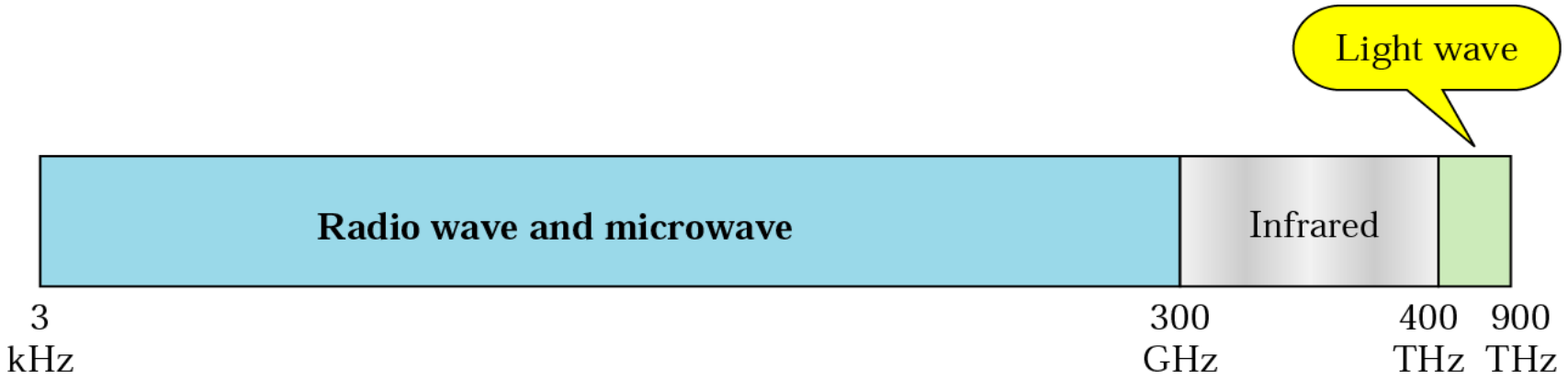
# 7.2 Unguided Media: Wireless

**Radio Waves**

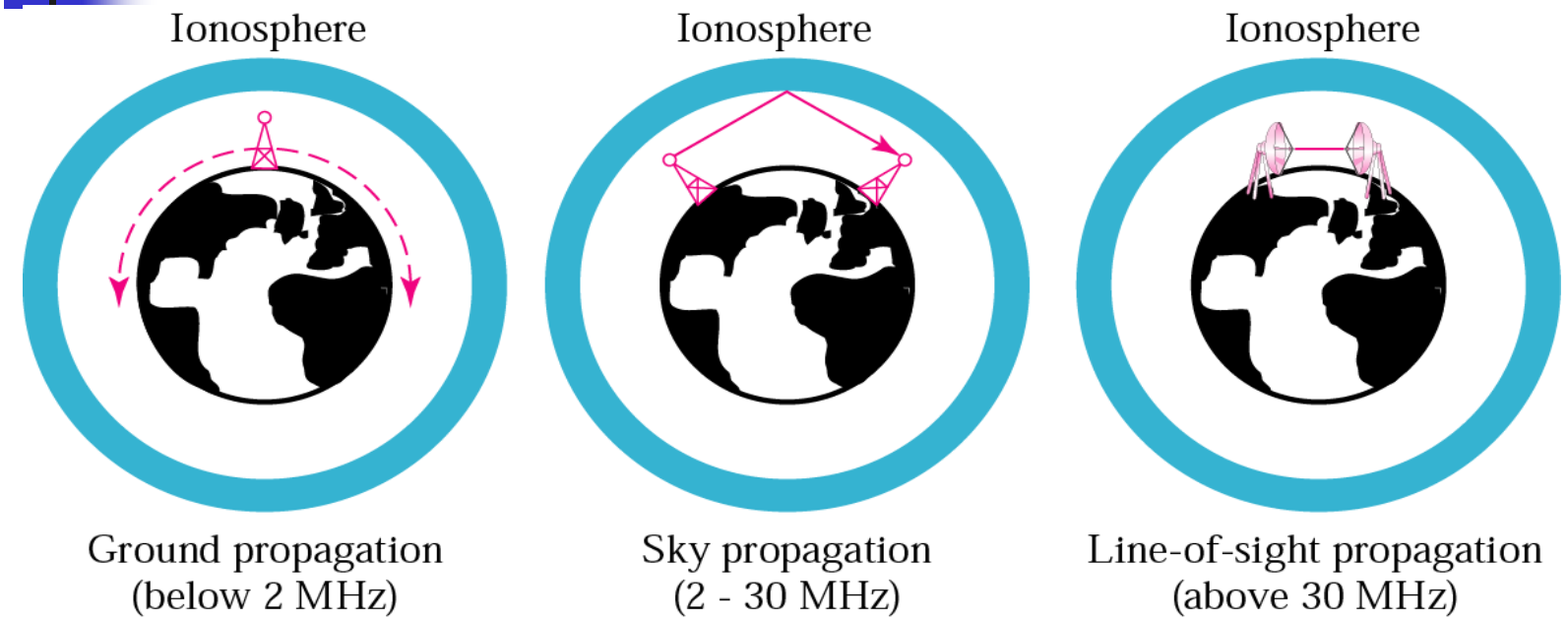
**Microwaves**

**Infrared**

# *Electromagnetic spectrum for wireless communication*



# *Propagation methods*




Ground propagation
(below 2 MHz)

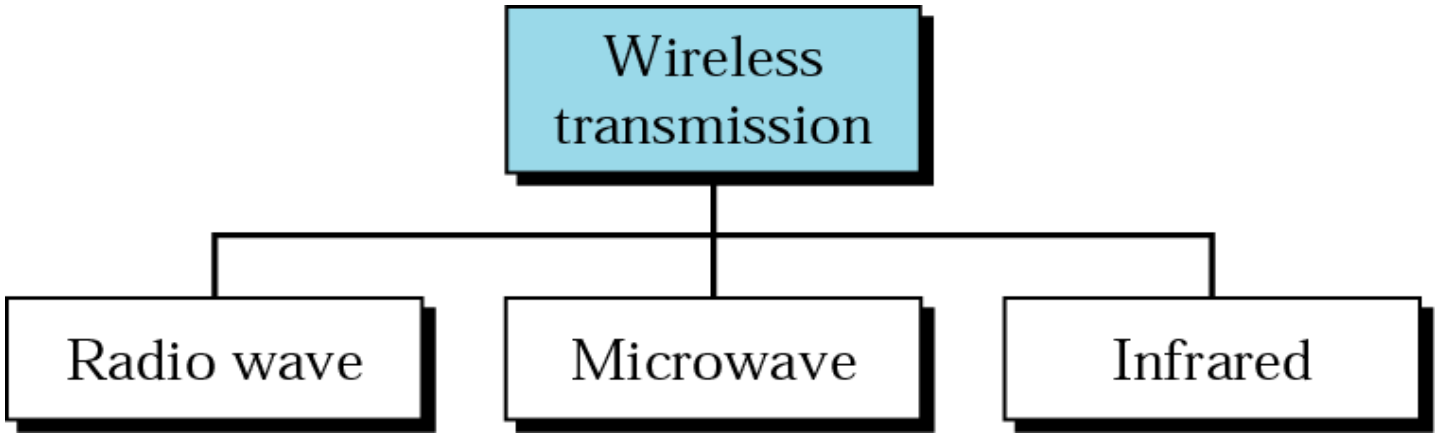
Sky propagation
(2 - 30 MHz)

Line-of-sight propagation
(above 30 MHz)

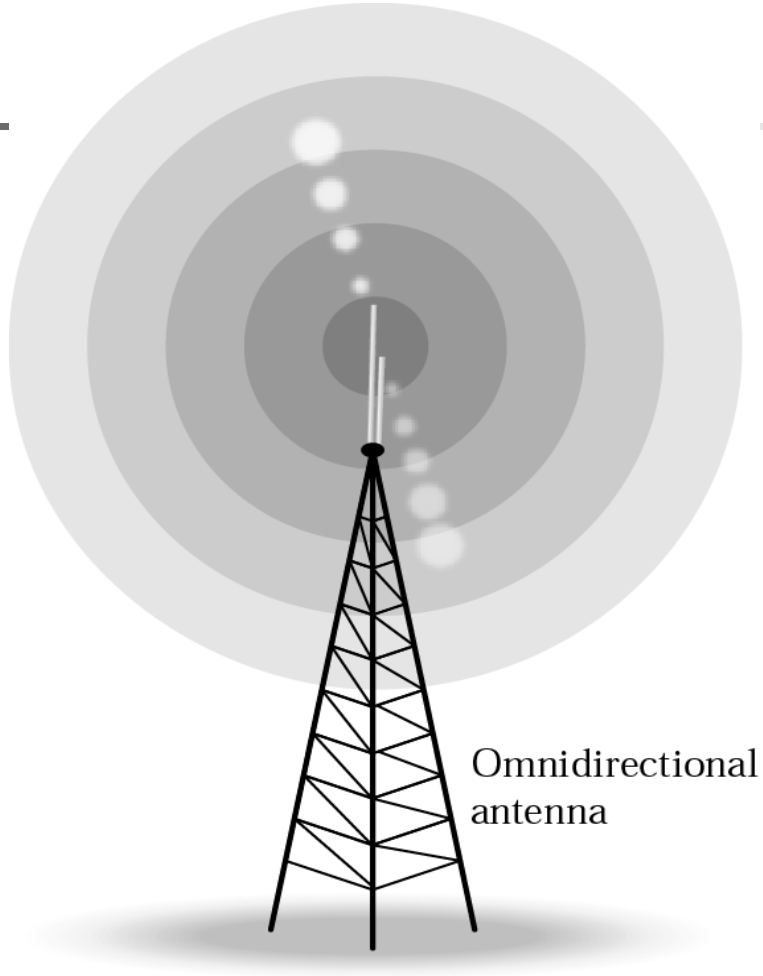
# *Bands*

| Band | Range | Propagation | Application |
|---|---|---|---|
| VLF | 3–30 KHz | Ground | Long-range radio navigation |
| LF | 30–300 KHz | Ground | Radio beacons and navigational locators |
| MF | 300 KHz–3 MHz | Sky | AM radio |
| HF | 3–30 MHz | Sky | Citizens band (CB), ship/aircraft communication |
| VHF | 30–300 MHz | Sky and line-of-sight | VHF TV, FM radio |
| UHF | 300 MHz–3 GHz | Line-of-sight | UHF TV, cellular phones, paging, satellite |
| SHF | 3–30 GHz | Line-of-sight | Satellite communication |
| EHF | 30–300 GHz | Line-of-sight | Long-range radio navigation |

## *Wireless transmission waves*



# *Omnidirectional antennas*

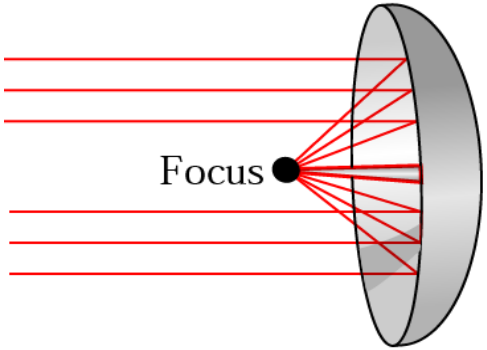




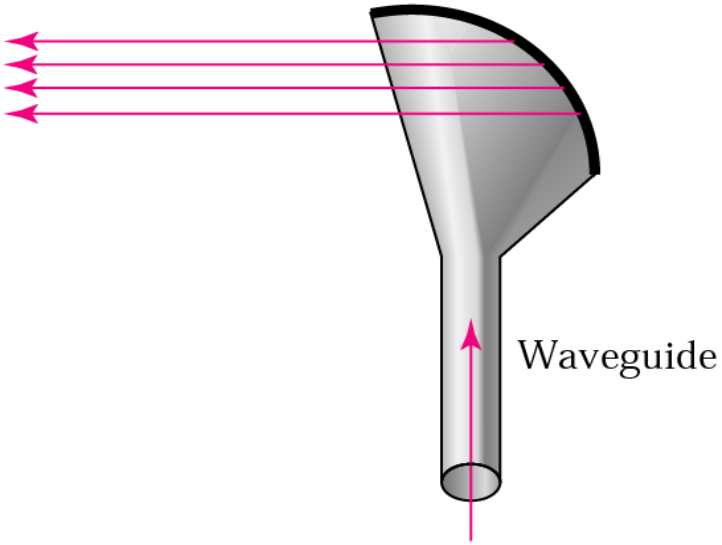
## Note:

***Radio waves are used for multicast communications, such as radio and television, and paging systems.***

# *Unidirectional antennas*




a. Dish antenna




b. Horn antenna



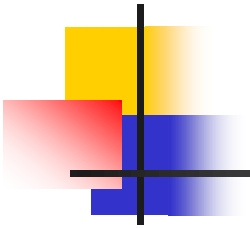
## Note:

***Microwaves are used for unicast communication such as cellular telephones, satellite networks, and wireless LANs.***



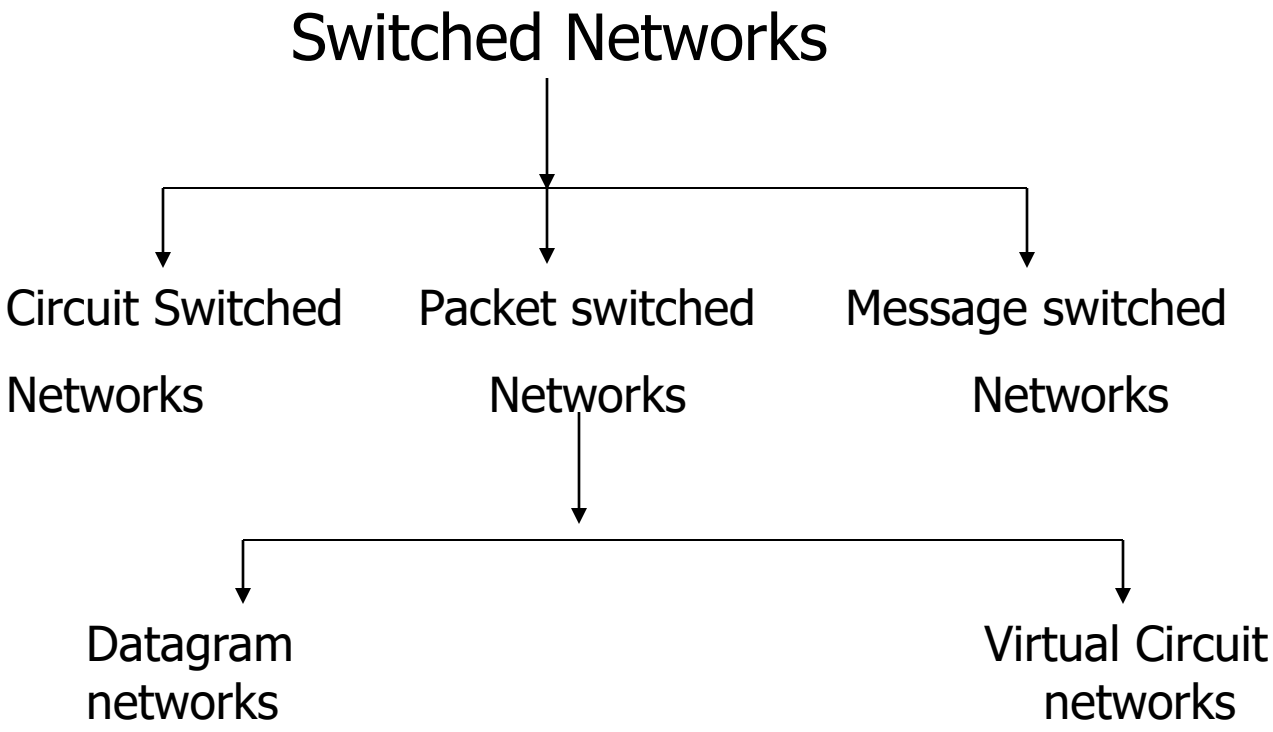
## Note:

***Infrared signals can be used for short-range communication in a closed area using line-of-sight propagation.***



# *Circuit Switching and Telephone Network*

# Taxonomy of switched networks



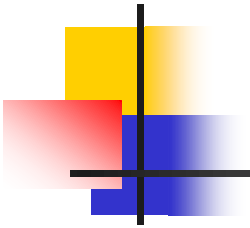
# 8.1 Circuit Switching

**Space-Division Switch**

**Time-Division Switch**

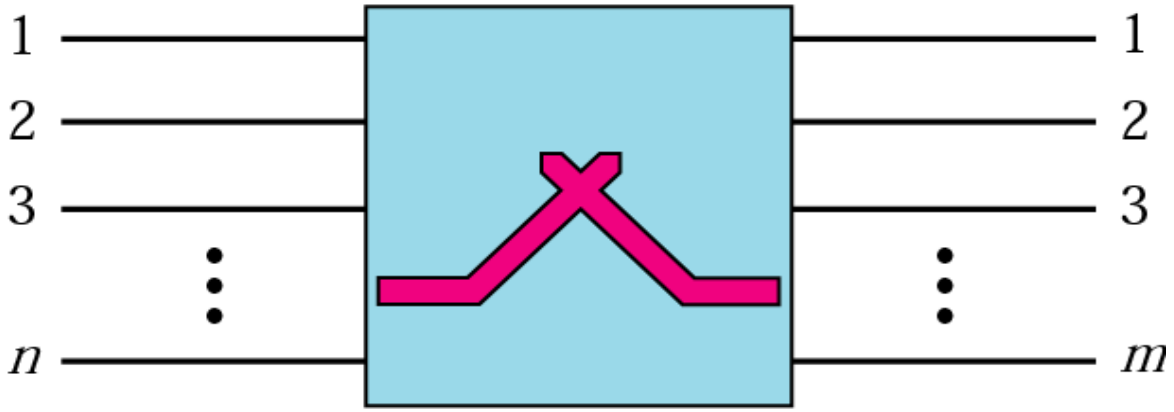
**TDM Bus**

**Combinations**



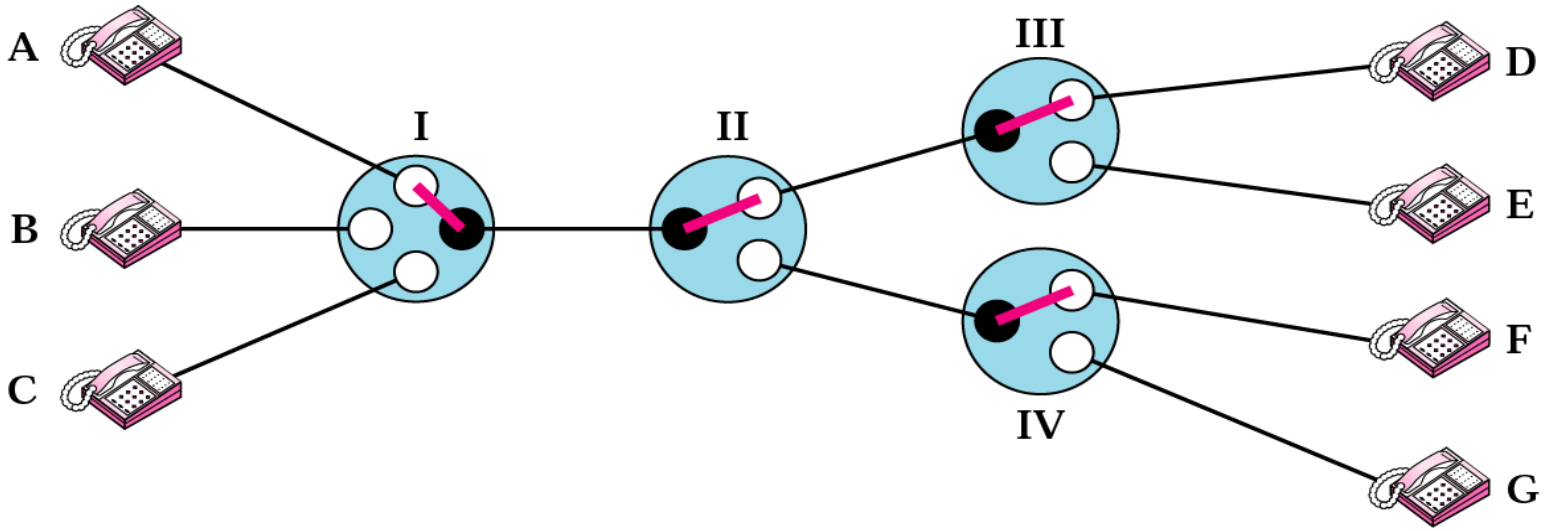
- Circuit Switching
- Store-and -Forward Networks
  - Message Switching
  - Packet Switching
    - connection-oriented vs connectionless
    - virtual circuit vs datagram

# *A circuit switch*




A circuit switched network is made up of a set of switches connected by a physical links, in which each link is divided into n channels

# *Circuit-switched network*





# Circuit Switching

- Seeking out and establishing a physical copper path from end-to-end [historic definition] .
- Circuit switching implies the need to first ***set up*** a dedicated, end-to-end path for the connection ***before*** the information transfer takes place.
- Once the connection is made the only delay is propagation time.

# Circuit switching continue…

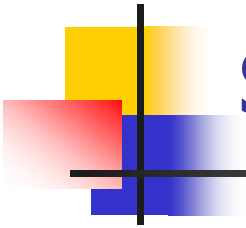
- Circuit switching takes place at the physical layer

- Before starting communication, the station must makes a reservation for the recourses to be used during the communications. These resources ,such as channel (BW in FDM and time slot in TDM) switch buffers, switch processing time and switch input/output ports, must remain dedicated during the entire duration of data transfer until the teardown phase

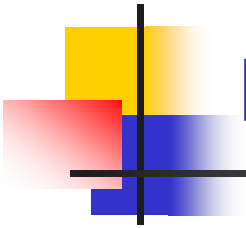
- Data transfer between the two stations are not packetized. The data are a continuous flow send by the source station and the receiving station, although there may be periods of silence

- There is no addressing involved during data transfer. The switches route the data based on their occupied band or slot.

# Three Phases in circuit switching

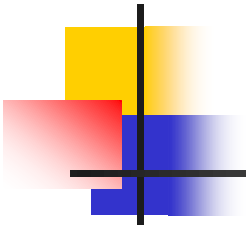


- setup phase
- Data transfer phase
- Teardown phase : when one of the parties needs to disconnect ,a signal is sent to each switch to release the resources



# Efficiency of circuit switched Network

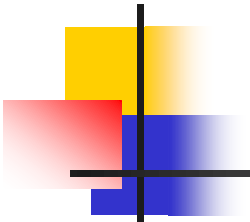
- Not efficient as compare to packet switching and message switching because resources are allocated during the entire duration of the connection. These resources are unavailable to other connections



# Delay

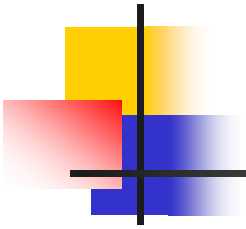
- Delay is mimimal.There is no waiting time at each switch
- The total delay is due to the time needed to create the connection, transfer data and disconnect the circuit.

# Store-and-Forward Networks



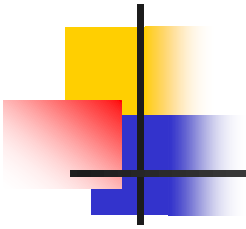
- Intermediate processors (nodes, routers, gateways, switches) along the path store the incoming block of data.

- Each block is received in its , inspected for errors, and retransmitted along the path to the destination. This implies buffering at the router and one transmission time per hop.



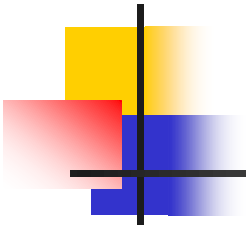
# Message Switching

- A store-and-forward network where the block of transfer is a complete
- Since messages can be quite large, this can cause:
  - buffering problems
  - high mean delay times



# Packet Switching

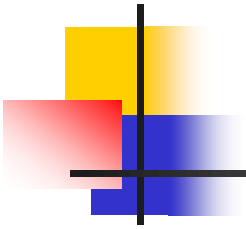
- A store-and-forward network where the block of transfer is a complete
- A packet is a variable length block of data with a tight upper bound.
  - Using packets improves mean message delay.



# Packet Switched Networks

## Connection-oriented Protocols

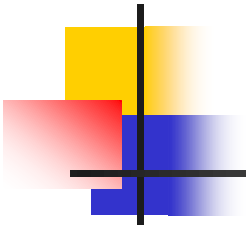
- A setup stage is used to determine the end-to-end path before a connection is established.
- Data flow streams are identified by some type of connection indicator (e.g. OSI, X.25, SNA).



# Packet Switched Networks

## Connectionless Protocols

- No set up is needed.
- Each packet contains information which allows the packet to be individually routed hop-by-hop through the network.



# Datagram vs Virtual Circuit

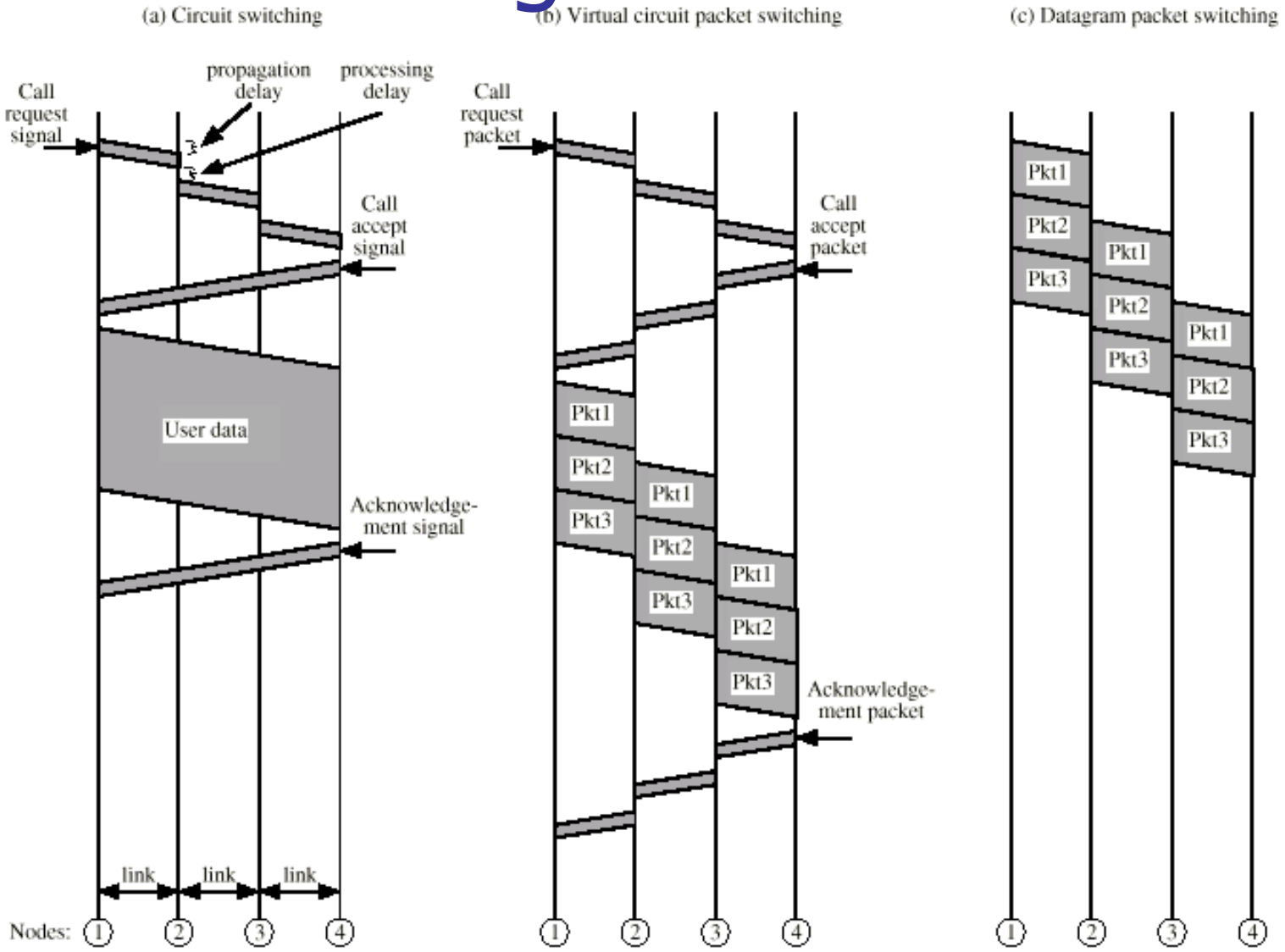
## Datagram

- Each datagram packet may be individually routed.

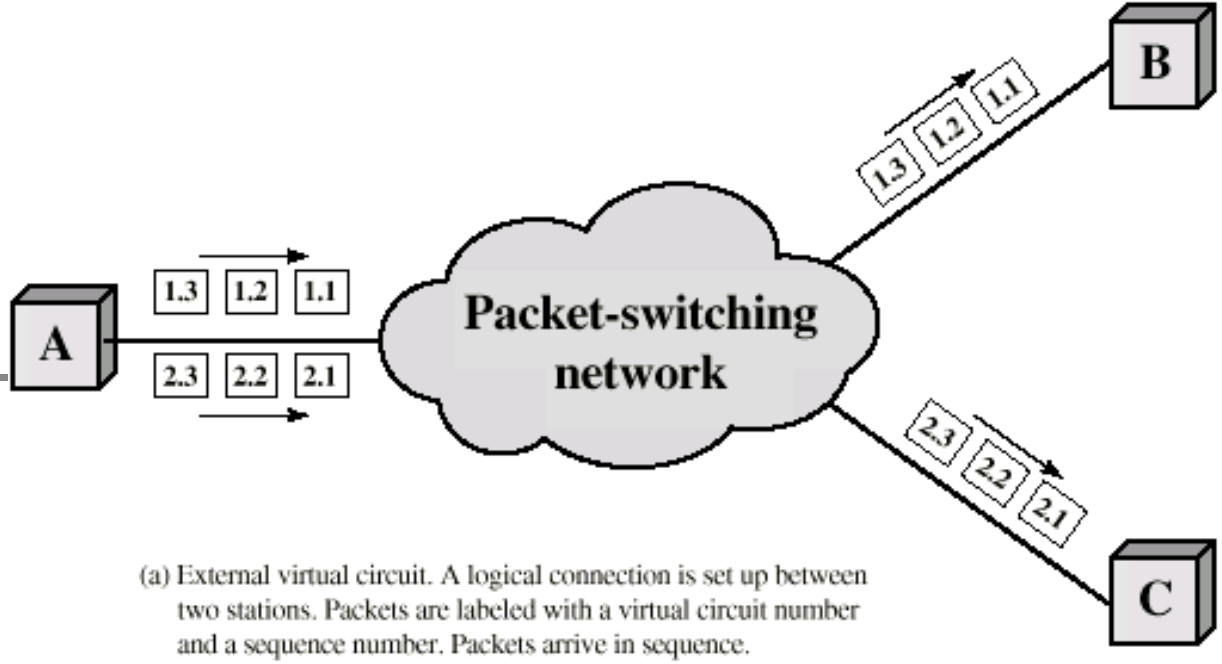
## Virtual Circuit

- Virtual circuit set up is required.
- All packets in a virtual circuit follow the same path.

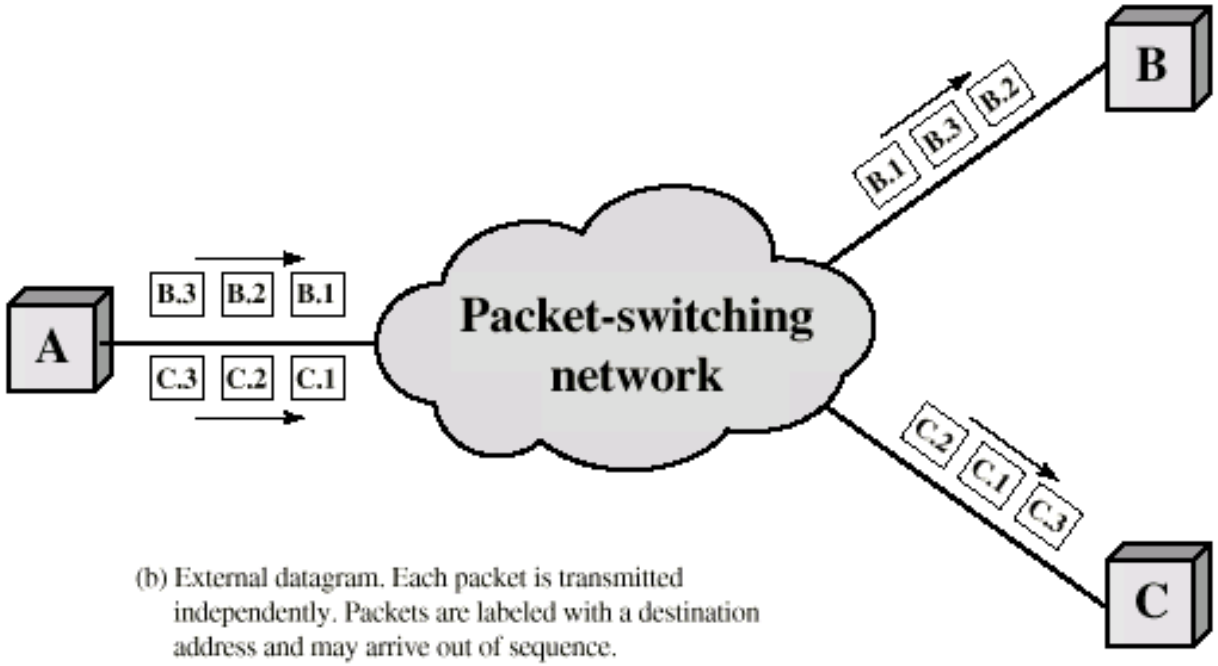
# Event Timing



# External Virtual Circuit And Datagram Operation

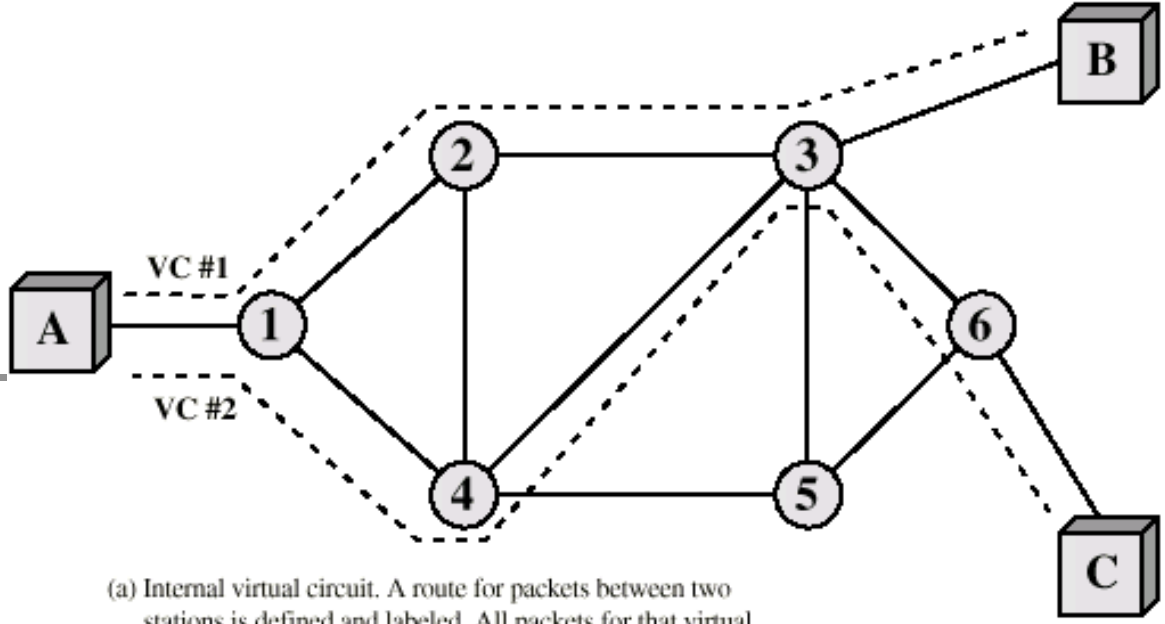



(a) External virtual circuit. A logical connection is set up between two stations. Packets are labeled with a virtual circuit number and a sequence number. Packets arrive in sequence.

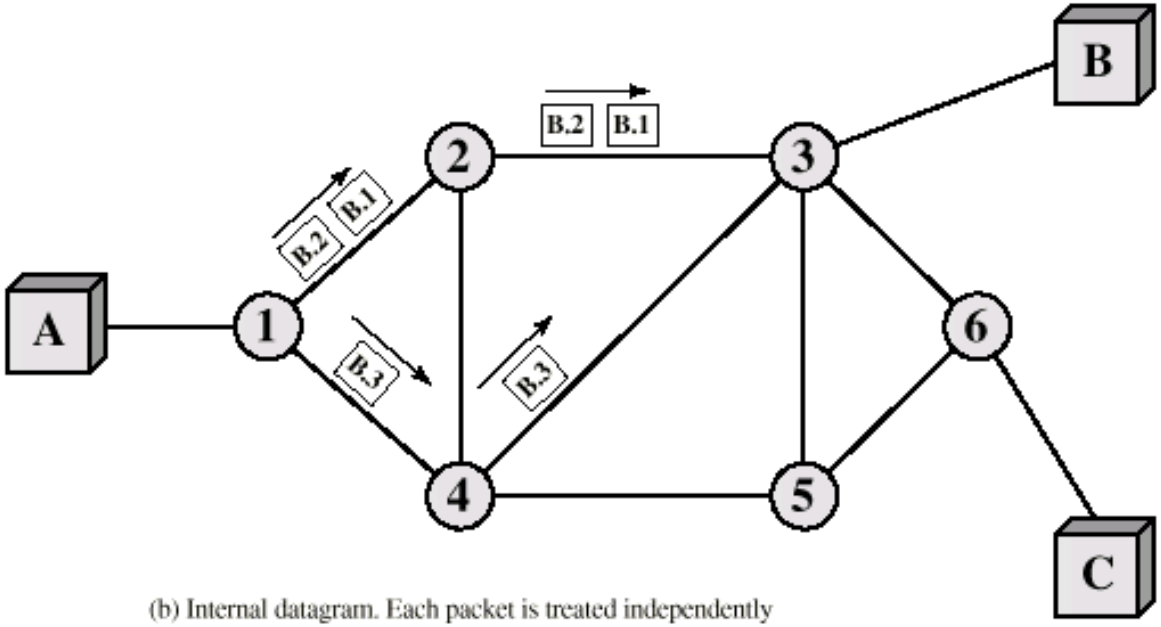



(b) External datagram. Each packet is transmitted independently. Packets are labeled with a destination address and may arrive out of sequence.

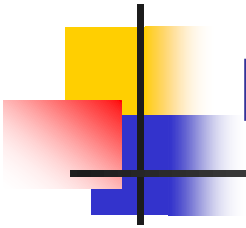
# Internal Virtual Circuit And Datagram Operation




(a) Internal virtual circuit. A route for packets between two stations is defined and labeled. All packets for that virtual circuit follow the same route and arrive in the same sequence.




(b) Internal datagram. Each packet is treated independently by the network. Packets are labeled with a destination address and may arrive at the destination node out of sequence.



# DATAGRAM NETWORKS

- In a packet switched network, there is no recourses reservation; resources are allocated on demand
- Connectionless networks
- Routing table
- Efficiency : Efficiency is better than circuit switched networks. Resources are allocated only when there are packets to be transferred
- Delay : There may be greater delay in a datagram network than in virtual circuit network. There are no setup and teardown phase each packet may experience a wait at a switch before it is forwarded

# Virtual-circuit networks

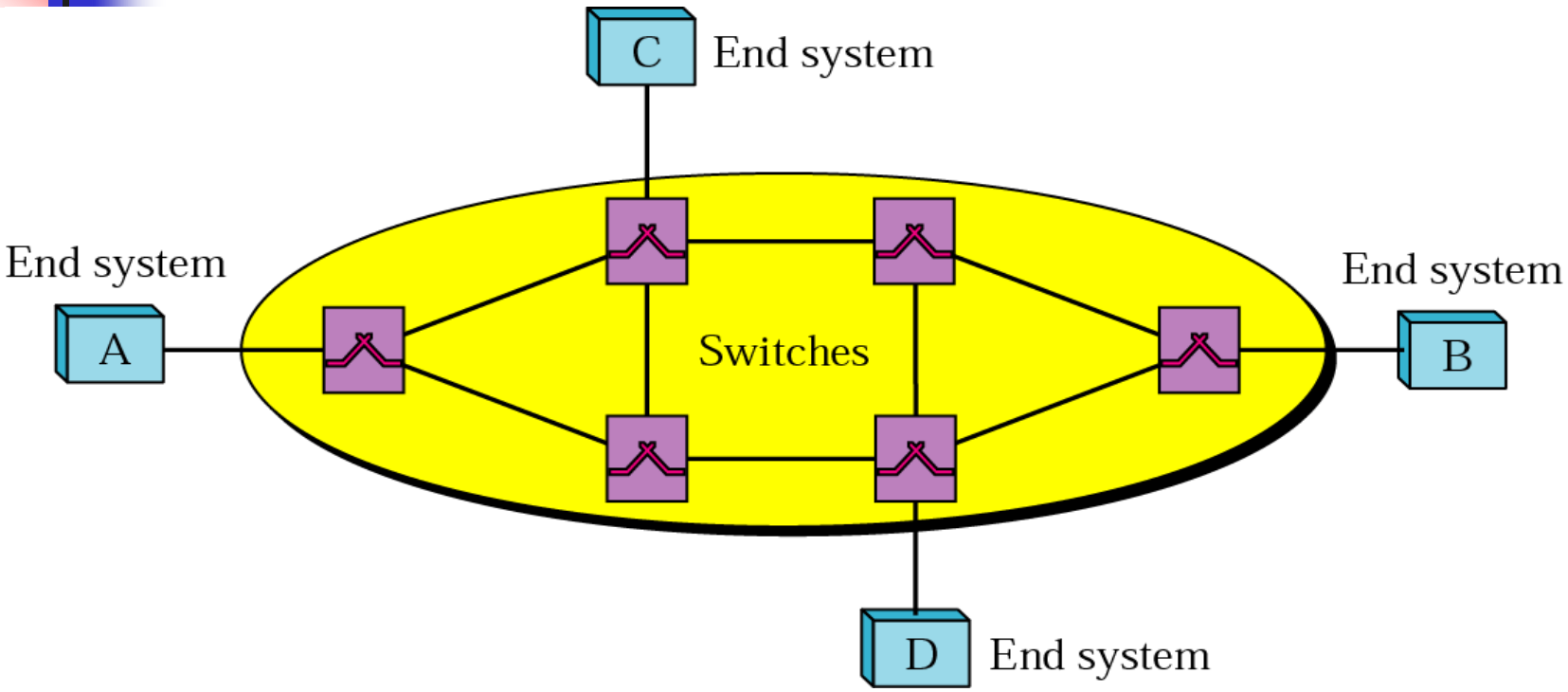
- It is cross between a circuit switched network and a datagram network
- It has some characteristics of both
  1. As in a circuit switched network, there are setup and teardown phase in addition to the data transfer phase
  2. Resources are allotted during the setup phase and as in a circuit switching and demand based in datagram approach

# Virtual Circuit networks

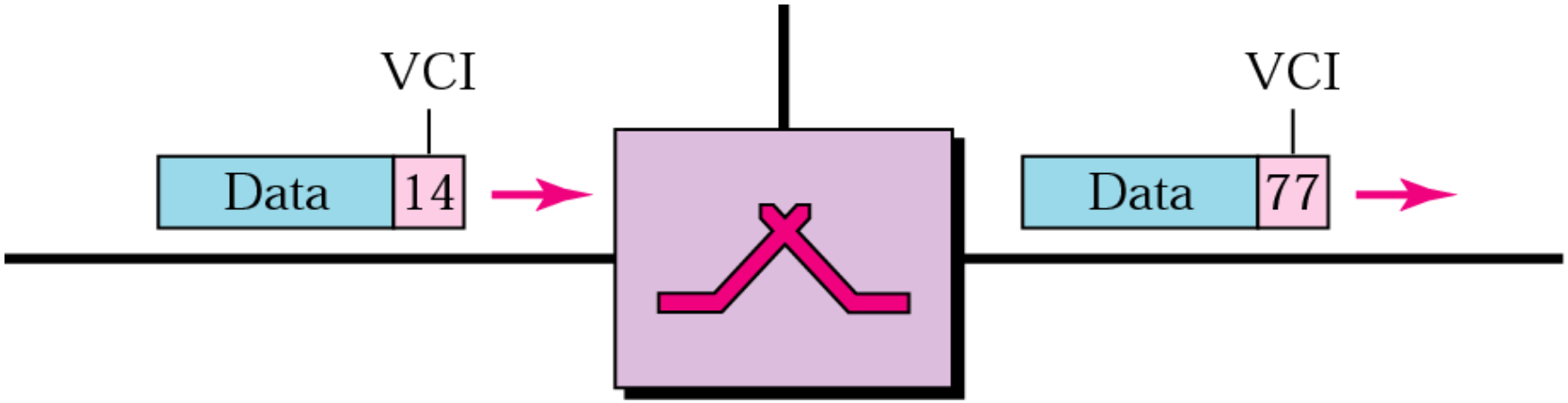
- It is cross between a circuit switched network and a datagram network
- Virtual circuit network is implemented in data link layer while a circuit switch network implemented in a physical layer and Datagram in network layer.
- Addressing ： 1) Global :unique in the scope of the n/w

  2) Local (Virtual circuit identifier) : identifier which is actually used for data transfer. It is a small number that has only switch scope. It is used by a frame between two switches. when a frame arrive it has a VCI. When it leaves it has different VCI.

**Figure 18.1** **Virtual circuit wide area network**



## Figure 18.2 VCI



## Figure 18.3 VCI phases

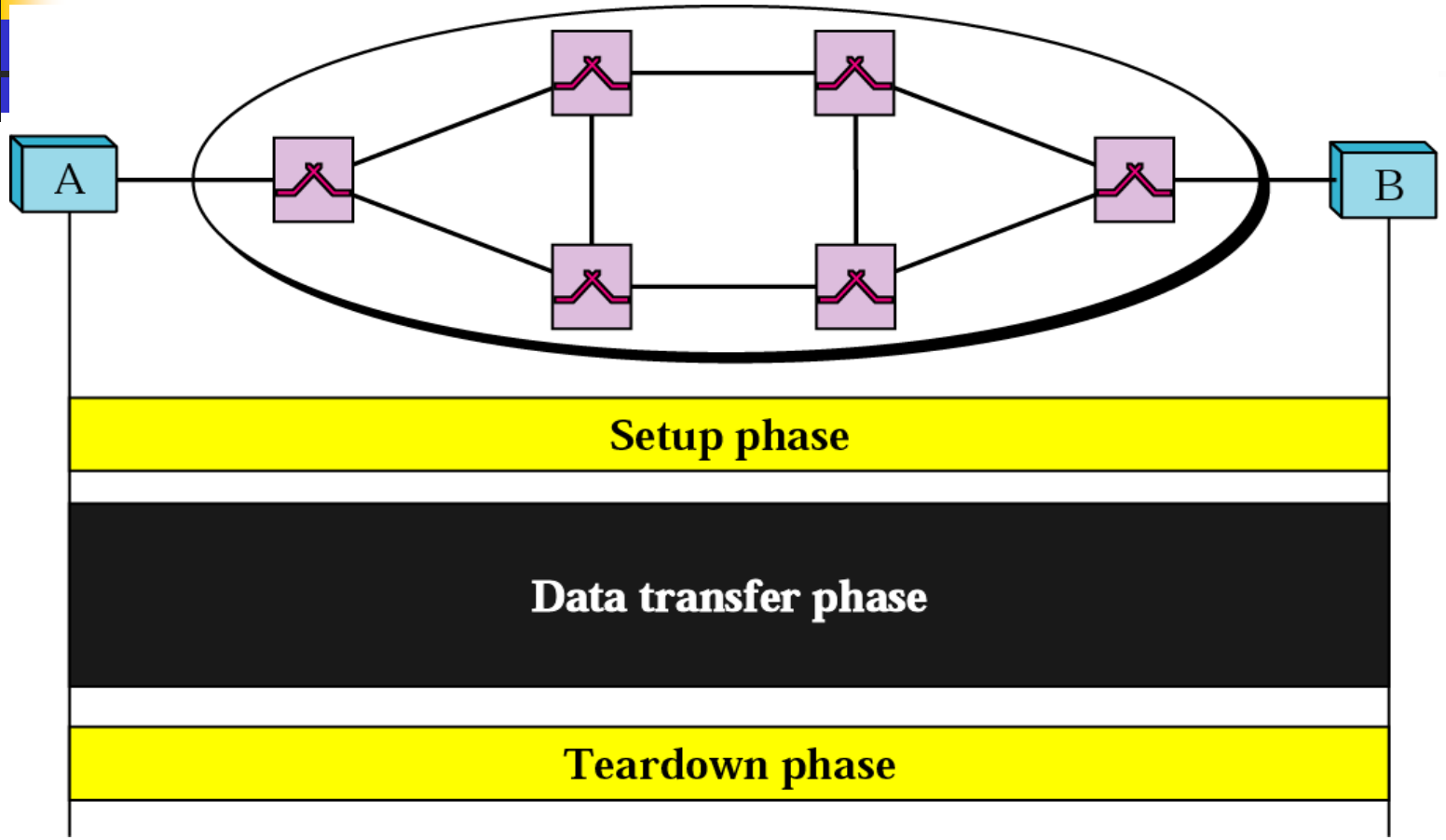


**Figure 18.4** Switch and table

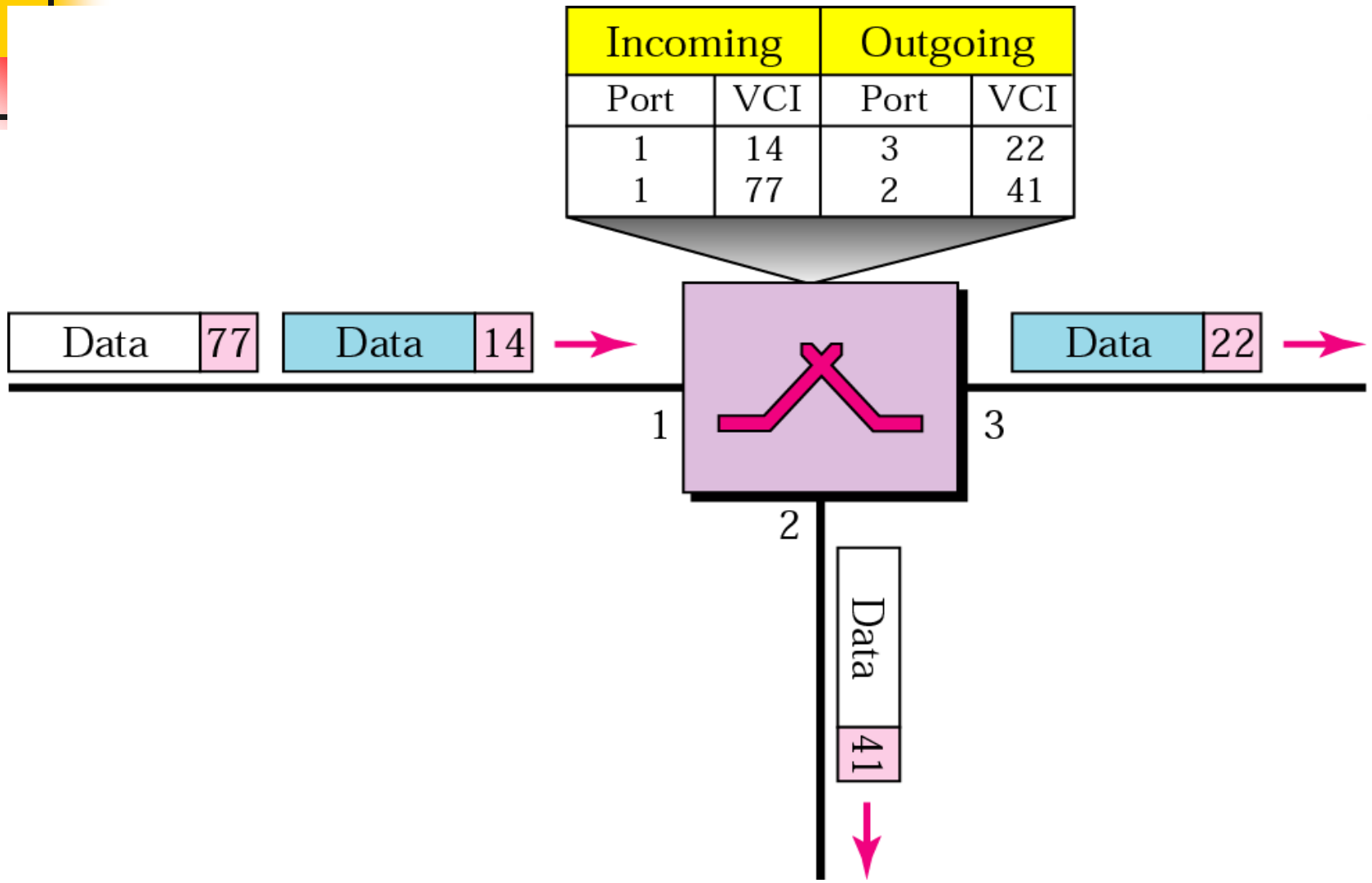
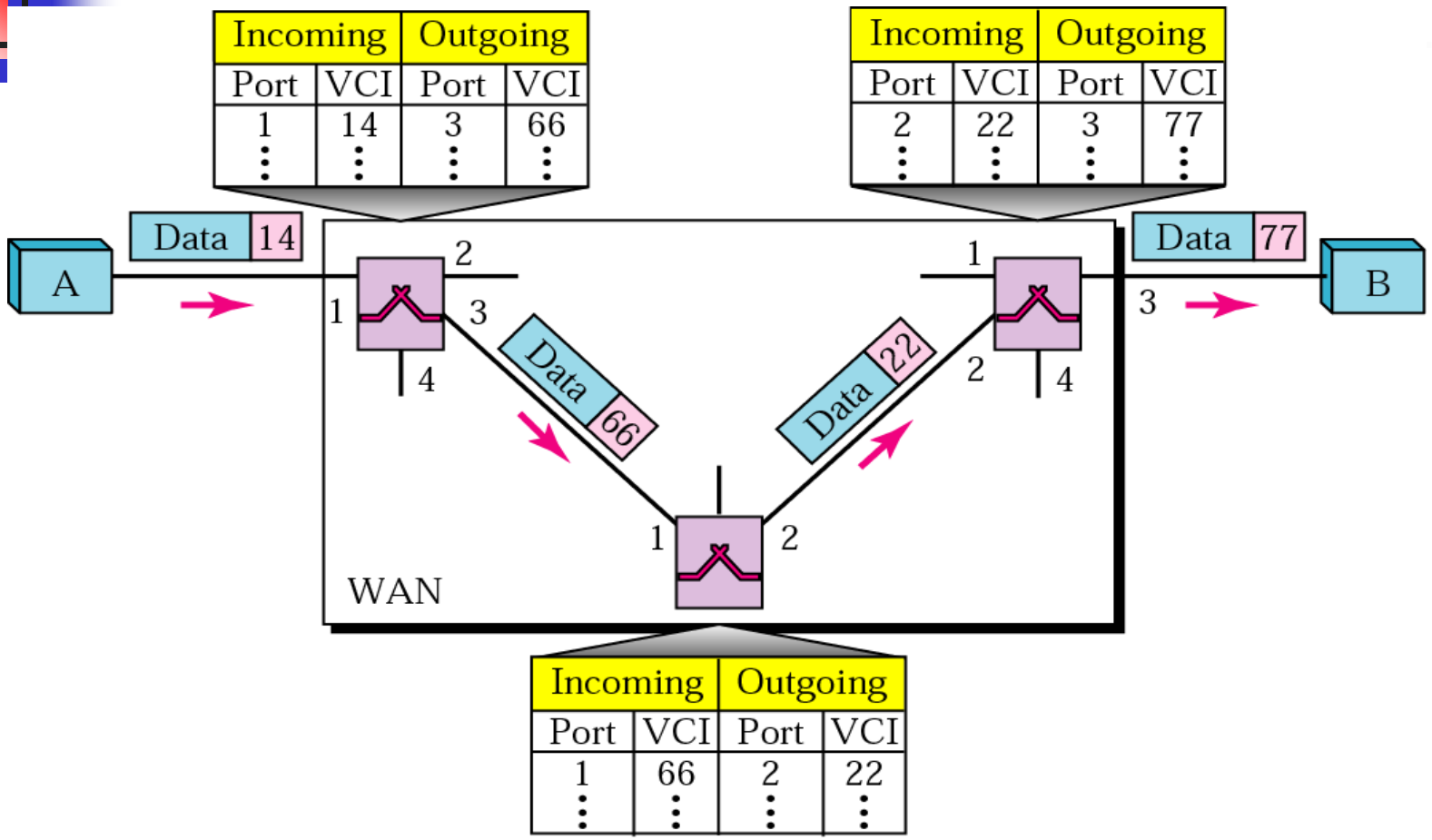
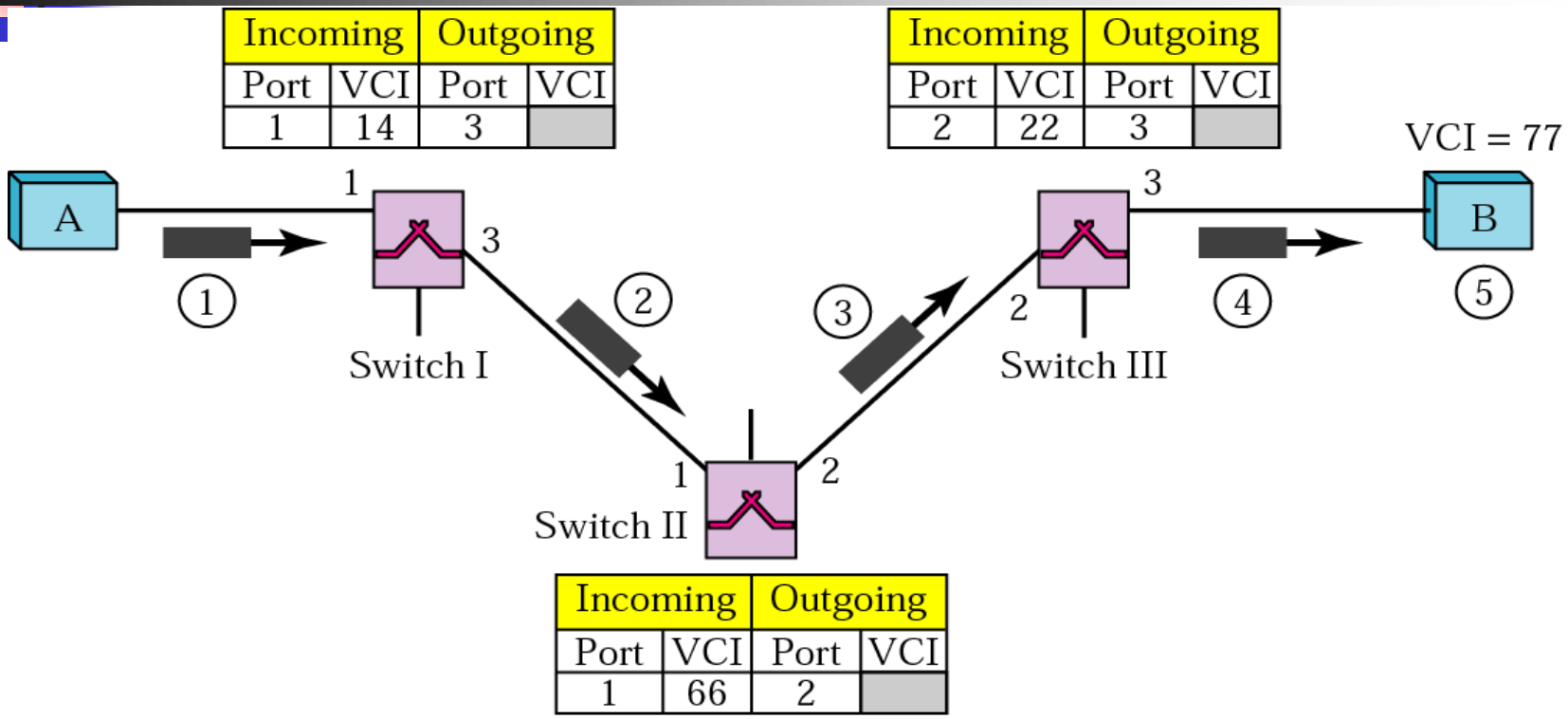


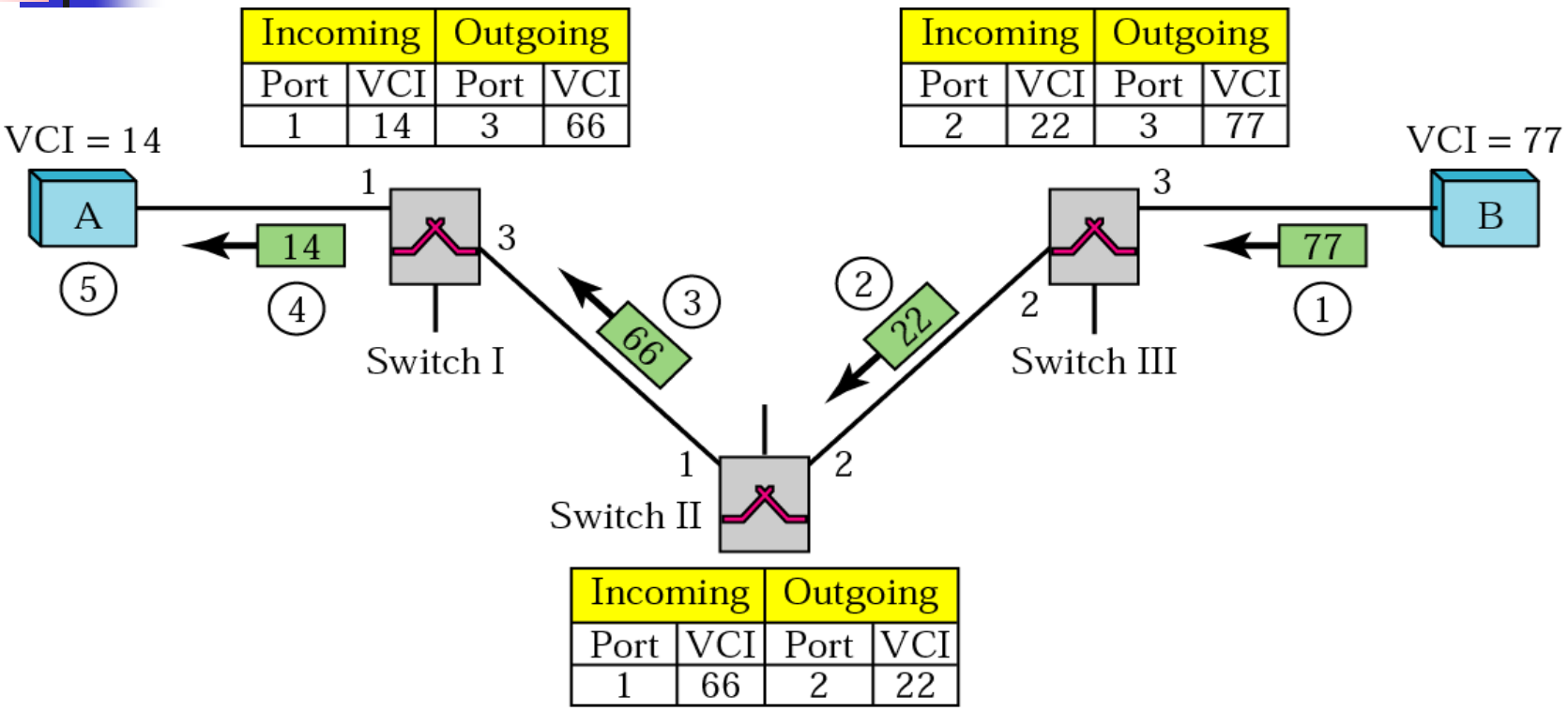
**Figure 18.5** Source-to-destination data transfer



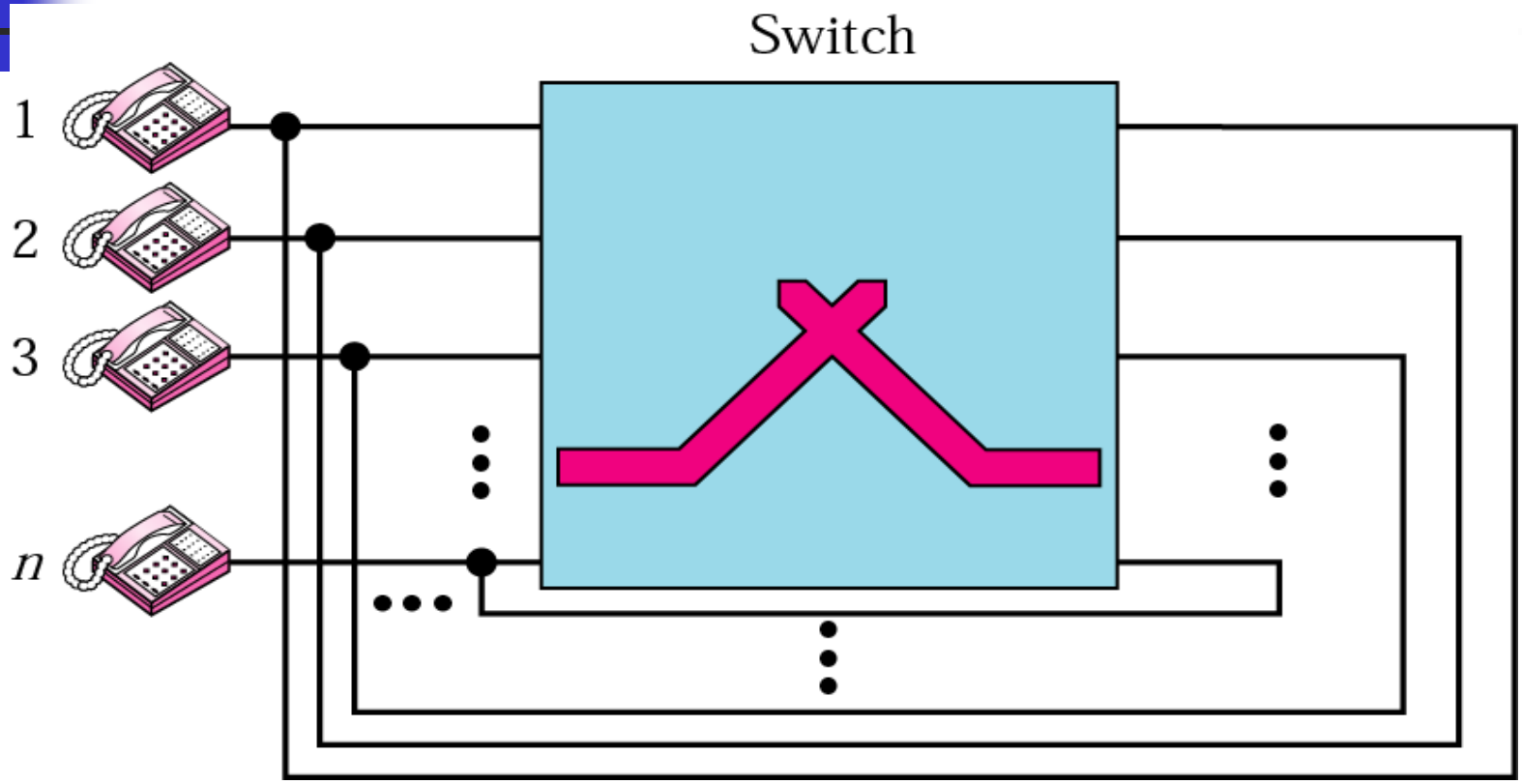
## Figure 18.6 SVC setup request



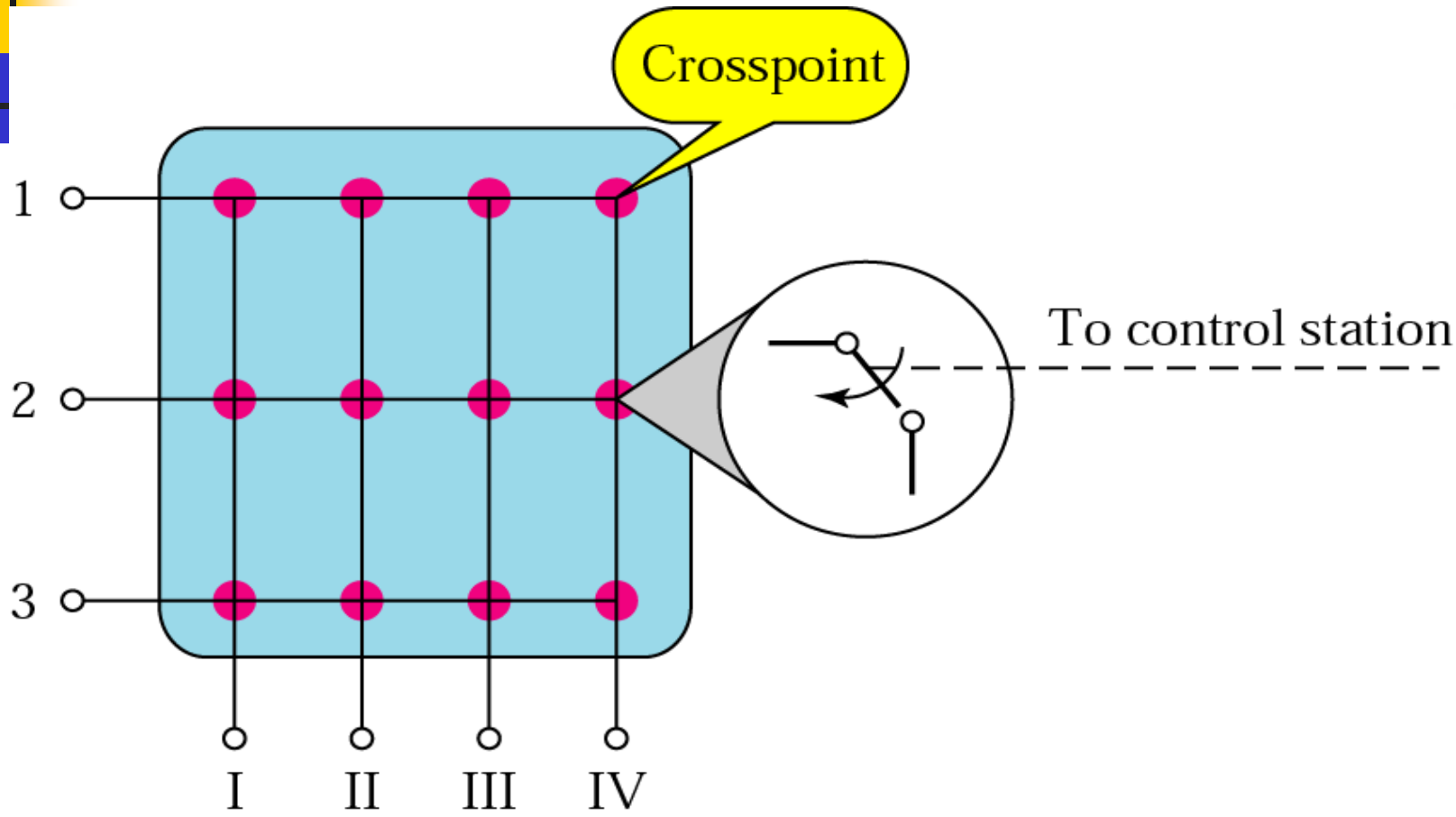
## Figure 18.7 SVC setup acknowledgment



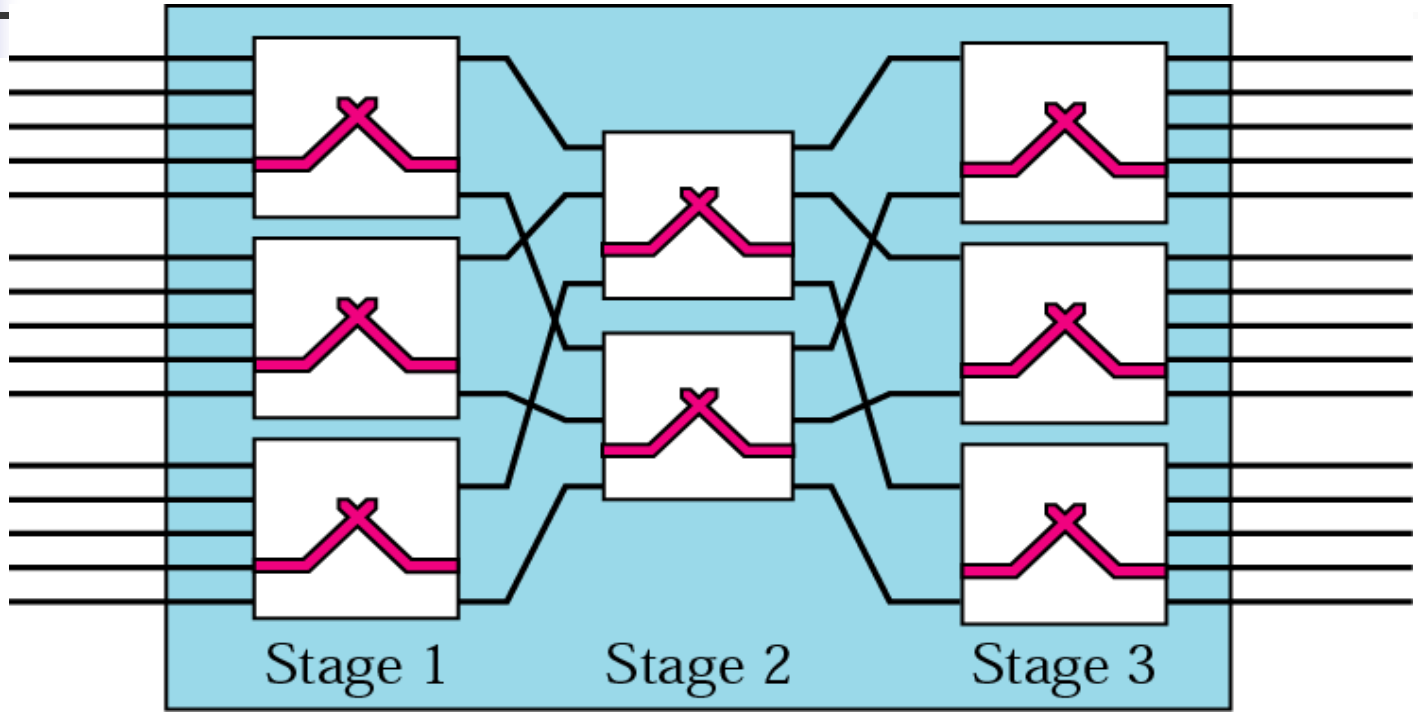
# *A folded switch*



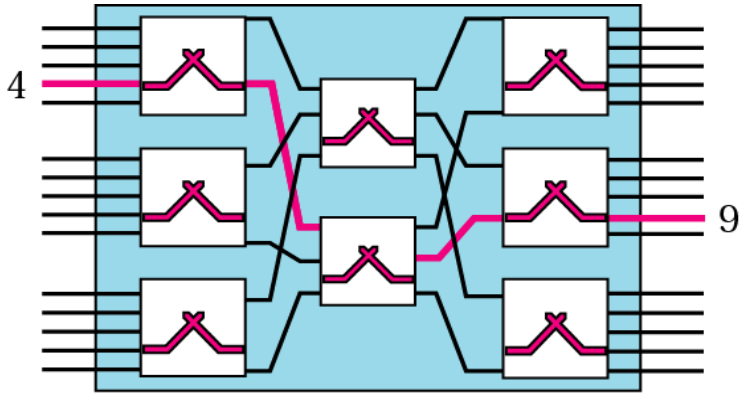
# *Crossbar switch*



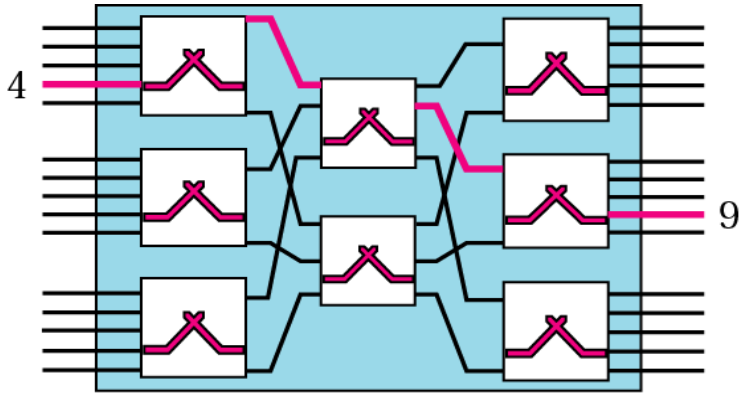
# *Multistage switch*



# *Switching path*

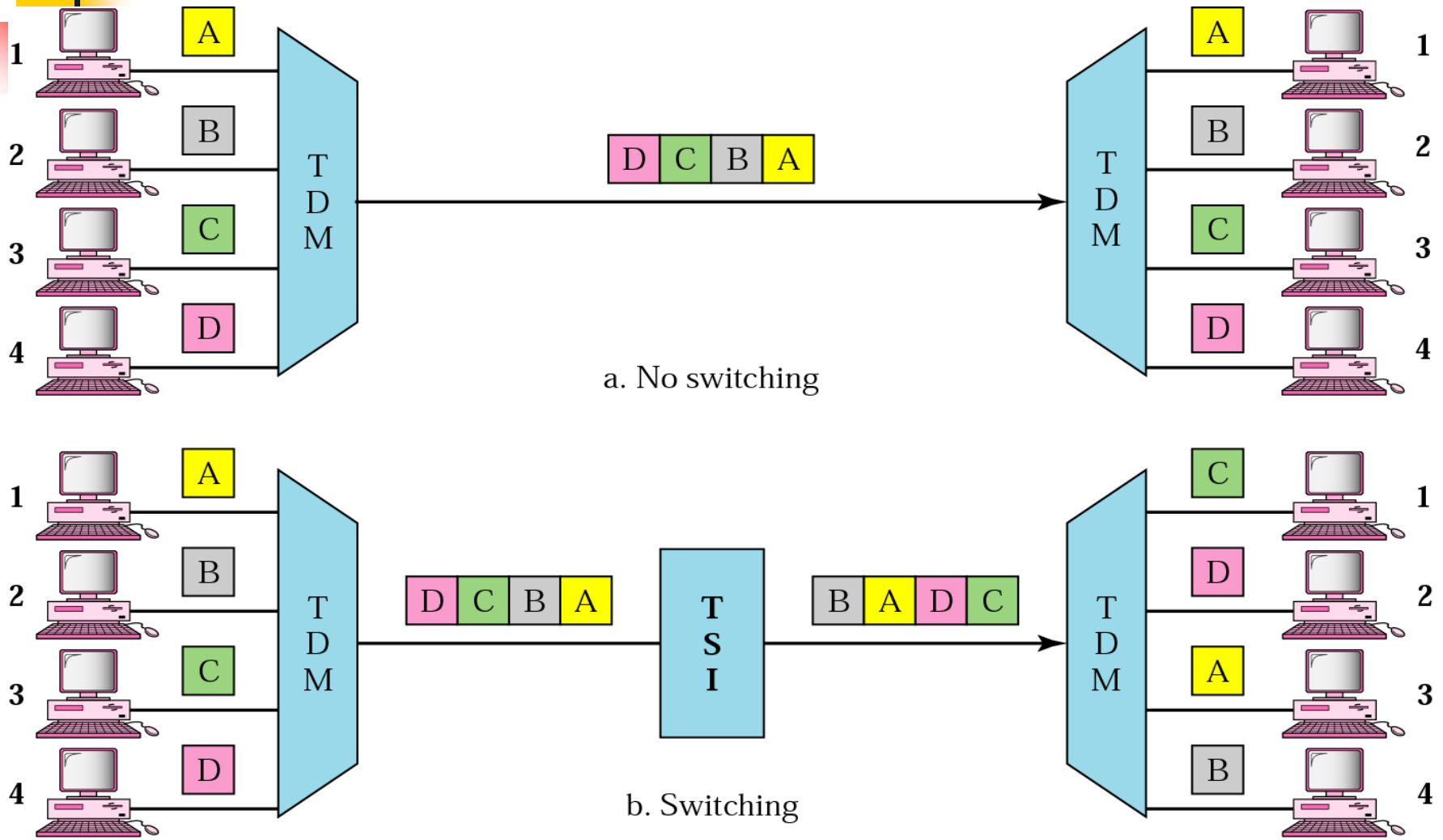



a. First option

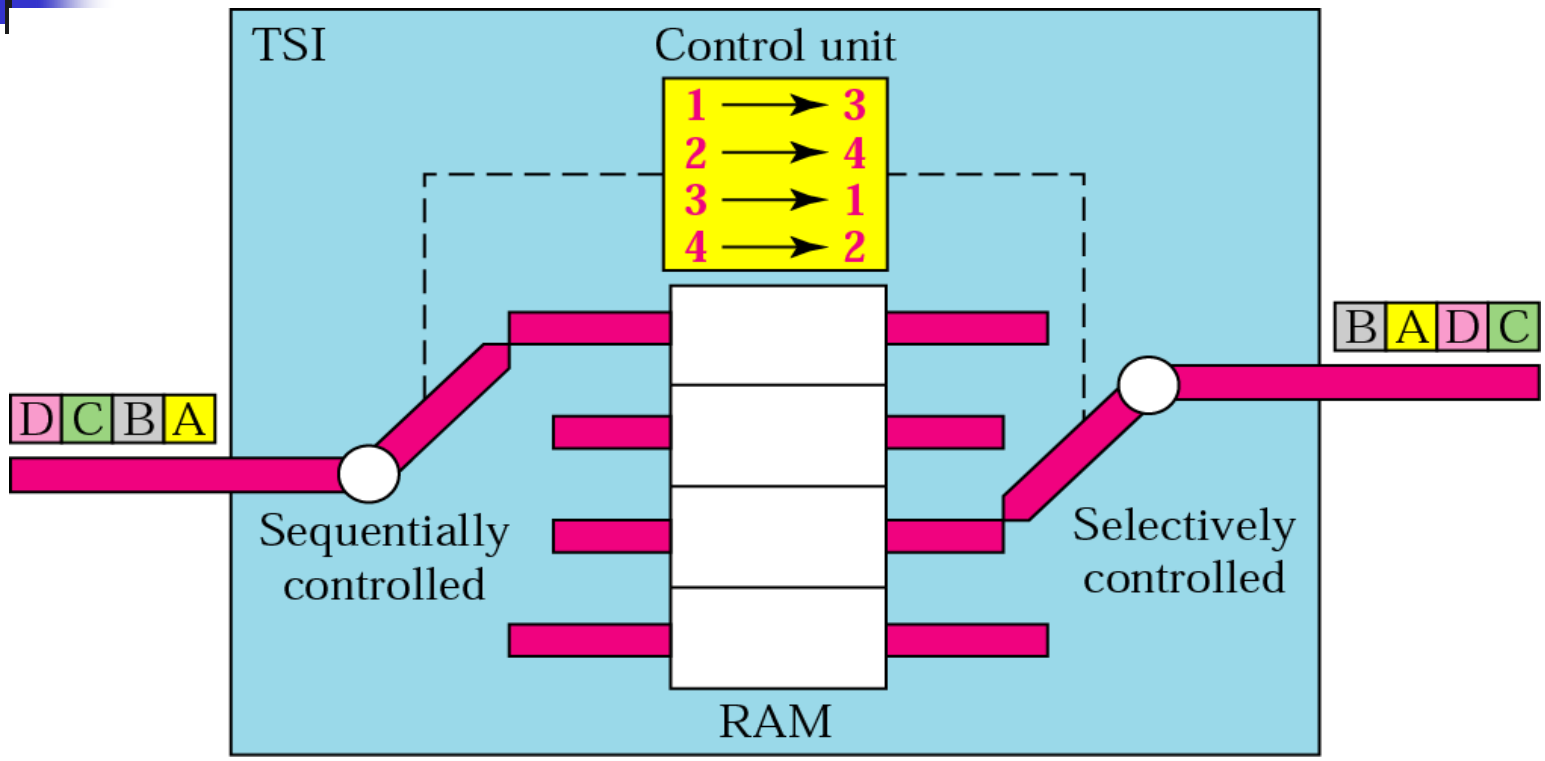



b. Second option

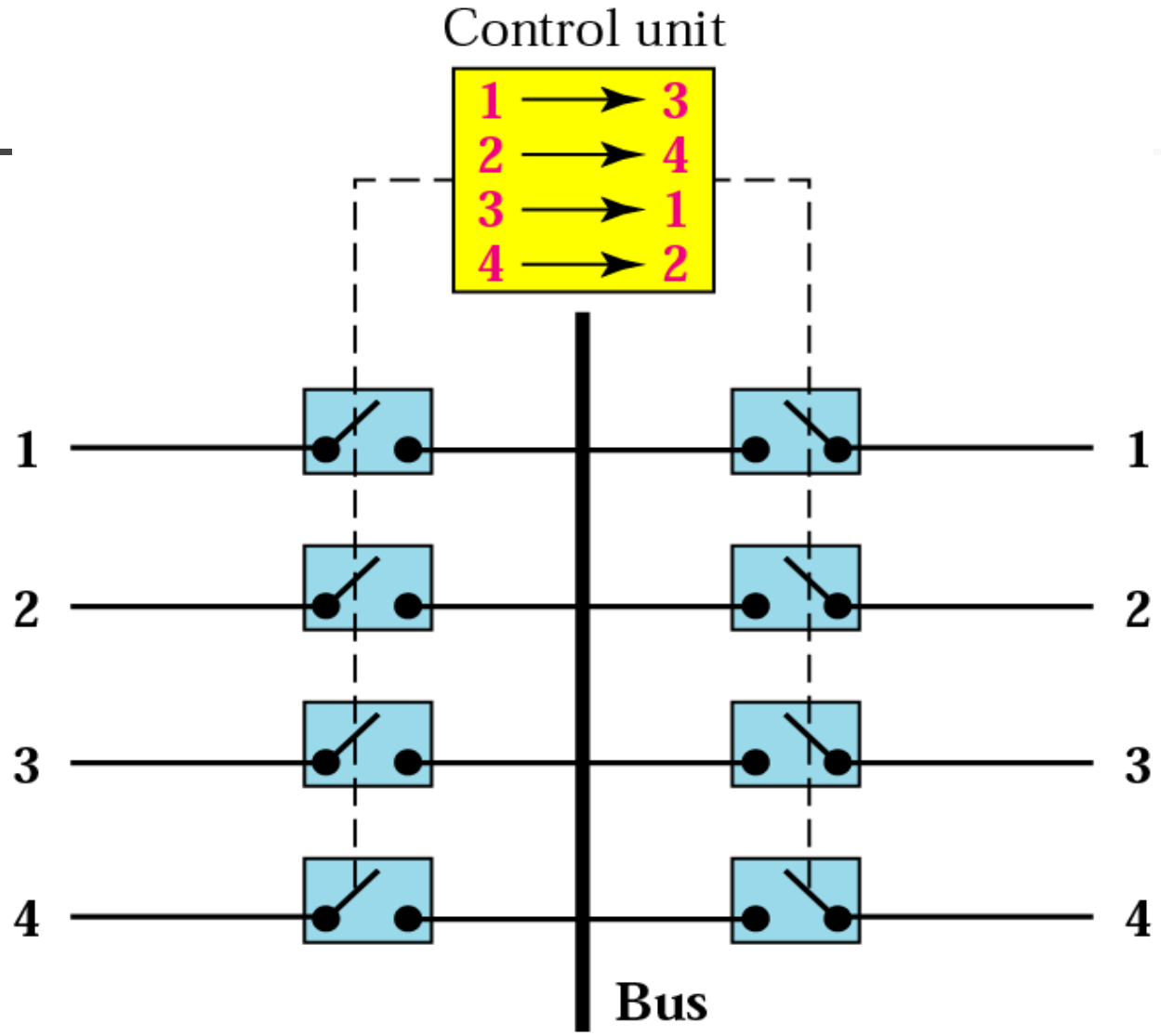
# *Time-division multiplexing, without and with a time-slot interchange*



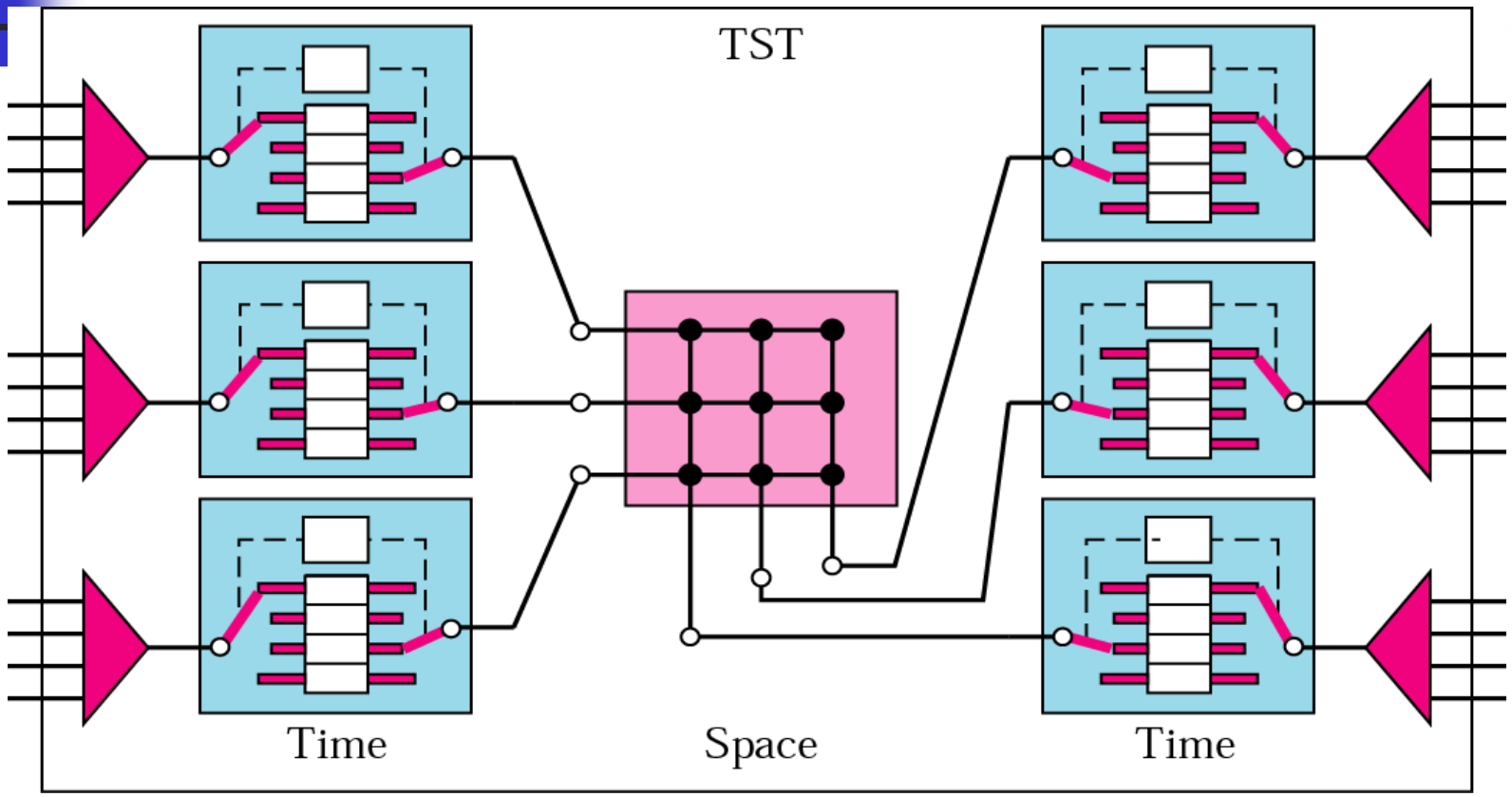
# *Time-slot interchange*



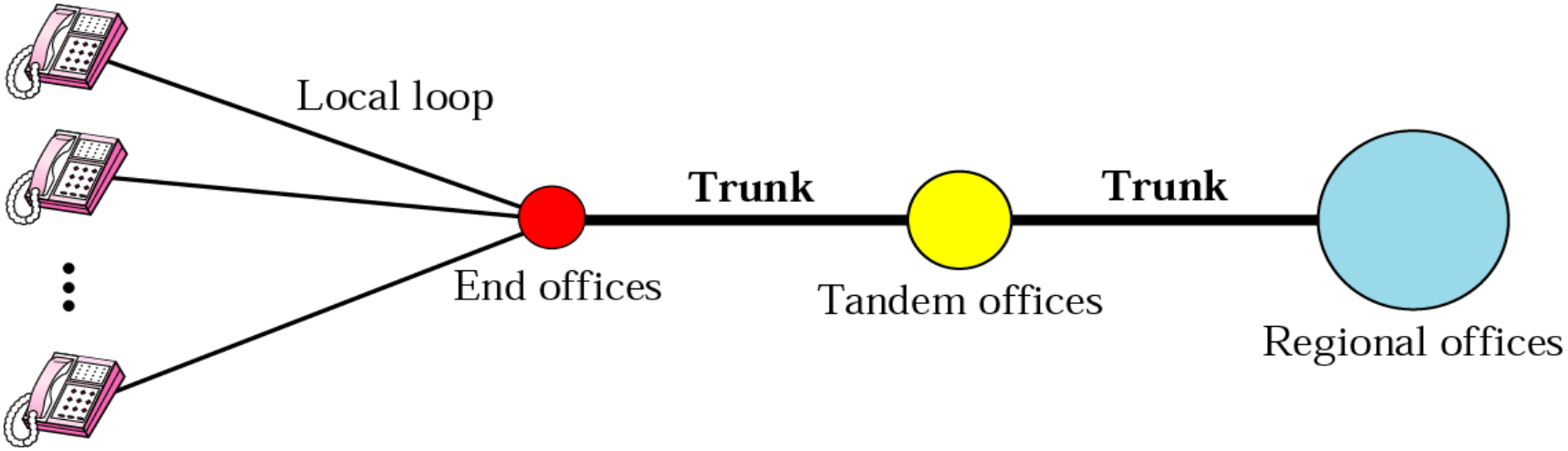
# *TDM bus*



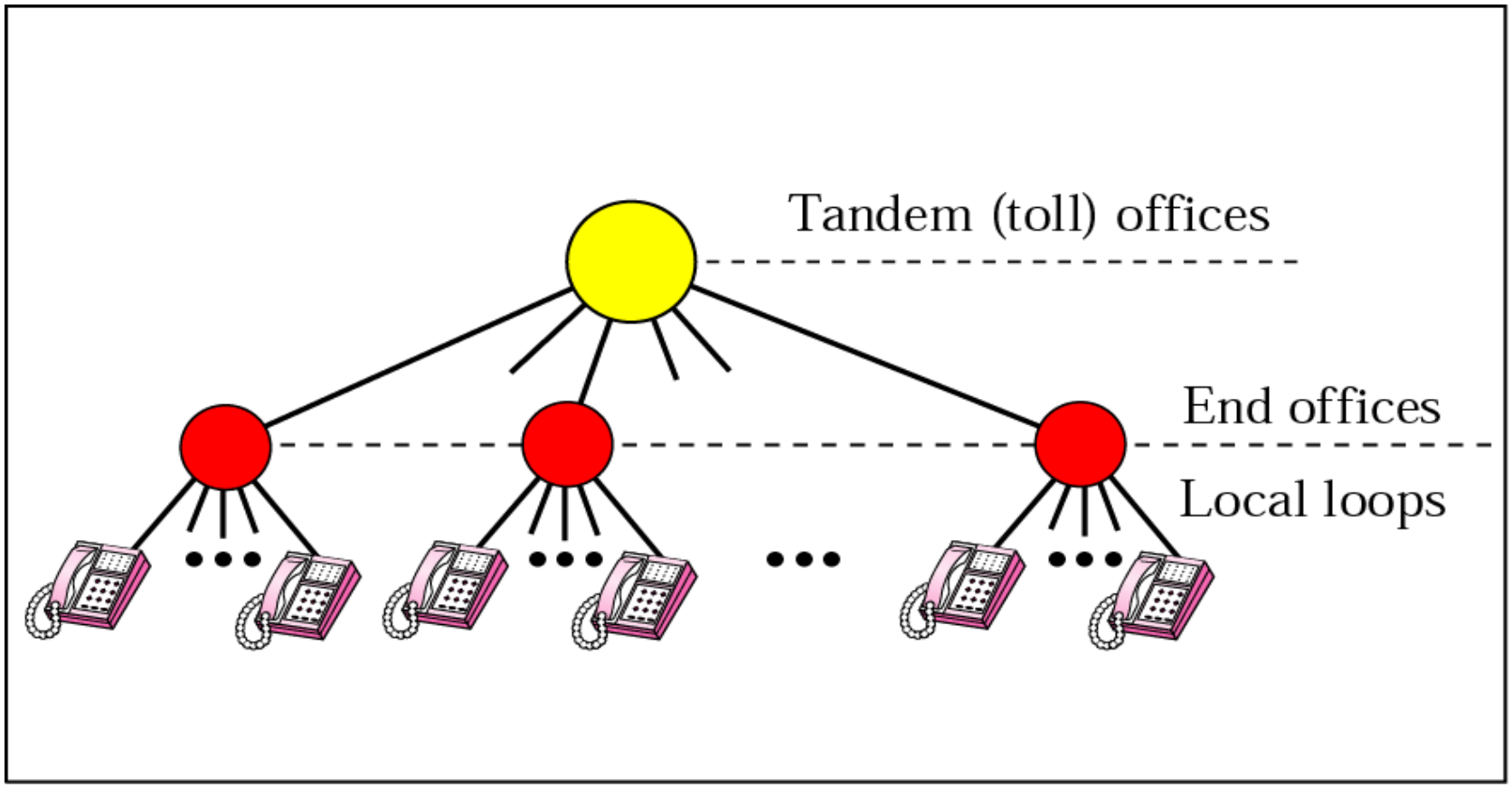
# *TST switch*



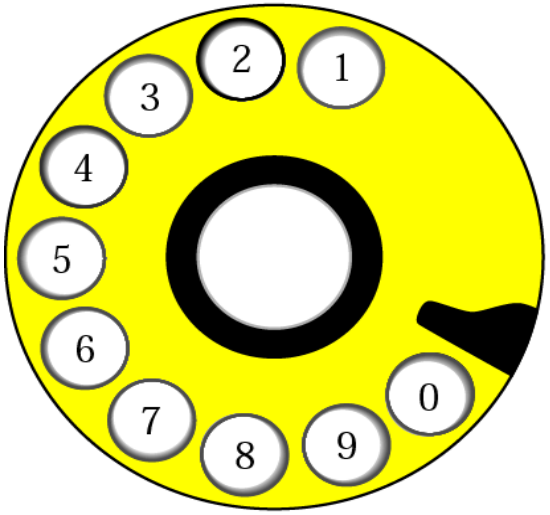
# A telephone system



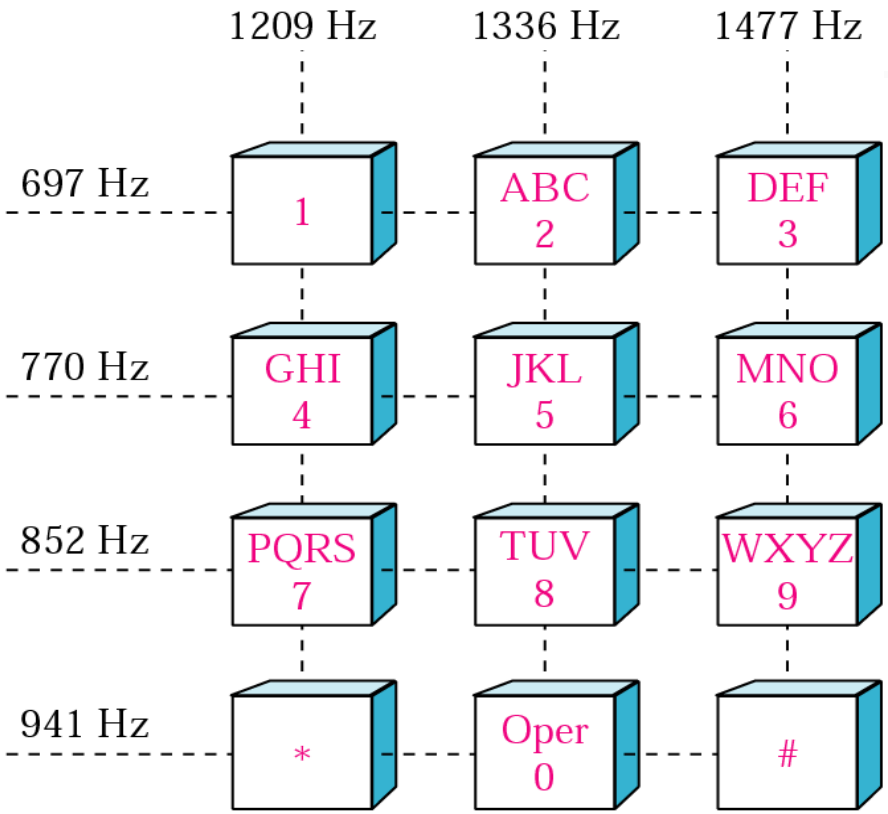
# *Switching offices in a LATA* (Local Access and Transportation Area)



# *Rotary and touch-tone dialing*




Rotary




Touch-Tone